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Closing: the two habits that matter most

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The two habits to carry into the exam.

Algebra / Calculus / Trig Quick Reference

1. Quadratic formula

For $ax^2 + bx + c = 0$ ($a \neq 0$):

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Discriminant $D = b^2 - 4ac$
- $D > 0 \rightarrow$ two distinct real roots
- $D = 0 \rightarrow$ one repeated real root, $x = -\frac{b}{2a}$
- $D < 0 \rightarrow$ two complex conjugate roots, $x = \frac{-b \pm i\sqrt{-D}}{2a}$

Related forms:

- Sum of roots: $x_1 + x_2 = -\frac{b}{a}$
- Product of roots: $x_1 x_2 = \frac{c}{a}$
- Vertex form via completing the square: $a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right)$

2. Transformation steps for $y = a \cdot f(b \cdot (x - h)) + k$

Parameter	Meaning	Effect on graph
a	vertical scale factor	stretch if $ a > 1$, compress if $0 < a < 1$; reflects over the x-axis if $a < 0$
b	horizontal scale factor (inside the function)	stretch if $0 < b < 1$, compress if $ b > 1$ (i.e. actual horizontal factor is $1/b$); reflects over the y-axis if $b < 0$
h	horizontal shift	shifts right if $h > 0$, left if $h < 0$
k	vertical shift	shifts up if $k > 0$, down if $k < 0$

Order of transformations (apply to the base graph $y = f(x)$)

Because the argument is written as $b \cdot (x - h)$ (b is factored out first), horizontal operations must be applied in this order:

1. Horizontal stretch/compression by factor $1/b$ (and reflect over y-axis if $b < 0$)
2. Horizontal shift by h (left/right)

3. Vertical stretch/compression by factor a (and reflect over x-axis if $a < 0$)
4. Vertical shift by k (up/down)

3. Techniques for evaluating limits

General order of attack: try direct substitution first; if it produces an indeterminate form, pick a technique below based on the shape of the expression.

Indeterminate forms

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \cdot \infty, 1^\infty, 0^0, \infty^0$$

Core techniques

1. **Direct substitution** - plug in the value; works whenever the function is continuous there.
2. **Factoring** - factor numerator/denominator and cancel the common factor causing $0/0$. - e.g.

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} = 4$$
3. **Rationalizing (multiply by the conjugate)** - for expressions with a single square root causing $0/0$; multiply top and bottom by the conjugate of the radical expression.
4. **Multiply by conjugate of both numerator and denominator** - when both numerator and denominator contain radicals; multiply by each conjugate in turn (or their product) to clear both radicals.
5. **Divide by the highest power of x in the denominator** - for limits as $x \rightarrow \pm\infty$ of rational functions. - degree(num) < degree(denom) $\rightarrow$ limit is 0 - degree(num) = degree(denom) $\rightarrow$ limit is ratio of leading coefficients - degree(num) > degree(denom) $\rightarrow$ limit is $\pm\infty$ (no horizontal asymptote)
6. **L'Hopital's Rule** - for $0/0$ or ∞/∞ : $\lim f(x)/g(x) = \lim f'(x)/g'(x)$, provided the new limit exists (can reapply repeatedly).
7. **Squeeze (Sandwich) Theorem** - if $g(x) \leq f(x) \leq h(x)$ near a point and $\lim g = \lim h = L$, then $\lim f = L$.
8. **Special trig limits** - $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$ - $\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x} = 0$ - $\lim_{x \rightarrow 0} \frac{\tan(x)}{x} = 1$
9. **One-sided limits** - evaluate left-hand and right-hand limits separately; the two-sided limit exists only if they agree. Required for piecewise functions and functions with absolute values or vertical asymptotes.
10. **Absolute value expressions** - split into cases based on the sign of the inner expression, then take the matching one-sided limit.
11. **Substitution / change of variable** - let $u = g(x)$ to turn an unfamiliar limit into a standard one (e.g. as $x \rightarrow a$, $u \rightarrow g(a)$).

12. **The e limit** - $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$, and more generally $\lim_{x \rightarrow 0} (1 + x)^{1/x} = e$. Useful for 1^∞ forms.

Workarounds for the harder indeterminate forms

- $\infty - \infty$: combine into a single fraction (common denominator), or rationalize if radicals are involved, then reduce to $0/0$ or ∞/∞ .
- $0 \cdot \infty$: rewrite one factor as a reciprocal to convert to $0/0$ or ∞/∞ , then apply L'Hopital or algebraic simplification.
- $1^\infty, 0^0, \infty^0$: take the natural log of the expression, evaluate $\lim \ln(y)$ (usually now $0 \cdot \infty$ or $0/0$), then exponentiate: $\lim y = e^{\lim \ln(y)}$.
- **Limits at infinity with nested radicals**: factor the highest power of x out from under each radical before simplifying.

4. Factoring formulas (powers 2 through 5)

$$a^2 - b^2 = (a - b)(a + b)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^4 - b^4 = (a - b)(a + b)(a^2 + b^2)$$

$$a^4 + b^4 = (a^2 + \sqrt{2}ab + b^2)(a^2 - \sqrt{2}ab + b^2)$$

$$a^5 - b^5 = (a - b)(a^4 + a^3b + a^2b^2 + ab^3 + b^4)$$

$$a^5 + b^5 = (a + b)(a^4 - a^3b + a^2b^2 - ab^3 + b^4)$$

$$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + b^{n-1})$$

5. Trigonometric identities

Pythagorean identities

$$\sin^2(x) + \cos^2(x) = 1$$

$$1 + \tan^2(x) = \sec^2(x)$$

$$1 + \cot^2(x) = \csc^2(x)$$

Reciprocal identities

$$\csc(x) = \frac{1}{\sin(x)} \quad \sec(x) = \frac{1}{\cos(x)} \quad \cot(x) = \frac{1}{\tan(x)}$$

Quotient identities

$$\tan(x) = \frac{\sin(x)}{\cos(x)} \quad \cot(x) = \frac{\cos(x)}{\sin(x)}$$

Even/odd (negative angle) identities

$$\sin(-x) = -\sin(x) \quad [\text{odd}]$$

$$\cos(-x) = \cos(x) \quad [\text{even}]$$

$$\tan(-x) = -\tan(x) \quad [\text{odd}]$$

Co-function identities

$$\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$$

$$\cos\left(\frac{\pi}{2} - x\right) = \sin(x)$$

$$\tan\left(\frac{\pi}{2} - x\right) = \cot(x)$$

Sum and difference identities

$$\sin(a \pm b) = \sin(a) \cos(b) \pm \cos(a) \sin(b)$$

$$\cos(a \pm b) = \cos(a) \cos(b) \mp \sin(a) \sin(b)$$

$$\tan(a \pm b) = \frac{\tan(a) \pm \tan(b)}{1 \mp \tan(a) \tan(b)}$$

Double angle identities

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$\cos(2x) = \cos^2(x) - \sin^2(x) = 1 - 2 \sin^2(x) = 2 \cos^2(x) - 1$$

$$\tan(2x) = \frac{2 \tan(x)}{1 - \tan^2(x)}$$

Power-reduction identities (rearranged double angle)

$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \iff 1 - \cos(2x) = 2 \sin^2(x)$$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2} \iff 1 + \cos(2x) = 2 \cos^2(x)$$

$$\tan^2(x) = \frac{1 - \cos(2x)}{1 + \cos(2x)}$$

Half-angle identities

$$\begin{aligned}\sin\left(\frac{x}{2}\right) &= \pm\sqrt{\frac{1-\cos(x)}{2}} \\ \cos\left(\frac{x}{2}\right) &= \pm\sqrt{\frac{1+\cos(x)}{2}} \\ \tan\left(\frac{x}{2}\right) &= \frac{1-\cos(x)}{\sin(x)} = \frac{\sin(x)}{1+\cos(x)}\end{aligned}$$

(the $\pm$ sign is chosen based on which quadrant $x/2$ falls in)

Product-to-sum identities

$$\begin{aligned}\sin(a)\cos(b) &= \frac{1}{2}[\sin(a+b) + \sin(a-b)] \\ \cos(a)\sin(b) &= \frac{1}{2}[\sin(a+b) - \sin(a-b)] \\ \cos(a)\cos(b) &= \frac{1}{2}[\cos(a-b) + \cos(a+b)] \\ \sin(a)\sin(b) &= \frac{1}{2}[\cos(a-b) - \cos(a+b)]\end{aligned}$$

Sum-to-product identities

$$\begin{aligned}\sin(a) + \sin(b) &= 2\sin\left(\frac{a+b}{2}\right)\cos\left(\frac{a-b}{2}\right) \\ \sin(a) - \sin(b) &= 2\cos\left(\frac{a+b}{2}\right)\sin\left(\frac{a-b}{2}\right) \\ \cos(a) + \cos(b) &= 2\cos\left(\frac{a+b}{2}\right)\cos\left(\frac{a-b}{2}\right) \\ \cos(a) - \cos(b) &= -2\sin\left(\frac{a+b}{2}\right)\sin\left(\frac{a-b}{2}\right)\end{aligned}$$

Table 1 of 2 - sine, cosine and tangent

The two tables below are the full exact-value reference for every multiple of 15° from 0° to 360° , split into two five-column tables so each fits a phone screen held sideways. The header marks **cos x** and **sec x** as the even functions, which is the first move on any negative angle.

Do not try to memorize 33 rows. Reduce the angle to a reference angle in $[0^\circ, 90^\circ]$, read the magnitude off the top of the table, then attach the sign from ASTC.

deg	rad	$\sin x$	$\cos x$ (even)	$\tan x$
0	0	0	1	0

deg	rad		(even)	
15	$\frac{\pi}{12}$	$\frac{\sqrt{6} - \sqrt{2}}{4}$	$\frac{\sqrt{6} + \sqrt{2}}{4}$	$2 - \sqrt{3}$
22.5	$\frac{\pi}{8}$	$\frac{\sqrt{2} - \sqrt{2}}{2}$	$\frac{\sqrt{2} + \sqrt{2}}{2}$	$\sqrt{2} - 1$
30	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
67.5	$\frac{3\pi}{8}$	$\frac{\sqrt{2} + \sqrt{2}}{2}$	$\frac{\sqrt{2} - \sqrt{2}}{2}$	$\sqrt{2} + 1$
75	$\frac{5\pi}{12}$	$\frac{\sqrt{6} + \sqrt{2}}{4}$	$\frac{\sqrt{6} - \sqrt{2}}{4}$	$2 + \sqrt{3}$
90	$\frac{\pi}{2}$	1	0	undefined
105	$\frac{7\pi}{12}$	$\frac{\sqrt{6} + \sqrt{2}}{4}$	$-\frac{\sqrt{6} - \sqrt{2}}{4}$	$-(2 + \sqrt{3})$
112.5	$\frac{5\pi}{8}$	$\frac{\sqrt{2} + \sqrt{2}}{2}$	$-\frac{\sqrt{2} - \sqrt{2}}{2}$	$-(\sqrt{2} + 1)$
120	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-(\sqrt{3})$
135	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	$-(1)$
150	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
157.5	$\frac{7\pi}{8}$	$\frac{\sqrt{2} - \sqrt{2}}{2}$	$-\frac{\sqrt{2} + \sqrt{2}}{2}$	$-(\sqrt{2} - 1)$
165	$\frac{11\pi}{12}$	$\frac{\sqrt{6} - \sqrt{2}}{4}$	$-\frac{\sqrt{6} + \sqrt{2}}{4}$	$-(2 - \sqrt{3})$
180	π	0	$-(1)$	0
195	$\frac{13\pi}{12}$	$-\frac{\sqrt{6} - \sqrt{2}}{4}$	$-\frac{\sqrt{6} + \sqrt{2}}{4}$	$2 - \sqrt{3}$
202.5	$\frac{9\pi}{8}$	$-\frac{\sqrt{2} - \sqrt{2}}{2}$	$-\frac{\sqrt{2} + \sqrt{2}}{2}$	$\sqrt{2} - 1$

deg	rad		(even)	
210	$\frac{7\pi}{6}$	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
225	$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	1
240	$\frac{4\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\sqrt{3}$
247.5	$\frac{11\pi}{8}$	$-\frac{\sqrt{2+\sqrt{2}}}{2}$	$-\frac{\sqrt{2-\sqrt{2}}}{2}$	$\sqrt{2}+1$
255	$\frac{17\pi}{12}$	$-\frac{\sqrt{6+\sqrt{2}}}{4}$	$-\frac{\sqrt{6-\sqrt{2}}}{4}$	$2+\sqrt{3}$
270	$\frac{3\pi}{2}$	-(1)	0	undefined
285	$\frac{19\pi}{12}$	$-\frac{\sqrt{6+\sqrt{2}}}{4}$	$\frac{\sqrt{6-\sqrt{2}}}{4}$	$-(2+\sqrt{3})$
292.5	$\frac{13\pi}{8}$	$-\frac{\sqrt{2+\sqrt{2}}}{2}$	$\frac{\sqrt{2-\sqrt{2}}}{2}$	$-(\sqrt{2}+1)$
300	$\frac{5\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-(\sqrt{3})$
315	$\frac{7\pi}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	-(1)
330	$\frac{11\pi}{6}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
337.5	$\frac{15\pi}{8}$	$-\frac{\sqrt{2-\sqrt{2}}}{2}$	$\frac{\sqrt{2+\sqrt{2}}}{2}$	$-(\sqrt{2}-1)$
345	$\frac{23\pi}{12}$	$-\frac{\sqrt{6-\sqrt{2}}}{4}$	$\frac{\sqrt{6+\sqrt{2}}}{4}$	$-(2-\sqrt{3})$
360	2π	0	1	0

Table 2 of 2 - cosecant, secant and cotangent

deg	rad	csc x	sec x (even)	cot x
0	0	undefined	1	undefined
15	$\frac{\pi}{12}$	$\sqrt{6+\sqrt{2}}$	$\sqrt{6-\sqrt{2}}$	$2+\sqrt{3}$
22.5	$\frac{\pi}{8}$	$\sqrt{4+2\sqrt{2}}$	$\sqrt{4-2\sqrt{2}}$	$\sqrt{2}+1$

deg	rad		(even)	
30	$\frac{\pi}{6}$	2	$\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
45	$\frac{\pi}{4}$	$\sqrt{2}$	$\sqrt{2}$	1
60	$\frac{\pi}{3}$	$\frac{2\sqrt{3}}{3}$	2	$\frac{\sqrt{3}}{3}$
67.5	$\frac{3\pi}{8}$	$\sqrt{4-2\sqrt{2}}$	$\sqrt{4+2\sqrt{2}}$	$\sqrt{2}-1$
75	$\frac{5\pi}{12}$	$\sqrt{6}-\sqrt{2}$	$\sqrt{6}+\sqrt{2}$	$2-\sqrt{3}$
90	$\frac{\pi}{2}$	1	undefined	0
105	$\frac{7\pi}{12}$	$\sqrt{6}-\sqrt{2}$	$-(\sqrt{6}+\sqrt{2})$	$-(2-\sqrt{3})$
112.5	$\frac{5\pi}{8}$	$\sqrt{4-2\sqrt{2}}$	$-(\sqrt{4+2\sqrt{2}})$	$-(\sqrt{2}-1)$
120	$\frac{2\pi}{3}$	$\frac{2\sqrt{3}}{3}$	$-(2)$	$-\frac{\sqrt{3}}{3}$
135	$\frac{3\pi}{4}$	$\sqrt{2}$	$-(\sqrt{2})$	$-(1)$
150	$\frac{5\pi}{6}$	2	$-\frac{2\sqrt{3}}{3}$	$-(\sqrt{3})$
157.5	$\frac{7\pi}{8}$	$\sqrt{4+2\sqrt{2}}$	$-(\sqrt{4-2\sqrt{2}})$	$-(\sqrt{2}+1)$
165	$\frac{11\pi}{12}$	$\sqrt{6}+\sqrt{2}$	$-(\sqrt{6}-\sqrt{2})$	$-(2+\sqrt{3})$
180	π	undefined	$-(1)$	undefined
195	$\frac{13\pi}{12}$	$-(\sqrt{6}+\sqrt{2})$	$-(\sqrt{6}-\sqrt{2})$	$2+\sqrt{3}$
202.5	$\frac{9\pi}{8}$	$-(\sqrt{4+2\sqrt{2}})$	$-(\sqrt{4-2\sqrt{2}})$	$\sqrt{2}+1$
210	$\frac{7\pi}{6}$	$-(2)$	$-\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
225	$\frac{5\pi}{4}$	$-(\sqrt{2})$	$-(\sqrt{2})$	1
240	$\frac{4\pi}{3}$	$-\frac{2\sqrt{3}}{3}$	$-(2)$	$\frac{\sqrt{3}}{3}$
247.5	$\frac{11\pi}{8}$	$-(\sqrt{4-2\sqrt{2}})$	$-(\sqrt{4+2\sqrt{2}})$	$\sqrt{2}-1$

deg	rad		(even)	
255	$\frac{17\pi}{12}$	$-(\sqrt{6} - \sqrt{2})$	$-(\sqrt{6} + \sqrt{2})$	$2 - \sqrt{3}$
270	$\frac{3\pi}{2}$	$-(1)$	undefined	0
285	$\frac{19\pi}{12}$	$-(\sqrt{6} - \sqrt{2})$	$\sqrt{6} + \sqrt{2}$	$-(2 - \sqrt{3})$
292.5	$\frac{13\pi}{8}$	$-(\sqrt{4 - 2\sqrt{2}})$	$\sqrt{4 + 2\sqrt{2}}$	$-(\sqrt{2} - 1)$
300	$\frac{5\pi}{3}$	$-\frac{2\sqrt{3}}{3}$	2	$-\frac{\sqrt{3}}{3}$
315	$\frac{7\pi}{4}$	$-(\sqrt{2})$	$\sqrt{2}$	$-(1)$
330	$\frac{11\pi}{6}$	$-(2)$	$\frac{2\sqrt{3}}{3}$	$-(\sqrt{3})$
337.5	$\frac{15\pi}{8}$	$-(\sqrt{4 + 2\sqrt{2}})$	$\sqrt{4 - 2\sqrt{2}}$	$-(\sqrt{2} + 1)$
345	$\frac{23\pi}{12}$	$-(\sqrt{6} + \sqrt{2})$	$\sqrt{6} - \sqrt{2}$	$-(2 + \sqrt{3})$
360	2π	undefined	1	undefined

How to read the two tables:

- The five rows to have cold are 0° , 30° , 45° , 60° , 90° .
- Everything else is one of those five with a sign change from ASTC, or a member of the 15-degree family or the 22.5-degree family.
- The 15-degree family uses only $\frac{\sqrt{6} + \sqrt{2}}{4}$ and $\frac{\sqrt{6} - \sqrt{2}}{4}$ for sine and cosine, and only $2 + \sqrt{3}$ and $2 - \sqrt{3}$ for tangent.
- The 22.5-degree family uses only $\frac{\sqrt{2 + \sqrt{2}}}{2}$ and $\frac{\sqrt{2 - \sqrt{2}}}{2}$, and only $\sqrt{2} + 1$ and $\sqrt{2} - 1$.
- Table 2 is Table 1 turned upside down, sign and all, because a reciprocal keeps the sign of what it came from.
- $|\csc x|$ and $|\sec x|$ are never less than 1, which is a free sanity check.
- $\tan x$ and $\sec x$ are undefined at 90° and 270° , where cosine is zero.
- $\cot x$ and $\csc x$ are undefined at 0° , 180° and 360° , where sine is zero.
- Tangent and cotangent repeat every 180° with no sign change, which is why 15° and 195° share $\tan x = 2 - \sqrt{3}$.

- For an angle outside $[0^\circ, 360^\circ]$, add or subtract full turns first: $\frac{25\pi}{12} = 375^\circ$ is coterminal with 15° .

Limits: Definitions and Laws

1. Formal definition of a limit

$$\lim_{x \rightarrow a} f(x) = L$$

2. One-sided limits

Notation	Meaning
$\lim_{x \rightarrow a^-} f(x) = L$	For every $\varepsilon > 0$ there is a $\delta > 0$ such that $a - \delta < x < a \Rightarrow f(x) - L < \varepsilon$
$\lim_{x \rightarrow a^+} f(x) = L$	For every $\varepsilon > 0$ there is a $\delta > 0$ such that $a < x < a + \delta \Rightarrow f(x) - L < \varepsilon$

Existence rule:

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L$$

If the two one-sided limits disagree, or either fails to exist, the two-sided limit does not exist.

3. Algebra of limits (limit laws)

Assume $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, both finite, and c is a constant.

Law	Formula
Sum	$\lim_{x \rightarrow a} [f(x) + g(x)] = L + M$
Difference	$\lim_{x \rightarrow a} [f(x) - g(x)] = L - M$
Constant multiple	$\lim_{x \rightarrow a} [c f(x)] = cL$
Product	$\lim_{x \rightarrow a} [f(x)g(x)] = LM$
Quotient	$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}$, provided $M \neq 0$
Power	$\lim_{x \rightarrow a} [f(x)]^n = L^n$ for any positive integer n
Root	$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{L}$, provided $L \geq 0$ when n is even
Constant function	$\lim_{x \rightarrow a} c = c$
Identity function	$\lim_{x \rightarrow a} x = a$

Law	Formula
Polynomial / direct substitution	$\lim_{x \rightarrow a} f(x) = f(a)$ whenever f is a polynomial (or any function continuous at a)
Composition	If g is continuous at L and $\lim_{x \rightarrow a} f(x) = L$, then $\lim_{x \rightarrow a} g(f(x)) = g(L)$

4. Limits involving infinity

Infinite limit (vertical asymptote), formal definition:

$$\lim_{x \rightarrow a} f(x) = \infty$$

Limit at infinity (horizontal asymptote), formal definition:

$$\lim_{x \rightarrow \infty} f(x) = L$$

Basic power-function limits:

Limit	Value	
$\lim_{x \rightarrow \infty} x^p \ (p > 0)$	∞	
$\lim_{x \rightarrow \infty} \frac{1}{x^p} \ (p > 0)$	0	
$\lim_{x \rightarrow -\infty} \frac{1}{x^p} \ (p > 0, \text{ defined})$	0	
$\lim_{x \rightarrow \infty} \frac{c}{x^p} \ (\text{any constant } c, p > 0)$	0	

Rational function at infinity (degree comparison), given

$$f(x) = \frac{a_n x^n + \cdots + a_0}{b_m x^m + \cdots + b_0} :$$

Degree comparison	$\lim_{x \rightarrow \pm\infty} f(x)$
$n < m$ (numerator degree smaller)	0
$n = m$ (equal degree)	$\frac{a_n}{b_m}$ (ratio of leading coefficients)
$n > m$ (numerator degree bigger)	$\pm\infty$, sign from the leading coefficients and the direction of approach

The standard technique is to divide numerator and denominator by x^m (the highest power in the denominator) before taking the limit; that mechanical step is what these three rows summarize.

5. Squeeze (sandwich) theorem

If $g(x) \leq f(x) \leq h(x)$

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L,$$

then

$$\lim_{x \rightarrow a} f(x) = L.$$

6. L'Hopital's rule

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, and f and g are differentiable near a with $g'(x) \neq 0$ near a (except possibly at a), then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

7. Continuity, stated as a limit equation

f is continuous at $x = a$ if and only if all three conditions hold:

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. $\lim_{x \rightarrow a} f(x) = f(a)$.

8. The derivative, defined as a limit

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

9. The definite integral, defined as a limit of Riemann sums

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x, \quad \Delta x = \frac{b-a}{n}$$

10. Limits of sequences

Formal definition:

$$\lim_{n \rightarrow \infty} a_n = L$$

means: for every $\varepsilon > 0$, there is an integer N such that $n > N \Rightarrow |a_n - L| < \varepsilon$.

Squeeze theorem for sequences:

if $b_n \leq a_n \leq c_n$ for all n

and $\lim b_n = \lim c_n = L$

then $\lim a_n = L$

Special Limits in Calculus

1. The trigonometric limits ($x \rightarrow 0$)

These are the ones almost every other trig limit reduces to.

Limit	Value	How it is normally proved
$\lim_{x \rightarrow 0} \frac{\sin x}{x}$	1	Squeeze theorem, using $\cos x \leq \frac{\sin x}{x} \leq 1$ near 0
$\lim_{x \rightarrow 0} \frac{x}{\sin x}$	1	Reciprocal of the above
$\lim_{x \rightarrow 0} \frac{\tan x}{x}$	1	$\frac{\tan x}{x} = \frac{\sin x}{x} \cdot \frac{1}{\cos x} \rightarrow 1 \cdot 1$
$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$	0	Multiply by the conjugate $\frac{1 + \cos x}{1 + \cos x}$
$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$	$\frac{1}{2}$	Same conjugate trick, or the Taylor series $\cos x \approx 1 - \frac{x^2}{2}$
$\lim_{x \rightarrow 0} \frac{\arcsin x}{x}$	1	Substitute $u = \arcsin x$, so $x = \sin u$, and reuse $\sin u/u \rightarrow 1$
$\lim_{x \rightarrow 0} \frac{\arctan x}{x}$	1	Substitute $u = \arctan x$, so $x = \tan u$, and reuse $\tan u/u \rightarrow 1$
$\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx}$	$\frac{a}{b}$ ($b \neq 0$)	The general pattern behind every "manufacture the matching denominator" trick

Limit	Value	
$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$	$\frac{1}{6}$	
$\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$	$\frac{1}{3}$	
$\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3}$	$-\frac{1}{2}$	

2. The exponential and logarithmic limits ($x \rightarrow 0$)

Limit	Value	Note
$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$	1	Derivative of e^x at 0
$\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$	$\ln a$ ($a > 0$)	Generalizes the row above ($a = e$ gives $\ln e = 1$)

Limit	Value	Note
$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$	1	Derivative of $\ln x$ at 1
$\lim_{x \rightarrow 0} \frac{\log_a(1+x)}{x}$	$\frac{1}{\ln a}$	Change-of-base version of the row above
$\lim_{x \rightarrow 0} (1+x)^{1/x}$	e	The defining limit for e

3. The number e as a limit at infinity

Limit	Value	
$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$	e	
$\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x$	e^a	
$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n$	e^{-1}	

4. Growth-rate hierarchy at infinity

Limit	Value	
$\lim_{x \rightarrow \infty} \frac{\ln x}{x}$	0	
$\lim_{x \rightarrow \infty} \frac{\ln x}{x^p} \ (p > 0)$	0	
$\lim_{x \rightarrow 0^+} x \ln x$	0	
$\lim_{x \rightarrow 0^+} x^p \ln x \ (p > 0)$	0	
$\lim_{x \rightarrow \infty} \frac{x^n}{e^x} \text{ (any fixed } n)$	0	
$\lim_{x \rightarrow \infty} \frac{e^x}{x^n} \text{ (any fixed } n)$	∞	
$\lim_{n \rightarrow \infty} \frac{a^n}{n!} \text{ (any fixed } a)$	0	
$\lim_{n \rightarrow \infty} \frac{n!}{n^n}$	0	

5. Root and power limits

Limit	Value	
$\lim_{x \rightarrow \infty} x^{1/x}$	1	
$\lim_{n \rightarrow \infty} n^{1/n}$	1	
$\lim_{n \rightarrow \infty} a^{1/n}$ ($a > 0$ fixed)	1	
$\lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n}$	e^{-1}	

6. Rationalizing and "hidden derivative" limits

Limit	Value	Note
$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$	$\frac{1}{2}$	Derivative of $\sqrt{x}$ at 1
$\lim_{x \rightarrow 0} \frac{\sqrt{a+x} - \sqrt{a}}{x}$ ($a > 0$)	$\frac{1}{2\sqrt{a}}$	General case of the row above
$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$	$\frac{1}{2}$	Multiply by the conjugate $\frac{\sqrt{x^2 + x} + x}{\sqrt{x^2 + x} + x}$
$\lim_{x \rightarrow \infty} (\sqrt{x^2 + ax + b} - x)$	$\frac{a}{2}$	General case of the row above

Internalize the pattern: limit of form $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ is, by definition, $f'(a)$.

7. Squeeze-theorem classics

Cases where the limit does not exist by substitution or algebra alone, only by trapping the expression between two functions that share the same limit.

Limit	Value	Squeeze used
$\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$	0	$- x \leq x \sin(1/x) \leq x $
$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$	0	$-x^2 \leq x^2 \sin(1/x) \leq x^2$
$\lim_{x \rightarrow 0} \frac{\sin x}{x}$	1	$\cos x \leq \frac{\sin x}{x} \leq 1$ (the proof behind section 1's first row)

The tell that a limit needs squeezing rather than algebra: a bounded-but-wildly-oscillating factor like **$\sin(1/x)$** or **$\cos(1/x)$** multiplied by a factor that shrinks to **0**.

Techniques for Evaluating Limits, Derivatives, and Integrals

Part 1 - Techniques for Evaluating Limits

The single most useful habit: **always substitute first.**

L1. Direct substitution (continuity)

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 4x + 1}{x + 3}$

Steps.

Calculation	Method
$\lim_{x \rightarrow 2} \frac{x^3 - 4x + 1}{x + 3} = \lim_{x \rightarrow 2} \frac{2^3 - 4 \cdot 2 + 1}{2 + 3}$	Direct substitution ($x = 2$ into every x)
$= \lim_{x \rightarrow 2} \frac{8 - 8 + 1}{5}$	Arithmetic: evaluate the powers and products
$= \lim_{x \rightarrow 2} \frac{1}{5}$	Arithmetic: combine the numerator
$= \frac{1}{5}$	The expression is now constant, so the limit is its value

MATH265 exam question (Q3.2c).

Evaluate $\lim_{x \rightarrow 0} \frac{6x - 9}{x^3 - 12x + 3}$

Calculation	Method
$\frac{6(0) - 9}{0 - 0 + 3} = \frac{-9}{3}$	Direct substitution - included because it looks like it needs a technique and does not
$= -3$	Arithmetic; the exercise is checking that you try substitution before reaching for machinery

MATH265 exam question (Q3.1a).

Evaluate $\lim_{x \rightarrow 2} (3x^2 - 2x + 1)$

Calculation	Method
$3(2)^2 - 2(2) + 1 = 12 - 4 + 1$	Direct substitution - a polynomial is continuous everywhere, so no diagnosis step is needed
$= 9$	Arithmetic

L2. Factor and cancel

Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x^2 - x - 6}$.

Steps.

Calculation	Method
$\frac{9 - 9}{9 - 3 - 6} = \frac{0}{0}$	Substitute $x = 3$: indeterminate, so $(x - 3)$ is a factor of both
$x^2 - 9 = (x - 3)(x + 3)$	Factor the numerator
$x^2 - x - 6 = (x - 3)(x + 2)$	Factor the denominator
$\frac{(x - 3)(x + 3)}{(x - 3)(x + 2)} = \frac{x + 3}{x + 2}$	Cancel the common $(x - 3)$, valid because $x \neq 3$ while the limit is being taken; now continuous at 3
$= \frac{6}{5}$	Substitute $x = 3$

Second example (higher degree).

$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^4 - 1}$

Calculation	Method
Both vanish at $x = 1$	Diagnose the $\frac{0}{0}$ form
$x^3 - 1 = (x - 1)(x^2 + x + 1),$ $x^4 - 1 = (x - 1)(x^3 + x^2 + x + 1)$	Difference-of-powers factorization
$\frac{1 + 1 + 1}{1 + 1 + 1 + 1} = \frac{3}{4}$	Cancel the shared $(x - 1)$ and substitute $x = 1$

Pattern worth memorizing.

$$\lim_{x \rightarrow 1} \frac{x^m - 1}{x^n - 1} = \frac{m}{n}$$

, which is the general form of the example above.

MATH265 exam question (Q3.2a).

Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^3 - 1}$.

Calculation	Method
$\frac{2}{3}$ by inspection	Matches the memorized pattern $\lim_{x \rightarrow 1} \frac{x^m - 1}{x^n - 1} = \frac{m}{n}$ with $m = 2, n = 3$
$x^2 - 1 = (x - 1)(x + 1),$ $x^3 - 1 = (x - 1)(x^2 + x + 1)$	Worked from scratch: factor both sides
$\frac{1 + 1}{1 + 1 + 1} = \frac{2}{3}$	Cancel the shared $(x - 1)$ and substitute $x = 1$

MATH265 exam question (Q4.1) – factor/cancel feeding an asymptote analysis.

Find the vertical and horizontal asymptotes of

$$f(x) = \frac{x^2 - 3x - 4}{x^2 - 16}.$$

Calculation	Method
$x^2 - 3x - 4 = (x - 4)(x + 1),$ $x^2 - 16 = (x - 4)(x + 4)$	Factor first, always
$f(x) = \frac{x + 1}{x + 4}$ for $x \neq 4$	Cancel the $(x - 4)$ factor: a cancelled factor is a hole , not an asymptote, sitting at $(4, \frac{5}{8})$
$x = -4$	Surviving denominator zero is a genuine vertical asymptote, confirmed because the numerator there is $-3 \neq 0$
$y = \frac{1}{1} = 1$	Horizontal asymptote from L8 (degree comparison): both original degrees are 2
VA $x = -4$, HA $y = 1$, hole at $(4, \frac{5}{8})$	Final result

The trap this question is built around. Reporting $x = 4$ as a vertical asymptote before checking whether the offending factor cancels.

L3. Conjugate multiplication (rationalizing)

Trigger. Substitution gives $\frac{0}{0}$ and a square root appears in the numerator or denominator.

Why it works. Multiplying by $\frac{\text{conjugate}}{\text{conjugate}}$ turns $(\sqrt{A} - \sqrt{B})(\sqrt{A} + \sqrt{B})$ into $A - B$, which removes the root and exposes the cancelling factor.

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$

Steps.

Calculation	Method
$\frac{2-2}{0} = \frac{0}{0}$	Substitute $x = 0$
Multiply top and bottom by $\sqrt{x+4} + 2$	Conjugate multiplication
$(x+4) - 4 = x$	Numerator becomes a difference of squares
$\frac{x}{x(\sqrt{x+4} + 2)} = \frac{1}{\sqrt{x+4} + 2}$ for $x \neq 0$	Cancel the common x
$\frac{1}{2+2} = \frac{1}{4}$	Substitute $x = 0$

Variant with the root on the bottom.

$\lim_{x \rightarrow 9} \frac{x-9}{\sqrt{x}-3}$

: multiply by $\frac{\sqrt{x}+3}{\sqrt{x}+3}$ to get $\sqrt{x} + 3 \rightarrow 6$.

Variant with two roots.

$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$

: the conjugate gives numerator $(1+x) - (1-x) = 2x$, so the value is $\frac{2}{1+1} = 1$.

Cube roots. Use the identity

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

instead, so the multiplier for $\sqrt[3]{A} - \sqrt[3]{B}$ is $\sqrt[3]{A^2} + \sqrt[3]{AB} + \sqrt[3]{B^2}$.

MATH265 exam question (Q3.1d) – a root on both top and bottom.

Evaluate $\lim_{x \rightarrow 2} \frac{\sqrt{6-x} - 2}{\sqrt{3-x} - 1}$.

Calculation	Method
$\frac{\sqrt{4} - 2}{\sqrt{1} - 1} = \frac{0}{0}$	Substitute; two separate roots need clearing
Multiply by $(\sqrt{6-x} + 2)$ on top and $(\sqrt{3-x} + 1)$ on the bottom	Both conjugates at once, so both differences of squares appear
Numerator: $(6-x) - 4 = 2-x$; denominator: $(3-x) - 1 = 2-x$	Conjugate products; the same factor appears on both sides
$\frac{\sqrt{3-x} + 1}{\sqrt{6-x} + 2}$	Cancel the shared $(2-x)$; now continuous at $x = 2$
$\frac{\sqrt{1} + 1}{\sqrt{4} + 2} = \frac{2}{4} = \frac{1}{2}$	Substitute $x = 2$

L4. Combine into a single fraction

Evaluate $\lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2}$.

Steps.

Calculation	Method
$\frac{0}{0}$	Substitute $x = 2$
$\frac{2-x}{2x}$	Combine the numerator over $2x$
$\frac{2-x}{2x(x-2)}$	Rewrite the whole expression as a single fraction
$\frac{-1}{2x}$	Note $2-x = -(x-2)$ and cancel

Calculation	Method
$-\frac{1}{4}$	Substitute $x = 2$

Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$.

Calculation	Method
Form is $\infty - \infty$	Each term blows up, with opposite signs from each side
$\frac{x - \sin x}{x \sin x}$, now $\frac{0}{0}$	Common denominator
$x - \sin x \approx \frac{x^3}{6}$, $x \sin x \approx x^2$, ratio $\approx \frac{x}{6}$	Apply L11 (series expansion)
0	Send $x \rightarrow 0$

L5. The standard trigonometric limits

Trigger. A trig function whose argument goes to 0, divided by something also going to 0.

The two facts everything reduces to.

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1, \quad \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2} = \frac{1}{2}.$$

Why it works. The first is the squeeze theorem applied to $\cos \theta \leq \frac{\sin \theta}{\theta} \leq 1$; the second follows by multiplying by the conjugate $1 + \cos \theta$.

Evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 3x}$.

Steps.

Calculation	Method
$\frac{0}{0}$	Substitute $x = 0$
$\frac{\sin 5x}{5x} \cdot \frac{3x}{\sin 3x} \cdot \frac{5x}{3x}$	Manufacture matching arguments by multiplying and dividing
First factor $\rightarrow 1$, second factor $\rightarrow 1$	Standard limit $\frac{\sin \theta}{\theta} \rightarrow 1$
Third factor = $\frac{5}{3}$ (constant)	Simplify the leftover ratio

Calculation	Method
$\frac{5}{3}$	Multiply the three factors' limits

The general pattern.

$$\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx} = \frac{a}{b}$$

Every problem in this family is the work of arranging the expression until each trig piece has its own matching denominator.

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$$

Calculation	Method
$\frac{(1 - \cos x)/x^2}{(\sin x)/x}$	Divide numerator and denominator by x^2
Numerator $\rightarrow \frac{1}{2}$, denominator $\rightarrow 1$	Apply the two standard trig limits
$\frac{1}{2}$	Divide the two resulting limits

MATH265 exam question (Q3.1e).

Evaluate $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x^2 - x}$

Calculation	Method
$\frac{\sin 0}{0 - 0} = \frac{0}{0}$	Substitute; the sine says build the special limit
$x^2 - x = x(x - 1)$	Factor the denominator
$\frac{\sin(3x)}{x(x - 1)} = \frac{\sin(3x)}{3x} \cdot \frac{3x}{x(x - 1)} = \frac{\sin(3x)}{3x} \cdot \frac{3}{x - 1}$	Force a matching $3x$ underneath the sine
First factor $\rightarrow 1$; second $\rightarrow \frac{3}{0 - 1} = -3$	Standard limit on the first factor, direct substitution on the second
-3	Multiply the two factors' limits

MATH265 exam question (Q3.2b) — a $1 - \cos$ form with a doubled argument.

Evaluate $\lim_{x \rightarrow 0} \frac{x^2}{1 - \cos(2x)}$

Calculation	Method
$\frac{0}{1-1} = \frac{0}{0}$	Substitute $x = 0$
$\frac{x^2}{1 - \cos(2x)} = \frac{x^2}{2 \sin^2 x}$	Identity $1 - \cos(2\theta) = 2 \sin^2 \theta$, which removes the cosine entirely instead of using the $1 - \cos$ limit directly
$\frac{1}{2} \left(\frac{x}{\sin x} \right)^2$	Rewrite as a squared standard limit
$\frac{1}{2}$	Bracket $\rightarrow 1$ by the standard trig limit

L6. The squeeze (sandwich) theorem

Trigger. A factor that oscillates but stays bounded ($\sin \frac{1}{x}$, $\cos \frac{1}{x}$, a fractional part) multiplied by something that goes to 0.

Why it works. If $g(x) \leq f(x) \leq h(x)$ near a and g, h share the limit L , then f has no room to do anything else.

Evaluate $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$.

Steps.

Calculation	Method
L'Hopital and substitution both fail	$\sin \frac{1}{x}$ has no limit at 0
$-1 \leq \sin \frac{1}{x} \leq 1$ for every $x \neq 0$	Bound the oscillating factor
$-x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$	Multiply through by $x^2 \geq 0$, which keeps the inequality direction
Both outer bounds $\rightarrow 0$ as $x \rightarrow 0$	Take the limit of the bounds
0	Squeeze theorem

MATH265 exam question (Q3.3a) — squeeze chained onto a degree comparison.

Evaluate $\lim_{x \rightarrow \infty} \left(\frac{3x^2 - 4}{2x^4 + 2x + 4} \right) \sin x$.

Calculation	Method
Handle $R(x) = \frac{3x^2 - 4}{2x^4 + 2x + 4}$ on its own first	$\sin x$ has no limit, so the product cannot be evaluated directly
$\lim_{x \rightarrow \infty} R(x) = 0$	By L8, the denominator's degree (4) beats the numerator's (2)
$- R(x) \leq R(x) \sin x \leq R(x) $	Bound the oscillating factor: $-1 \leq \sin x \leq 1$
0	Both outer bounds $\rightarrow 0$ because $R(x) \rightarrow 0$, so the middle is squeezed to 0

This is the general pattern behind L16 as well: a bounded oscillating factor is harmless once it is multiplying something that already dies.

L7. Divide by the dominant power (limits at infinity)

Trigger. $x \rightarrow \pm\infty$ with a ratio of polynomials or polynomial-like expressions.

Why it works. $\frac{1}{x^k} \rightarrow 0$ for every $k > 0$, so dividing by the highest power turns every subordinate term into a term you can substitute 0 into.

Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 7}{5x^2 + x - 1}$.

Steps.

Calculation	Method
Highest power present: x^2	Identify the dominant power
$\frac{3 - \frac{2}{x} + \frac{7}{x^2}}{5 + \frac{1}{x} - \frac{1}{x^2}}$	Divide every term, top and bottom, by x^2
$\frac{3 - 0 + 0}{5 + 0 - 0}$	Send each $\frac{1}{x^k}$ to 0
$\frac{3}{5}$	Simplify

L8. Degree comparison (the shortcut for L7)

For $\lim_{x \rightarrow \pm\infty} \frac{P(x)}{Q(x)}$ with $\deg P = m$ and $\deg Q = n$:

Case	Limit
$m < n$	0
$m = n$	ratio of the leading coefficients
$m > n$	$\pm\infty$, sign from the leading coefficients and the parity of $m - n$

Evaluate $\lim_{x \rightarrow \infty} \frac{4x^3 + x}{2x^3 - 9}$.

Steps.

Calculation	Method
$\deg P = 3, \deg Q = 3$	Read the degrees
$m = n$	Degrees are equal, so the middle row of the L8 table applies
$\frac{4}{2}$	Ratio of the leading coefficients
2	Simplify

(bottom heavier) - Evaluate $\lim_{x \rightarrow -\infty} \frac{x^2 + 1}{x^3}$.

Steps.

Calculation	Method
$\deg P = 2, \deg Q = 3$	Read the degrees
$m < n$	First row of the L8 table applies; the denominator wins
0, from either direction	Conclusion

(top heavier, sign matters) Evaluate $\lim_{x \rightarrow -\infty} \frac{x^3}{x^2 + 1}$.

Calculation	Method
$m = 3 > n = 2$	Read the degrees; magnitude goes to infinity and only the sign is in question
Ratio behaves like x for large $ x $; $m - n = 1$ is odd	Third row of the L8 table (top heavier)
$-\infty$	As $x \rightarrow -\infty$, x is negative

Use it as a check, not as the work. On an exam that asks you to show your work, present L7 and use L8 to confirm you did not slip.

MATH265 exam question (Q3.2d).

Evaluate $\lim_{x \rightarrow \infty} \frac{5 - 2x^3}{x^2 + 2}$.

Calculation	Method
Numerator degree 3 , denominator degree 2 : $m > n$	Read the degrees; the top wins
$\frac{\frac{5}{x^2} - 2x}{1 + \frac{2}{x^2}}$, numerator $\rightarrow -\infty$, denominator $\rightarrow 1$	By L7, divide by x^2
$-\infty$	Negative leading coefficient and $m - n = 1$ (odd) send the ratio to $-\infty$
No horizontal asymptote, only the answer $-\infty$	Final result

L9. Radicals at infinity, and the $|x|$ trap

Trigger. A square root of a polynomial as $x \rightarrow \pm\infty$, or a $\sqrt{\quad} - \sqrt{\quad}$ difference at infinity.

The rule that gets missed. $\sqrt{x^2} = |x|$, which is x when $x \rightarrow +\infty$ but $-x$ when $x \rightarrow -\infty$. Pulling x out of a root as $x \rightarrow -\infty$ without the sign flip is the single most common error in this family.

Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + x}}{x}$.

Steps.

Calculation	Method
$\sqrt{4x^2 + x} = \sqrt{x^2} \sqrt{4 + \frac{1}{x}} = x \sqrt{4 + \frac{1}{x}}$	Factor x^2 out of the root
$ x = -x$	Because $x \rightarrow -\infty$ (the $ x $ trap)
$\frac{-x \sqrt{4 + \frac{1}{x}}}{x} = -\sqrt{4 + \frac{1}{x}}$	Substitute $ x = -x$ and simplify
-2	Send $\frac{1}{x} \rightarrow 0$

The same limit as $x \rightarrow +\infty$ is $+2$, and the only difference in the whole calculation is step 2.

Example ($\infty - \infty$ with roots).

Evaluate $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 3x} - x \right)$.

Calculation	Method
Form is $\infty - \infty$	Combine using the conjugate $\sqrt{x^2 + 3x} + x$
$(x^2 + 3x) - x^2 = 3x$	Numerator after conjugate multiplication
$\frac{3x}{\sqrt{x^2 + 3x} + x}$, now $\frac{\infty}{\infty}$	Rewrite as a single fraction
$\frac{3}{\sqrt{1 + \frac{3}{x}} + 1}$	Divide top and bottom by x (positive, so $\sqrt{x^2 + 3x}/x = \sqrt{1 + \frac{3}{x}}$)
$\frac{3}{2}$	Send $\frac{3}{x} \rightarrow 0$

Generalization worth carrying. $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + bx + c} - x \right) = \frac{b}{2}$, independent of c .

L10. L'Hopital's rule

Trigger. Substitution gives exactly $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Statement. If f, g are differentiable near a , $g' \neq 0$ near a , and $\frac{f}{g}$ has form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ at a , then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the right-hand limit exists.

Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.

Steps.

Calculation	Method
$\frac{1 - 1 - 0}{0} = \frac{0}{0}$	Substitute $x = 0$; L'Hopital's rule applies
$\frac{e^x - 1}{2x}$	Differentiate top and bottom separately (this is not the quotient rule)
Still $\frac{0}{0}$	Substitute again; apply the rule a second time
$\frac{e^x}{2}$	Differentiate top and bottom again
$\frac{1}{2}$	Substitute $x = 0$

The three ways this rule is misused.

1. Applying it when the form is not indeterminate. $\lim_{x \rightarrow 0} \frac{\cos x}{x^2}$ is $\frac{1}{0}$, not $\frac{0}{0}$, and the rule gives a wrong answer if you use it.
2. Using the quotient rule instead of differentiating the parts separately.
3. Looping forever. $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+1}}{x}$ cycles back to itself under the rule; L7 solves it in one line (= 1).

When it applies but is the wrong tool. For $\frac{x - \sin x}{x^3}$ the rule needs three passes; the series in L11 needs none.

L11. Taylor / Maclaurin expansion

Trigger. A $\frac{0}{0}$ where L'Hopital would need two or more passes, or where the derivatives get messy.

The expansions to know. Each is valid near 0.

Function	Expansion
e^x	$1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$
$\sin x$	$x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$
$\cos x$	$1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots$
$\ln(1 + x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$
$\tan x$	$x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$

Function	Expansion
$(1+x)^\alpha$	$1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \dots$

Evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$.

Steps.

Calculation	Method
$\sin x = x - \frac{x^3}{6} + O(x^5)$	Expand the numerator far enough to survive the cancellation
$x - \sin x = \frac{x^3}{6} + O(x^5)$	Subtract
$\frac{1}{6} + O(x^2)$	Divide by x^3
$\frac{1}{6}$	Send $x \rightarrow 0$

How far to expand. Expand to the order of the denominator. Here the denominator is x^3 , so keep the numerator through x^3 ; stopping at x would leave $0/x^3$ and lose the answer.

Second example.

$$\lim_{x \rightarrow 0} \frac{\cos x - 1 + \frac{x^2}{2}}{x^4} = \frac{1}{24}, \text{ read straight off the } x^4 \text{ term of } \cos x.$$

L12. Change of variable (substitution)

Trigger. A root of the variable, a nested argument, or a limit at infinity you would rather see at 0.

Why it works. A limit is preserved under any substitution that is continuous and sends the old target to the new one.

Evaluate $\lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{x - 1}$.

Steps.

Calculation	Method
Let $u = \sqrt[3]{x}$, so $x = u^3$	Change of variable
$u \rightarrow 1$	As $x \rightarrow 1$
$\frac{u-1}{u^3-1} = \frac{u-1}{(u-1)(u^2+u+1)}$	Rewrite in terms of u and factor

Calculation	Method
$\frac{1}{3}$	Cancel and substitute $u = 1$

$$\lim_{x \rightarrow \infty} x \sin \frac{1}{x}.$$

Calculation	Method
Let $t = \frac{1}{x}$, so $x = \frac{1}{t}$ and $t \rightarrow 0^+$	Change of variable (infinity to zero)
$\frac{\sin t}{t}$	Rewrite the expression in terms of t
1	By L5, the standard trig limit

L13. One-sided limits and piecewise functions

Trigger. An absolute value, a piecewise definition, an even root, or a denominator that hits zero without a matching zero on top.

The rule. $\lim_{x \rightarrow a} f(x)$ exists exactly when both one-sided limits exist and agree.

Evaluate $\lim_{x \rightarrow 0} \frac{|x|}{x}.$

Steps.

Calculation	Method
$ x = x$ for $x > 0$, $-x$ for $x < 0$	Split the absolute value by sign
$\frac{x}{x} = 1$	Right-hand limit
$\frac{-x}{x} = -1$	Left-hand limit
Limit does not exist	One-sided limits disagree ($1 \neq -1$)

Example (pole).

$$\lim_{x \rightarrow 2} \frac{1}{x - 2}$$

: from the right the denominator is a small positive, giving $+\infty$; from the left a small negative, giving $-\infty$. The two-sided limit does not exist, and this is not an indeterminate form, so L'Hopital never enters the picture.

Example (matching a piecewise function). Find a making

$$f(x) = \begin{cases} x^2 + a & x \leq 1 \\ 3x - 1 & x > 1 \end{cases}$$

continuous at 1.

Calculation	Method
$1 + a$	Left limit ($x^2 + a$ at $x = 1$)
2	Right limit ($3x - 1$ at $x = 1$)
$1 + a = 2 \implies a = 1$	Set the two one-sided limits equal for continuity

The two-step diagnosis this family always needs. Substitute first; if the result is $\frac{k}{0}$ with $k \neq 0$ (nothing to cancel, unlike L2's $\frac{0}{0}$), go straight to the one-sided limits rather than hunting for an algebra step that does not exist.

MATH265 exam question (Q3.1c) – direct $k/0$, no cancelling.

Evaluate $\lim_{x \rightarrow 3} \frac{2x^2 - x + 1}{x - 3}$.

Calculation	Method
Numerator = $16 \neq 0$, denominator = 0 : form $\frac{16}{0}$	Substitute $x = 3$; nothing to factor
$\frac{16}{0^-} \rightarrow -\infty$	From the left, $x - 3 \rightarrow 0^-$
$\frac{16}{0^+} \rightarrow +\infty$	From the right, $x - 3 \rightarrow 0^+$
Limit does not exist	$x = 3$ is a vertical asymptote with opposite behavior on the two sides

MATH265 exam question (Q3.1b) – a $\frac{0}{0}$ that turns into $k/0$ after cancelling.

Evaluate $\lim_{x \rightarrow -2} \frac{3x^2 - 2x - 16}{(x + 2)^2}$.

Calculation	Method
$\frac{0}{0}$; numerator factors as $(x + 2)(3x - 8)$	Substitute, then factor (L2)
$\frac{3x - 8}{x + 2}$	Cancel one $(x + 2)$
$\frac{-14}{0}$, a genuine $k/0$	Re-diagnose at $x = -2$: the remaining $(x + 2)$ cannot be cancelled, so the one-sided check takes over
Left: $\frac{-14}{0^-} \rightarrow +\infty$. Right: $\frac{-14}{0^+} \rightarrow -\infty$	One-sided limits
Limit does not exist	The two sides disagree

MATH265 exam question (Q3.2e) — the $k/0$ disguised as a trig special limit.

Evaluate $\lim_{x \rightarrow -\pi/3} \frac{\tan(2x)}{3x + \pi}$.

Calculation	Method
$\tan\left(-\frac{2\pi}{3}\right) = \sqrt{3} \neq 0$	Check the numerator first, rather than assuming $\frac{\tan u}{u} \rightarrow 1$ applies
$\frac{\sqrt{3}}{0}$, not $\frac{0}{0}$	The special limit does not apply at all
Left: $\frac{\sqrt{3}}{0^-} \rightarrow -\infty$. Right: $\frac{\sqrt{3}}{0^+} \rightarrow +\infty$	One-sided limits as $3x + \pi \rightarrow 0^\mp$
Limit does not exist	The two sides disagree

MATH265 exam question (Q3.2f) — an even-power denominator forces the same sign from both sides.

Evaluate $\lim_{x \rightarrow 2} \frac{\cos(\pi x)}{(x - 2)^2}$.

Calculation	Method
Numerator $\cos(2\pi) = 1$; $(x - 2)^2 \rightarrow 0^+$ from both sides	A square is never negative
Both one-sided limits $\frac{1}{0^+} \rightarrow +\infty$	They agree
$\lim_{x \rightarrow 2} \frac{\cos(\pi x)}{(x - 2)^2} = +\infty$	Does not exist as a finite number, but the sides agree well enough to name the shared infinite behavior

MATH265 exam question (Q3.3b) — a coterminal angle at a tangent asymptote.

Evaluate $\lim_{x \rightarrow 5\pi/2} \tan x$.

Calculation	Method
$\frac{5\pi}{2} = 2\pi + \frac{\pi}{2}$, coterminal with $\frac{\pi}{2}$, where $\cos x = 0$	Reduce the angle
$\tan x = \frac{\sin x}{\cos x}$, $\sin\left(\frac{5\pi}{2}\right) = 1$	Rewrite tangent as sine over cosine
Left: $\tan x \rightarrow +\infty$. Right: $\tan x \rightarrow -\infty$	$\cos x \rightarrow 0^+$ from the left, $\cos x \rightarrow 0^-$ from the right
Limit does not exist	$x = \frac{5\pi}{2}$ is a vertical asymptote of the tangent

L14. The logarithm trick for indeterminate powers

Trigger. The form is 1^∞ , 0^0 , or ∞^0 .

Why it works. $\ln$ turns an exponent into a product, converting the power form into $0 \cdot \infty$, which L15 then turns into a quotient that L10 can finish.

Method. Set $L = \lim f(x)^{g(x)}$, compute $\ln L = \lim g(x) \ln f(x)$, then exponentiate: $L = e^{\ln L}$. The final exponentiation is the step most often forgotten.

Example (0^0).

Evaluate $\lim_{x \rightarrow 0^+} x^x$.

Steps.

Calculation	Method
$\ln L = \lim_{x \rightarrow 0^+} x \ln x$	Let L be the limit and take logs
$0 \cdot (-\infty) \rightarrow \frac{\ln x}{1/x}, \text{ now } \frac{-\infty}{\infty}$	Rewrite the product as a quotient (L15)
$\frac{1/x}{-1/x^2} = -x$	L'Hopital's rule
$\ln L = 0$	Send $x \rightarrow 0^+$
$L = e^0 = 1$	Exponentiate

Example (1^∞).

Evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^{2x}$.

Calculation	Method
$\ln L = \lim 2x \ln \left(1 + \frac{3}{x}\right)$	Take logs of both sides
$\ln L = \lim_{t \rightarrow 0^+} \frac{2 \ln(1 + 3t)}{t}$	Substitute $t = \frac{1}{x} \rightarrow 0^+$
$\frac{6/(1 + 3t)}{1} \rightarrow 6$	Form is $\frac{0}{0}$; apply L'Hopital's rule
$L = e^6$	Exponentiate

Shortcut for the 1^∞ family.

$$\lim \left(1 + \frac{a}{x}\right)^{bx} = e^{ab}.$$

L15. Rewriting a $0 \cdot \infty$ product

Trigger. A product where one factor goes to 0 and the other to $\pm\infty$.

Method. Move one factor into the denominator as its reciprocal, producing $\frac{0}{0}$ or $\frac{\infty}{\infty}$. Choose the direction that gives derivatives you actually want to compute.

Evaluate $\lim_{x \rightarrow 0^+} x \ln x$.

Steps.

Calculation	Method
Form is $0 \cdot (-\infty)$	Diagnose the indeterminate product
Two rewrites available: $\frac{\ln x}{1/x}$ or $\frac{x}{1/\ln x}$	Move one factor into the denominator as its reciprocal
Choose $\frac{\ln x}{1/x}$	Differentiating $1/\ln x$ is worse than differentiating $1/x$
$\frac{1/x}{-1/x^2} = -x \rightarrow 0$	L'Hopital's rule
0	Send $x \rightarrow 0^+$

The general fact behind it.

Powers beat logarithms: $x^a \ln x \rightarrow 0$ as $x \rightarrow 0^+$ for every $a > 0$.

L16. Bounded over unbounded

Trigger. A bounded numerator (a sine, a cosine, anything you can cap) over something going to infinity.

Evaluate $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$.

Steps.

Calculation	Method
L'Hopital is not available	$\sin x$ has no limit, so the form is not $\frac{\infty}{\infty}$
$\left \frac{\sin x}{x} \right \leq \frac{1}{x}$ for $x > 0$	Bound the numerator, which is bounded over an unbounded denominator
0	$\frac{1}{x} \rightarrow 0$, so by squeeze the limit is 0

Contrast.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

but

$$\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0.$$

Same expression, different technique, because the *form* at the two points is different.

L17. Epsilon-delta proof

Trigger. The question says "prove", "using the definition", or "show that".

Definition. $\lim_{x \rightarrow a} f(x) = L$ means: for every $\varepsilon > 0$ there is a $\delta > 0$ such that
 $0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$.

Prove $\lim_{x \rightarrow 3} (2x + 1) = 7$.

Steps.

Calculation	Method
$ (2x + 1) - 7 = 2x - 6 = 2 x - 3 $	Scratch work: find δ
$2 x - 3 < \varepsilon \iff x - 3 < \frac{\varepsilon}{2}$	Force this below ε
Let $\varepsilon > 0$, choose $\delta = \frac{\varepsilon}{2}$	Proof: write it forwards
If $0 < x - 3 < \delta$ then $ (2x + 1) - 7 = 2 x - 3 < 2\delta = \varepsilon$	Apply the definition
Limit is 7. ■	Conclusion

For nonlinear f . Bound the extra factor first. For $\lim_{x \rightarrow 2} x^2 = 4$: $|x^2 - 4| = |x - 2||x + 2|$, and restricting to $\delta \leq 1$ gives $|x + 2| < 5$, so $\delta = \min\left(1, \frac{\varepsilon}{5}\right)$ works.

L18. Showing a limit does not exist

Trigger. You suspect there is no limit, or the expression oscillates.

Method (single variable). Exhibit two sequences $x_n \rightarrow a$ whose function values converge to different numbers.

Show $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ **does not exist.**

Steps.

Calculation	Method
$x_n = \frac{1}{n\pi} \rightarrow 0,$ $\sin \frac{1}{x_n} = \sin(n\pi) = 0$	First test sequence
$y_n = \frac{1}{2n\pi + \pi/2} \rightarrow 0, \sin \frac{1}{y_n} = 1$	Second test sequence
Limits 0 and 1	Two sequences approaching 0 give different function-value limits
No limit exists	A limit, if it existed, would be unique

Method (two variables). Approach along different paths.

Show $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ does not exist.

Calculation	Method
0 for every $x \neq 0$	Approach along $y = 0$
$\frac{x^2}{2x^2} = \frac{1}{2}$	Approach along $y = x$
Limit does not exist	Different paths give different values

Warning about paths. Agreement along every straight line does **not** prove existence. For $\frac{x^2y}{x^4+y^2}$, every line through the origin gives 0 but the parabola $y = x^2$ gives $\frac{1}{2}$. To prove a multivariable limit exists, convert to polar coordinates and squeeze, or bound $|f - L|$ by a function of r alone.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{x^2 + y^2}.$$

Calculation	Method
$x = r \cos \theta, y = r \sin \theta; x^2 + y^2 = r^2$	Substitute polar coordinates
$\frac{r^3 \cos^3 \theta}{r^2} = r \cos^3 \theta$	Simplify the expression
$ r \cos^3 \theta \leq r, r \rightarrow 0$ independently of θ	Bound and squeeze
0	Conclusion

L19. Recognizing a Riemann sum

Trigger. $\lim_{n \rightarrow \infty}$ of a sum whose terms contain $\frac{i}{n}$ and an overall $\frac{1}{n}$.

The template. $\lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(\frac{i}{n}\right) \frac{1}{n} = \int_0^1 f(x) dx.$

Evaluate $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{n}{n^2 + i^2}.$

Steps.

Calculation	Method
$\frac{n}{n^2 \left(1 + \frac{i^2}{n^2}\right)} = \frac{1}{n} \cdot \frac{1}{1 + \left(\frac{i}{n}\right)^2}$	Factor n^2 out of the denominator
$f(x) = \frac{1}{1+x^2}$, sample points $x_i = \frac{i}{n}$, width $\frac{1}{n}$ on $[0, 1]$	Read off the Riemann sum template
$\int_0^1 \frac{dx}{1+x^2} = \arctan(1) - \arctan(0)$	Convert the sum's limit to a definite integral
$\frac{\pi}{4}$	Evaluate the integral

Second example. $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \sqrt{\frac{i}{n}} = \int_0^1 \sqrt{x} dx = \frac{2}{3}.$

L20. Sequence-specific tools

Sequences allow techniques that functions of a real variable do not.

(a) Monotone convergence. A bounded monotone sequence converges.

$$a_1 = \sqrt{2}, a_{n+1} = \sqrt{2 + a_n}.$$

Calculation	Method
If $a_n < 2$ then $a_{n+1} = \sqrt{2 + a_n} < \sqrt{4} = 2$	Show it is bounded above by 2 , by induction
$a_{n+1} > a_n \iff (a_n - 2)(a_n + 1) < 0$, true for $0 < a_n < 2$	Show it is increasing
Limit L exists	Bounded and increasing (monotone convergence)
$L = \sqrt{2 + L} \implies L^2 - L - 2 = 0 \implies L = 2$ or $L = -1$	Take limits in the recursion

Calculation	Method
$L = 2$	The sequence is positive, so the negative root is rejected

Step 3 is the load-bearing one. Solving the fixed-point equation first proves nothing, because a divergent sequence can have a fixed point too.

(b) Growth hierarchy. As $n \rightarrow \infty$, each term is dominated by the next:

$$\ln n \ll n^a \ll b^n \ll n! \ll n^n \quad (a > 0, b > 1).$$

Any ratio across this chain has limit 0 downward and ∞ upward.

(c) Stolz-Cesaro (the discrete L'Hopital). If b_n is strictly increasing and unbounded, then

$$\lim \frac{a_n}{b_n} = \lim \frac{a_{n+1} - a_n}{b_{n+1} - b_n}$$

when the right side exists.

$$\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \cdots + \frac{1}{n}}{\ln n}.$$

Calculation	Method
$a_n = H_n, b_n = \ln n$, with b_n increasing and unbounded	Set up Stolz-Cesaro (the discrete L'Hopital)
$a_{n+1} - a_n = \frac{1}{n+1}, b_{n+1} - b_n = \ln\left(1 + \frac{1}{n}\right)$	Take differences
$\frac{1/(n+1)}{\ln(1 + 1/n)}$, and $\ln(1 + \frac{1}{n}) \approx \frac{1}{n}$ for large n	Form the ratio and approximate
1	Ratio $\rightarrow 1$

(d) The ratio test for a sequence limit. If $\left| \frac{a_{n+1}}{a_n} \right| \rightarrow r < 1$ then $a_n \rightarrow 0$. This settles things like $\frac{3^n}{n!} \rightarrow 0$ in one line.

Part 2 - Techniques for Evaluating Derivatives

D1. The definition (first principles)

Trigger. The question says "using the definition", or the function is not built from table entries.

Definition.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Differentiate $f(x) = \sqrt{x}$ from first principles.

Steps.

Calculation	Method
$\frac{\sqrt{x+h} - \sqrt{x}}{h}$	Write the difference quotient
$= \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})}$	Substituting $h = 0$ gives $\frac{0}{0}$ with a root present, so use L3: multiply by the conjugate $\sqrt{x+h} + \sqrt{x}$
$= \frac{h}{h(\sqrt{x+h} + \sqrt{x})} = \frac{1}{\sqrt{x+h} + \sqrt{x}}$	Numerator collapses to $(x+h) - x = h$; cancel the common factor of h
$= \frac{1}{2\sqrt{x}}$	Let $h \rightarrow 0$

Example ($1/x$).

$$\frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \frac{x - (x+h)}{hx(x+h)} = \frac{-1}{x(x+h)} \rightarrow -\frac{1}{x^2}.$$

The alternate form, useful for a derivative at one specific point:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

Differentiability check. A function is differentiable at a only if the left and right difference quotients agree. $f(x) = |x|$ is continuous at 0 but its quotients give $+1$ and -1 (this is L13 in disguise), so it is not differentiable there.

MATH265 exam question (Q7.1) – a trig derivative from the definition.

Use the limit definition to find

$$\frac{d}{dx} \cot x$$

Calculation	Method
$\frac{d}{dx} \cot x = \lim_{h \rightarrow 0} \frac{\cot(x+h) - \cot x}{h}$	Write the definition
$= \lim_{h \rightarrow 0} \frac{\cos(x+h) \sin x - \cos x \sin(x+h)}{h \sin(x+h) \sin x}$	Convert to sine and cosine and combine over a common denominator
$= \lim_{h \rightarrow 0} \frac{-\sin h}{h \sin(x+h) \sin x}$	Recognize the numerator as the sine difference identity $\sin(A - B)$ with $A = x$, $B = x + h$: it collapses to $\sin(-h) = -\sin h$
$= \lim_{h \rightarrow 0} \left(-\frac{\sin h}{h} \right) \cdot \frac{1}{\sin(x+h) \sin x} = (-1) \cdot \frac{1}{\sin^2 x}$	Split off the special limit $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$
$\frac{d}{dx} \cot x = -\frac{1}{\sin^2 x} = -\csc^2 x$	Result, matching the D6 table entry

The sign trap. Writing $\sin(x - (x + h)) = \sin(h)$ loses the minus sign (sine is odd) and turns the answer into $+\csc^2 x$.

D2. Power, constant multiple, and sum rules

$$\frac{d}{dx} x^n = nx^{n-1}, \quad (cf)' = cf', \quad (f \pm g)' = f' \pm g'.$$

The power rule holds for every real n , not just integers, which is what makes rewriting worthwhile.

Differentiate $f(x) = 3x^4 - \frac{2}{x^3} + 5\sqrt[3]{x} - 7$.

Steps.

Calculation	Method
$3x^4 - \frac{2}{x^3} + 5\sqrt[3]{x} - 7 = 3x^4 - 2x^{-3} + 5x^{1/3} - 7$	Rewrite every term as a power
$f'(x) = 12x^3 + 6x^{-4} + \frac{5}{3}x^{-2/3}$	Apply the power rule term by term; the constant -7 differentiates to 0
$f'(x) = 12x^3 + \frac{6}{x^4} + \frac{5}{3\sqrt[3]{x^2}}$	Restore radical form if the answer must match a textbook

Step 1 is the whole technique. Most "hard" derivatives in this family are easy ones written in a disguise that the power rule cannot see.

MATH265 exam question (Q8.1a).

Differentiate $y = 4x^5 + 3x^4 - 6x^3 + 6$.

Calculation	Method
$y' = 20x^4 + 12x^3 - 18x^2$	Multiply each coefficient by its exponent and drop the exponent by one; the constant 6 differentiates to 0

D3. The product rule

$$(fg)' = f'g + fg'.$$

Differentiate $y = x^2e^x$.

Steps.

Calculation	Method
$f = x^2, g = e^x$	Identify the two factors
$f' = 2x, g' = e^x$	Differentiate each factor separately
$y' = 2xe^x + x^2e^x$	Assemble with the product rule $f'g + fg'$
$y' = e^x(x^2 + 2x)$	Factor for a usable form

Three factors. $(fgh)' = f'gh + fg'h + fgh'$: differentiate each factor in turn, leaving the others alone.

Example. $\frac{d}{dx}[x \sin x \cos x] = \sin x \cos x + x \cos^2 x - x \sin^2 x$.

MATH265 exam question (Q8.3a) — a product whose second factor is three chain-rule layers deep.

Differentiate $y = \sin x \cdot \cos(\sin x^2)$.

Calculation	Method
$f = \sin x, g = \cos(\sin x^2)$	The outer structure is a product, so the product rule runs first, leaving the hard part (differentiating the second factor) for after
$g' = -\sin(\sin x^2) \cdot \cos(x^2) \cdot 2x$	Differentiate $\cos(\sin x^2)$ through its three layers - cosine, then sine, then x^2
$y' = \cos x \cos(\sin x^2) - \sin x \sin(\sin x^2) \cos(x^2)(2x)$	Assemble the product rule $f'g + fg'$

The notation trap.

$\sin x^2$ means $\sin(x^2)$, not $(\sin x)^2$

; the inner derivative is

$\cos(x^2) \cdot 2x$, not $2 \sin x \cos x$.

D4. The quotient rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

The order in the numerator matters, unlike the product rule. The mnemonic "low d-high minus high d-low, over low squared" fixes it.

Differentiate $y = \frac{2x + 1}{x^2 + 3}$.

Steps.

Calculation	Method
$f = 2x + 1, f' = 2; g = x^2 + 3, g' = 2x$	Name the pieces and differentiate each
$2(x^2 + 3) - (2x + 1)(2x) = 2x^2 + 6 - 4x^2 - 2x$	Assemble the quotient rule numerator $f'g - fg'$
$= -2x^2 - 2x + 6$	Simplify
$y' = \frac{-2x^2 - 2x + 6}{(x^2 + 3)^2}$	Divide by g^2

When not to use it. If the denominator is a single power, rewrite instead. $\frac{x^3+1}{x} = x^2 + x^{-1}$ differentiates to $2x - x^{-2}$ with no rule at all.

MATH265 exam question (Q8.1b) — cancel a common factor after the quotient rule, before expanding.

Differentiate $y = \frac{2x - 16}{(x + 3)^2}.$

Calculation	Method
$f = 2x - 16, f' = 2; g = (x + 3)^2, g' = 2(x + 3)$	Name the pieces; g' needs the chain rule
$y' = \frac{2(x + 3)^2 - (2x - 16) \cdot 2(x + 3)}{(x + 3)^4}$	Assemble the quotient rule
$y' = \frac{2(x + 3) - 2(2x - 16)}{(x + 3)^3}$	Every term in the numerator carries a factor of $(x + 3)$: cancel one against the $(x + 3)^4$ in the denominator before expanding, dropping it to $(x + 3)^3$
$y' = \frac{38 - 2x}{(x + 3)^3}$	Simplify the numerator

Doing the cancellation in step 3 first avoids expanding a $(x + 3)^2$ that was only going to be thrown away.

MATH265 exam question (Q8.4a) — naming the rules as you stack them.

Differentiate $\frac{d}{dx} \frac{\tan(2x)}{\sqrt{x}}$

Calculation	Method
$f = \tan(2x), f' = 2 \sec^2(2x); g = x^{1/2},$ $g' = \frac{1}{2\sqrt{x}}$	Quotient rule on the outside; chain rule on $\tan(2x)$; power rule on $\sqrt{x} = x^{1/2}$ - name the pieces and differentiate each
$\frac{d}{dx} \frac{\tan(2x)}{\sqrt{x}} = \frac{2 \sec^2(2x)\sqrt{x} - \tan(2x) \cdot \frac{1}{2\sqrt{x}}}{x}$	Assemble the quotient rule, using $g^2 = x$ in the denominator

MATH265 exam question (Q8.2c) — a quotient whose two pieces each need their own product/chain-rule work first.

Differentiate $\frac{d}{dx} \frac{x^2 - \cos(3x)}{x \sin(2x)}$

Calculation	Method
$f = x^2 - \cos(3x), g = x \sin(2x)$	Name the pieces
$f' = 2x + 3 \sin(3x); g' = \sin(2x) + 2x \cos(2x)$	Differentiate each on its own before assembling: the derivative of $-\cos(3x)$ is $+3 \sin(3x)$; g' needs the product rule
$\frac{(2x + 3 \sin(3x))(x \sin(2x)) - (x^2 - \cos(3x))(\sin(2x) + 2x \cos(2x))}{(x \sin(2x))^2}$	Assemble the quotient rule; no simplification is required

MATH265 exam question (Q8.2d) — a quotient rule evaluated at a point from function values alone.

Find $\frac{d}{dx} \frac{f(x)}{g(x)} \bigg|_{x=1}$ where $f(1) = 3, g(1) = 6, f'(x) = \sqrt{x},$
 $g'(x) = 3 \sin\left(\frac{\pi x}{3}\right).$

Calculation	Method
$\left(\frac{f}{g}\right)' \bigg _{x=1} = \frac{f'(1)g(1) - f(1)g'(1)}{(g(1))^2}$	Write the quotient rule at the point

Calculation	Method
$f'(1) = \sqrt{1} = 1; g'(1) = 3 \sin\left(\frac{\pi}{3}\right) = 3 \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$	Evaluate the two derivatives at $x = 1$ - read in radians, not degrees
$\frac{(1)(6) - (3)\left(\frac{3\sqrt{3}}{2}\right)}{36} = \frac{6 - \frac{9\sqrt{3}}{2}}{36} = \frac{4 - 3\sqrt{3}}{24} \approx -0.04984$	Substitute all four numbers

MATH265 exam question (Q8.3d) – a root sitting inside a quotient rule.

Differentiate $y = \frac{\sqrt{x^2 - 1}}{x^2 - 2x - 8}$.

Calculation	Method
$f = \sqrt{x^2 - 1}, g = x^2 - 2x - 8$	Name the pieces
$f' = \frac{2x}{2\sqrt{x^2 - 1}} = \frac{x}{\sqrt{x^2 - 1}}; g' = 2x - 2$	Differentiate f with the chain rule
$y' = \frac{\frac{x}{\sqrt{x^2 - 1}}(x^2 - 2x - 8) - \sqrt{x^2 - 1}(2x - 2)}{(x^2 - 2x - 8)^2}$	Assemble the quotient rule directly, without clearing the inner root

D5. The chain rule

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x).$$

The procedure. Differentiate the outer function leaving the inside untouched, then multiply by the derivative of the inside. Repeat for every layer.

Differentiate $y = \sin(3x^2 + 1)$.

Steps.

Calculation	Method
Outer $\sin(\square)$, inner $\square = 3x^2 + 1$	Identify the layers
$\cos(3x^2 + 1)$	Derivative of the outer, inside unchanged
$6x$	Derivative of the inner
$y' = 6x \cos(3x^2 + 1)$	Multiply the two layers

Example (three layers).

Differentiate $y = \sqrt{1 + \tan 2x}$.

Calculation	Method
Layers, outermost first: square root, then $1 + \tan(\cdot)$, then $2x$	Identify the layers
$\frac{1}{2\sqrt{1 + \tan 2x}}$	Square root layer
$\sec^2 2x$	Tangent layer
2	Innermost layer
$y' = \frac{2 \sec^2 2x}{2\sqrt{1 + \tan 2x}} = \frac{\sec^2 2x}{\sqrt{1 + \tan 2x}}$	Multiply all three layers

The most common slip. Forgetting the innermost factor.

$\frac{d}{dx} \sin(2x)$ is $2 \cos 2x$, **not** $\cos 2x$.

MATH265 exam question (Q8.3c) — a root nested inside a root.

Differentiate $y = \sqrt{1 + \sqrt{1 + x}}$.

Calculation	Method
Layers, outermost first: a square root, then $1 + \sqrt{\cdot}$, then $1 + x$	Identify the layers
$y' = \frac{1}{2\sqrt{1 + \sqrt{1 + x}}} \cdot v'$ where $v = 1 + \sqrt{1 + x}$	Peel the outer root
$v' = \frac{1}{2\sqrt{1 + x}}$	Differentiate the inner root (the constant 1 contributes nothing)
$y' = \frac{1}{4\sqrt{1 + x} \sqrt{1 + \sqrt{1 + x}}}$	Multiply the two layers

MATH265 exam question (Q8.4b) — chain rule evaluated at a point using exact trig values.

Find $\frac{d}{dx} \cos^3(x^2) \Big _{x=\sqrt{\pi}/2}$.	
Calculation	Method
$\cos^3(x^2) = [\cos(x^2)]^3$	Read the notation first: three layers (cube, cosine, x^2)
$3[\cos(x^2)]^2 \cdot (-\sin(x^2)) \cdot 2x = -6x \cos^2(x^2) \sin(x^2)$	Differentiate through all three layers
$x^2 = \left(\frac{\sqrt{\pi}}{2}\right)^2 = \frac{\pi}{4}$	Find the inner value <i>before</i> substituting the trig functions, which is what makes the arithmetic clean
$\cos^2 \frac{\pi}{4} = \frac{1}{2}, \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$	Use the exact values at $\frac{\pi}{4}$
$-6 \cdot \frac{\sqrt{\pi}}{2} \cdot \frac{1}{2} \cdot \frac{\sqrt{2}}{2} = -\frac{3\sqrt{2\pi}}{4} \approx -1.87997$	Substitute

MATH265 exam question (Q8.1c) – plain two-layer chain rule.

Differentiate $y = \sin(2x^2 - x + 1)$.

Calculation	Method
Outer $\sin(\cdot) \rightarrow \cos(\cdot)$; inner $2x^2 - x + 1 \rightarrow 4x - 1$	Identify and differentiate each layer
$y' = \cos(2x^2 - x + 1) \cdot (4x - 1)$	Multiply the two layers

MATH265 exam question (Q8.1d) – the power rule wrapping a polynomial.

Differentiate $y = (-4x^3 - x^2 + 3x + 7)^4$.

Calculation	Method
Outer $(\cdot)^4 \rightarrow 4(\cdot)^3$; inner derivative $-12x^2 - 2x + 3$	Identify and differentiate each layer
$y' = 4(-4x^3 - x^2 + 3x + 7)^3 (-12x^2 - 2x + 3)$	Multiply the two layers

MATH265 exam question (Q8.3b) – a fractional outer power wrapping a quotient rule.

Differentiate $y = \left(\frac{1 + x^3}{1 - x^2} \right)^{1/3}$.

Calculation	Method
$u = \frac{1 + x^3}{1 - x^2}, y = u^{1/3}, y' = \frac{1}{3}u^{-2/3}u'$	Substitute u and apply the chain rule
$u' = \frac{3x^2(1 - x^2) - (1 + x^3)(-2x)}{(1 - x^2)^2} = \frac{-x^4 + 3x^2 + 2x}{(1 - x^2)^2}$	Differentiate u with the quotient rule, then expand
$y' = \frac{1}{3} \left(\frac{1 + x^3}{1 - x^2} \right)^{-2/3} \cdot \frac{-x^4 + 3x^2 + 2x}{(1 - x^2)^2}$	Assemble

MATH265 exam question (Q8.3e) — a square wrapping a secant wrapping a quotient.

Differentiate $y = \sec^2 \left(\frac{x + 1}{x - 2} \right)$.

Calculation	Method
$y = (\sec u)^2$ with $u = \frac{x + 1}{x - 2}$	Read the layers - a square, then a secant, then a quotient
$u' = \frac{(1)(x - 2) - (x + 1)(1)}{(x - 2)^2} = \frac{-3}{(x - 2)^2}$	Differentiate the inner quotient
$\frac{d}{dx}(\sec u)^2 = 2 \sec u \cdot \sec u \tan u \cdot u' = 2 \sec^2 u \tan u \cdot u'$	Differentiate the outer layers
$y' = \frac{-6 \sec^2 \left(\frac{x+1}{x-2} \right) \tan \left(\frac{x+1}{x-2} \right)}{(x - 2)^2}$	Substitute u and u' back in

D6. The named-function tables

Trigonometric.

f	f'	f	f'
$\sin x$	$\cos x$	$\csc x$	$-\csc x \cot x$
$\cos x$	$-\sin x$	$\sec x$	$\sec x \tan x$
$\tan x$	$\sec^2 x$	$\cot x$	$-\csc^2 x$

Every co-function derivative carries a minus sign, which is the only thing that needs memorizing once you know the first column.

Exponential and logarithmic.

f	f'
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$\frac{1}{x}$
$\log_a x$	$\frac{1}{x \ln a}$
$\ln u $	$\frac{u'}{u}$

Example.

Differentiate $y = 5^{x^2}$.

Calculation	Method
a^u with $a = 5$, $u = x^2$	Recognize the form
$5^{x^2} \ln 5 \cdot 2x$	Table entry $a^u \ln a$ plus the chain rule on u'
$y' = 2x \ln 5 \cdot 5^{x^2}$	Rearrange for the final form

Example. $\frac{d}{dx} \ln(x^2 + 1) = \frac{2x}{x^2 + 1}$, which is the $\frac{u'}{u}$ row.

MATH265 exam question (Q8.2b) — the secant row plus the chain rule.

Differentiate $\sec(x^2 - 3x)$.

Calculation	Method
$\frac{d}{dx} \sec u = \sec u \tan u$, with $u = x^2 - 3x$, $u' = 2x - 3$	Table entry plus the chain rule
$\sec(x^2 - 3x) \tan(x^2 - 3x)(2x - 3)$	Substitute u and u'

D7. Inverse trigonometric and hyperbolic derivatives

f	f'	f	f'
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	$\sinh x$	$\cosh x$

$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$	$\cosh x$	$\sinh x$
$\arctan x$	$\frac{1}{1+x^2}$	$\tanh x$	$\operatorname{sech}^2 x$
$\operatorname{arcsec} x$	$\frac{1}{ x \sqrt{x^2-1}}$	$\operatorname{arcsinh} x$	$\frac{1}{\sqrt{x^2+1}}$

Note $\frac{d}{dx} \cosh x = +\sinh x$: unlike the circular case, no minus sign appears.

Example.

Differentiate $y = \arctan(x^2)$.

Calculation	Method
$\frac{1}{1+(x^2)^2} = \frac{1}{1+x^4}$	Outer derivative with the inside intact
$2x$	Inner derivative
$y' = \frac{2x}{1+x^4}$	Multiply the two layers

Deriving one, in case the table is unavailable. For $y = \arcsin x$: write $\sin y = x$, differentiate implicitly to get $\cos y y' = 1$, so $y' = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$.

D8. Implicit differentiation

Trigger. y is defined by an equation you cannot (or do not want to) solve for y .

Method. Differentiate both sides with respect to x , treating y as a function of x so every y term picks up a $\frac{dy}{dx}$ by the chain rule, then solve algebraically for $\frac{dy}{dx}$.

Example.

Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$.

Steps.

Calculation	Method
$2x + 2y \frac{dy}{dx} = 0$	Differentiate both sides; the $2y \frac{dy}{dx}$ comes from the chain rule, because y^2 is a composite of squaring with $y(x)$

Calculation	Method
$\frac{dy}{dx} = -\frac{x}{y}$	Solve algebraically for $\frac{dy}{dx}$

Example (folium of Descartes).

Find $\frac{dy}{dx}$ for $x^3 + y^3 = 6xy$.

Calculation	Method
$3x^2 + 3y^2y' = 6y + 6xy'$	Differentiate both sides (the right side needs the product rule)
$3y^2y' - 6xy' = 6y - 3x^2$	Collect the y' terms on one side
$y'(3y^2 - 6x) = 6y - 3x^2$	Factor out y'
$y' = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$	Solve and simplify

Second derivative implicitly. Differentiate y' again, then substitute the expression for y' back in.
For the circle:

$$y'' = \frac{d}{dx} \left(-\frac{x}{y} \right) = -\frac{y - xy'}{y^2} = -\frac{y + \frac{x^2}{y}}{y^2} = -\frac{x^2 + y^2}{y^3} = -\frac{25}{y^3}.$$

MATH265 exam question (Q10.1) – every term needs the product or chain rule.

Find y' for $x^3y + xy^2 = 4xy + 7$.

Calculation	Method
$\frac{d}{dx}(x^3y) = 3x^2y + x^3y'; \frac{d}{dx}(xy^2) = y^2 + 2xyy';$ $\frac{d}{dx}(4xy) = 4y + 4xy'$	Differentiate term by term (product rule; product then chain; product rule); the constant 7 vanishes
$3x^2y + x^3y' + y^2 + 2xyy' = 4y + 4xy'$	Collect the differentiated terms
$y'(x^3 + 2xy - 4x) = 4y - 3x^2y - y^2$	Move every y' term to one side and factor
$y' = \frac{4y - 3x^2y - y^2}{x^3 + 2xy - 4x}$	Solve

The trap. Differentiating xy^2 as just $2xyy'$ and dropping the y^2 term that the product rule also produces.

Find the tangent line to $y^3 + yx^2 + x^2 = 3y^2$ at $(1, 1)$.

Calculation	Method
$3 = 3$ at $(1, 1)$	Confirm the point lies on the curve first
$3y^2y' + (y'x^2 + 2xy) + 2x = 6yy'$	Differentiate implicitly
$3y' + y' + 2 + 2 = 6y'$	Because only one point is needed, substitute $x = 1, y = 1$ now rather than solving for y' symbolically first
$4y' + 4 = 6y' \implies y' = 2$	Solve the now-numerical equation
$y - 1 = 2(x - 1)$, so $y = 2x - 1$	Point-slope form

Substituting early is faster and loses fewer terms than carrying the full symbolic y' formula to the end.

D9. Logarithmic differentiation

Trigger. The variable is in the exponent *and* in the base, or the expression is a long product/quotient/power stack.

Method. Take $\ln$ of both sides, use log laws to break the expression apart, differentiate implicitly, then multiply by y .

Example (variable in both places).

Differentiate $y = x^{\sin x}$.

Steps.

Calculation	Method
$\ln y = \sin x \ln x$	Neither the power rule (exponent is not constant) nor the a^x rule (base is not constant) applies, so take logs of both sides
$\frac{y'}{y} = \cos x \ln x + \frac{\sin x}{x}$	Differentiate both sides, product rule on the right
$y' = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$	Multiply by y and restore it

Example (product/quotient stack).

Differentiate $y = \frac{x^3\sqrt{x^2+1}}{(3x+2)^5}$.

Calculation	Method
$\ln y = 3 \ln x + \frac{1}{2} \ln(x^2 + 1) - 5 \ln(3x + 2)$	Take logs and expand with log laws
$\frac{y'}{y} = \frac{3}{x} + \frac{x}{x^2 + 1} - \frac{15}{3x + 2}$	Differentiate implicitly
$y' = \frac{x^3\sqrt{x^2+1}}{(3x+2)^5} \left(\frac{3}{x} + \frac{x}{x^2+1} - \frac{15}{3x+2} \right)$	Multiply back by y

The quotient and product rules would give the same answer after considerably more algebra.

D10. The inverse function derivative

Trigger. You need $(f^{-1})'$ at a point but inverting f is hard or impossible in closed form.

Formula.

If $f(a) = b$ then

$$(f^{-1})'(b) = \frac{1}{f'(a)}.$$

Example.

For $f(x) = x^3 + 2x + 1$, **find** $(f^{-1})'(4)$.

Steps.

Calculation	Method
$f(1) = 1 + 2 + 1 = 4$, so $a = 1$	Find the a with $f(a) = 4$ by trying small integers - solved by inspection or a solver, never by inverting the cubic
$f'(x) = 3x^2 + 2$	Differentiate f
$f'(1) = 5$	Evaluate at $a = 1$
$(f^{-1})'(4) = \frac{1}{5}$	Reciprocate

D11. Parametric and polar derivatives

Parametric.

For $x = x(t)$, $y = y(t)$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{dx/dt}.$$

The second-derivative formula is *not* $\frac{y''(t)}{x''(t)}$, which is the standard error here.

Example.

For $x = t^2$, $y = t^3 - 3t$, find both derivatives at $t = 2$.

Steps.

Calculation	Method
$\frac{dx}{dt} = 2t, \frac{dy}{dt} = 3t^2 - 3$	Differentiate x and y with respect to t
$\frac{dy}{dx} = \frac{3t^2 - 3}{2t}$, which at $t = 2$ is $\frac{9}{4}$	Divide dy/dt by dx/dt
$\frac{3}{2}t - \frac{3}{2}t^{-1} \rightarrow \frac{3}{2} + \frac{3}{2}t^{-2}$	Differentiate dy/dx with respect to t
$\frac{d^2y}{dx^2} = \frac{\frac{3}{2} + \frac{3}{2t^2}}{2t}$	Divide by $dx/dt = 2t$
$\frac{\frac{3}{2} + \frac{3}{8}}{4} = \frac{15/8}{4} = \frac{15}{32}$	Evaluate at $t = 2$

Polar. With $r = r(\theta)$, use $x = r \cos \theta$, $y = r \sin \theta$ as a parametrization in θ :

$$\frac{dy}{dx} = \frac{r' \sin \theta + r \cos \theta}{r' \cos \theta - r \sin \theta}.$$

Example.

For $r = 1 + \cos \theta$ at $\theta = \frac{\pi}{2}$: $r = 1$, $r' = -\sin \theta = -1$, so

$$\frac{dy}{dx} = \frac{(-1)(1) + (1)(0)}{(-1)(0) - (1)(1)} = \frac{-1}{-1} = 1.$$

D12. Higher-order derivatives and the Leibniz rule

Trigger. An n th derivative, a pattern, or a Taylor coefficient.

Method for a pattern. Compute the first three or four derivatives, write them in a form that exposes the structure, then state and check the general term.

Example.

Find the n th derivative of $f(x) = xe^x$.

Steps.

Calculation	Method
$f' = e^x + xe^x = (x+1)e^x$	First derivative, via the product rule
$f'' = e^x + (x+1)e^x = (x+2)e^x$	Second derivative
$f''' = (x+3)e^x$	Third derivative, exposing the pattern
$f^{(n)} = (x+n)e^x$	Conjecture the general term
$e^x + (x+n)e^x = (x+n+1)e^x$	Confirm by induction: differentiating $(x+n)e^x$ gives the formula at $n+1$

Leibniz rule (the product rule iterated):

$$(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}.$$

It is most useful when one factor is a polynomial, because that factor's derivatives terminate.

Example.

Find the 10th derivative of $x^2 \sin x$.

Calculation	Method
$f = x^2$: only $k = 0, 1, 2$ contribute	Choose f as the polynomial factor, since $f''' = 0$ truncates the Leibniz sum

Calculation	Method
$\binom{10}{0}x^2g^{(10)} + \binom{10}{1}(2x)g^{(9)} + \binom{10}{2}(2)g^{(8)}$, with $g = \sin x$	Write out the surviving Leibniz terms
$g^{(8)} = \sin x, g^{(9)} = \cos x, g^{(10)} = -\sin x$	Use the cyclic pattern of sin 's derivatives
$-x^2 \sin x + 20x \cos x + 90 \sin x$	Substitute and simplify

MATH265 exam question (Q8.2a) — a second derivative via the chain rule twice.

Find $\frac{d^2}{dx^2} \cot(2x)$.

Calculation	Method
$\frac{d}{dx} \cot(2x) = -2 \csc^2(2x)$	First derivative: table entry $\frac{d}{dx} \cot u = -\csc^2 u$ with $u = 2x$
$\frac{d}{dx} \csc^2(2x) = 2 \csc(2x) \cdot (-2 \csc(2x) \cot(2x)) = -4 \csc^2(2x) \cot(2x)$	Differentiate again: write $\csc^2(2x) = [\csc(2x)]^2$ and chain twice
$-2 \cdot (-4 \csc^2(2x) \cot(2x)) = 8 \csc^2(2x) \cot(2x)$	Apply the leading -2 from the first derivative
$\frac{d^2}{dx^2} \cot(2x) = 8 \csc^2(2x) \cot(2x)$	Result, positive despite the two minus signs along the way

D13. Related rates

Trigger. Two or more quantities changing in time, with one rate given and another asked for.

The procedure that prevents the usual mistakes.

1. Draw and label; name every varying quantity as a function of t .
2. Write the relation between the quantities, valid for **all** t .
3. Differentiate the relation with respect to t (implicitly).
4. **Only now** substitute the instantaneous values.
5. Solve for the wanted rate and attach units.

Example.

A 10 ft ladder leans on a wall. The base slides out at 2 ft/s. How fast is the top falling when the base is 6 ft from the wall?

Steps.

Calculation	Method
x = base distance, y = height; $\frac{dx}{dt} = 2$, want $\frac{dy}{dt}$	Name every varying quantity as a function of t
$x^2 + y^2 = 100$	Write the relation, true at every instant
$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$	Differentiate the relation with respect to t (implicitly)
$x = 6 \Rightarrow y = \sqrt{100 - 36} = 8$	Only now find the instantaneous values - substituting $x = 6$ before differentiating would zero out its derivative and collapse the answer to nonsense
$2(6)(2) + 2(8) \frac{dy}{dt} = 0 \Rightarrow 24 + 16 \frac{dy}{dt} = 0$	Substitute the instantaneous values
$\frac{dy}{dt} = -\frac{3}{2} \text{ ft/s}$	Solve and attach units; the negative sign says the top is descending

MATH265 exam question (Q11.1) — eliminate the second variable before differentiating.

Gravel forms a cone whose base diameter always equals its height; $\frac{dV}{dt} = 0.5 \text{ m}^3/\text{min}$. Find $\frac{dh}{dt}$ when $h = 4 \text{ m}$.

Calculation	Method
$V = \frac{1}{3}\pi r^2 h$, with $2r = h$ so $r = \frac{h}{2}$	Write the geometry and the constraint "diameter equals height"
$V = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}$	Use the constraint to eliminate r before differentiating, since there is no given rate for r
$\frac{dV}{dt} = \frac{\pi h^2}{4} \frac{dh}{dt}$	Differentiate in t
$0.5 = 4\pi \frac{dh}{dt}$	Substitute $h = 4$ and $\frac{dV}{dt} = 0.5$
$\frac{dh}{dt} = \frac{1}{8\pi} \approx 0.0398 \text{ m/min}$	Solve

The trap. Reading "diameter equals height" as $r = h$ instead of $r = \frac{h}{2}$; that makes the final rate four times too small.

MATH265 exam question (Q11.2) – the hypotenuse is the one that changes.

A rocket rises vertically, tracked by radar 5 mi from the pad. Find $\frac{dy}{dt}$ when the height $y = 4$ mi and the line-of-sight distance z changes at $\frac{dz}{dt} = 2000$ mi/h.

Calculation	Method
$z^2 = 5^2 + y^2$	Write the relation, with the 5 a fixed constant, which is what makes the problem solvable
$z = \sqrt{25 + 16} = \sqrt{41}$	Find the missing side at the instant of interest
$2z \frac{dz}{dt} = 2y \frac{dy}{dt}$, i.e. $z \frac{dz}{dt} = y \frac{dy}{dt}$	Differentiate in t
$\sqrt{41}(2000) = 4 \frac{dy}{dt}$, so $\frac{dy}{dt} = 500\sqrt{41} \approx 3201.6$ mi/h	Substitute the instantaneous values and solve
$3201.6 > 2000$	Sanity check: the rocket's true speed must exceed the line-of-sight rate, since only part of its motion is along the sightline

The trap. Swapping the leg and the hypotenuse (using $z = 4$, $y = \sqrt{41}$); the hypotenuse is always the longest side.

D14. Differentials and linear approximation

Trigger. Estimate a function value near a point you know exactly, or propagate a small error.

Formulas. $dy = f'(x) \, dx$, and the tangent-line approximation

$$f(x) \approx f(a) + f'(a)(x - a).$$

Example.

Approximate $\sqrt{4.1}$.

Steps.

Calculation	Method
$f(x) = \sqrt{x}, a = 4$	Take the nearby exact point
$f(4) = 2, f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f'(4) = \frac{1}{4}$	Evaluate f and f' at a
$\sqrt{4.1} \approx 2 + \frac{1}{4}(0.1) = 2.025$	Apply $f(x) \approx f(a) + f'(a)(x - a)$
true value 2.0248456..., error $\approx 1.5 \times 10^{-4}$	Compare against the true value

Error propagation. If a sphere's radius is measured as **10 cm** with error up to **0.05 cm**, then from $V = \frac{4}{3}\pi r^3$ and $dV = 4\pi r^2 dr$ the volume error is at most $4\pi(100)(0.05) \approx 62.8 \text{ cm}^3$. The *relative* error is cleaner: $\frac{dV}{V} = 3\frac{dr}{r}$, so a **0.5%** radius error gives a **1.5%** volume error.

MATH265 exam question (Q12.1) – a square root near a perfect square.

Use differentials to approximate $\sqrt{9.2}$.

Calculation	Method
$f(x) = \sqrt{x}, a = 9$ (nearest perfect square), $dx = 0.2$	Choose the base point and step
$f(9) = 3, f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f'(9) = \frac{1}{6}$	Evaluate f and f' at a
$\sqrt{9.2} \approx 3 + \frac{1}{6}(0.2) = \frac{91}{30} \approx 3.0333$	Apply the linear approximation
true value 3.03315..., high by about 2×10^{-4}	Compare - consistent with $\sqrt{x}$ being concave down (its tangent line lies above the curve)

MATH265 exam question (Q12.2) – an angle, so the step must be converted to radians first.

Use linearization to estimate $\sin(62^\circ)$.

Calculation	Method
$a = 60^\circ = \frac{\pi}{3}, dx = 2^\circ = \frac{\pi}{90} \approx 0.0349066$	Base point is the nearest angle with exact values; converting the step to radians is what decides the whole answer, since the derivative formulas are only valid in radians
$\sin(a + dx) \approx \sin a + \cos a dx = \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{\pi}{90}$	Linearization
$\approx 0.8660254 + 0.0174533 = 0.8834787$	Evaluate

Calculation	Method
true value 0.8829476 , high by about 5×10^{-4}	Compare; the companion question (MATH265 Q12.3a, cos 62°) uses the same dx but the <i>opposite</i> sign flips which side the estimate lands on, since cosine is decreasing where sine is increasing here

The trap common to both. Using $dx = 2$ instead of $dx = \frac{\pi}{90}$, which gives an impossible value like **1.87** for a sine.

MATH265 exam question (Q12.3) — the two four-decimal-place estimates promised above.

(a) - Estimate $\cos(62^\circ)$. Base point $a = \frac{\pi}{3}$, $dx = \frac{\pi}{90}$; $f'(x) = -\sin x$ so $f'(\frac{\pi}{3}) = -\frac{\sqrt{3}}{2} \cdot \cos(62^\circ) \approx \frac{1}{2} - \frac{\sqrt{3}}{2} \cdot \frac{\pi}{90} \approx 0.5 - 0.0302300 = 0.4698$ (four decimal places: 0.4698). The true value is 0.4694716 — the estimate is now high, the error sign having flipped relative to Q12.2 because cosine is decreasing here.

(b) - Estimate $\sqrt{16.4}$. Base point $a = 16$, $dx = 0.4$; $f'(16) = \frac{1}{2\sqrt{16}} = \frac{1}{8}$. $\sqrt{16.4} \approx 4 + \frac{1}{8}(0.4) = 4.0500$ (four decimal places, so the trailing zero is required). The true value is 4.0496913, high by about 0.0003.

D15. Partial derivatives

Trigger. A function of more than one variable.

Method. Differentiate with respect to one variable while treating every other variable as a constant.

Example.

For $f(x, y) = x^3y^2 + e^{xy}$, find f_x , f_y , and f_{xy} .

Steps.

Calculation	Method
$f_x = 3x^2y^2 + ye^{xy}$	Treat y as constant: $x^3y^2 \rightarrow 3x^2y^2$ and $e^{xy} \rightarrow ye^{xy}$ (chain rule, inner derivative y)

Calculation	Method
$f_y = 2x^3y + xe^{xy}$	Treat x as constant
$\partial_y (ye^{xy}) = e^{xy} + xye^{xy}$, so $f_{xy} = 6x^2y + e^{xy}(1 + xy)$	$f_{xy} = \partial_y(f_x)$: differentiate $3x^2y^2 + ye^{xy}$ in y , needing the product rule on the second term
$\partial_x (2x^3y + xe^{xy}) = 6x^2y + e^{xy} + xye^{xy}$	Check with Clairaut's theorem by computing f_{yx} from f_y - it matches

Clairaut's theorem ($f_{xy} = f_{yx}$ when both are continuous) is a free correctness check on every mixed partial you compute.

D16. Multivariable chain rule, gradient, and directional derivative

Chain rule.

For $z = f(x, y)$ with $x = x(t)$, $y = y(t)$:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Draw the dependency tree, then sum over every path from z down to t , multiplying along each path.

Example.

$z = x^2y$ with $x = \cos t$, $y = \sin t$; find $\frac{dz}{dt}$ at $t = 0$.

Calculation	Method
$f_x = 2xy$, $f_y = x^2$	Compute the partial derivatives of z
$x' = -\sin t$, $y' = \cos t$	Compute the derivatives of $x(t)$, $y(t)$
$\frac{dz}{dt} = 2xy(-\sin t) + x^2 \cos t$	Combine with the multivariable chain rule
$x = 1$, $y = 0 \Rightarrow 0 + 1 = 1$	Evaluate at $t = 0$

Gradient and directional derivative.

$$\nabla f = \langle f_x, f_y, f_z \rangle, \quad D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} \quad (\|\mathbf{u}\| = 1).$$

Example.

For $f(x, y) = x^2 + 3xy$ at $(1, 2)$ in the direction of $\langle 3, 4 \rangle$.

Calculation	Method
$\nabla f = \langle 2x + 3y, 3x \rangle$, so at $(1, 2)$: $\langle 8, 3 \rangle$	Compute the gradient and evaluate at the point
$\ \langle 3, 4 \rangle\ = 5$, so $\mathbf{u} = \langle \frac{3}{5}, \frac{4}{5} \rangle$	Normalize the direction - skipping this is the standard error, and it silently scales the answer
$8 \cdot \frac{3}{5} + 3 \cdot \frac{4}{5} = \frac{24+12}{5} = \frac{36}{5}$	Dot ∇f with $\mathbf{u}$

Implicit differentiation, multivariable form.

If $F(x, y) = 0$ defines y implicitly, then $\frac{dy}{dx} = -\frac{F_x}{F_y}$. For $x^3 + y^3 - 6xy = 0$: $F_x = 3x^2 - 6y$, $F_y = 3y^2 - 6x$, giving $-\frac{3x^2-6y}{3y^2-6x} = \frac{2y-x^2}{y^2-2x}$

D17. Differentiating an integral

Trigger. The variable appears in a limit of integration or inside the integrand as a parameter.

Fundamental Theorem of Calculus, part 1.

$$\frac{d}{dx} \int_a^x f(t) dt = f(x).$$

With a variable upper limit that is a function. Chain rule applies:

$$\frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x))g'(x).$$

Example.

$$\frac{d}{dx} \int_1^{x^2} \sqrt{1+t^3} dt.$$

Calculation	Method
$\sqrt{1+(x^2)^3} = \sqrt{1+x^6}$	The integrand evaluated at the upper limit

Calculation	Method
$2x\sqrt{1+x^6}$	Multiply by the derivative of the upper limit, $2x$ (chain rule on the FTC-1 form)

Both limits variable. Split at a constant c and apply the rule twice, with a sign flip on the lower one:

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f(v)v' - f(u)u'.$$

Example.

$$\frac{d}{dx} \int_x^{x^2} \frac{dt}{t} = \frac{1}{x^2}(2x) - \frac{1}{x}(1) = \frac{2}{x} - \frac{1}{x} = \frac{1}{x}. \text{ Sanity check: the integral equals } \ln x^2 - \ln x = \ln x, \text{ whose derivative is indeed } \frac{1}{x}.$$

Leibniz's general rule (parameter inside the integrand too):

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(x, t) dt = f(x, v)v' - f(x, u)u' + \int_u^v \frac{\partial f}{\partial x} dt.$$

This rule read backwards is the integration technique I20.

MATH265 exam question (Q18.1) – both limits variable, worked from the split rule directly.

Evaluate $\frac{d}{dx} \int_{2x}^x \sin(t^2) dt.$

Calculation	Method
$\int_{2x}^x = \int_c^x - \int_c^{2x}$	Both limits depend on x , so the basic FTC-1 form does not apply directly; split at any constant c
$\sin(x^2)$	Differentiate the first piece (upper limit is exactly x)
$\sin((2x)^2) \cdot 2 = 2 \sin(4x^2)$	Differentiate the second piece with the chain rule (upper limit is $2x$): evaluate the integrand at $2x$ and multiply by $\frac{d}{dx}(2x) = 2$

Calculation	Method
$\frac{d}{dx} \int_{2x}^x \sin(t^2) dt = \sin(x^2) - 2 \sin(4x^2)$	Subtract

The trap. Writing $(2x)^2$ as $2x^2$ instead of $4x^2$, and dropping the chain-rule factor of 2 — both errors hide inside the same term.

D18. Numerical differentiation

Trigger. No formula: only sampled data, or a function available solely as a black box.

Forward difference.

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}, \text{ with error } O(h).$$

Central difference.

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}, \text{ with error } O(h^2), \text{ so it is the default choice.}$$

Second derivative.

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}.$$

Example.

Estimate $f'(1)$ for $f(x) = \ln x$ with $h = 0.01$.

Calculation	Method
$f(1.01) = 0.00995033 \dots$ $f(0.99) = -0.01005034 \dots$	Sample $f = \ln x$ at $x \pm h$
$\frac{0.00995033 - (-0.01005034)}{0.02} = \frac{0.02000067}{0.02}$	Apply the central difference formula
1.0000333, against the exact $f'(1) = 1$	Compare against the exact value
error $\approx 3.3 \times 10^{-5}$	Consistent with the $O(h^2)$ prediction $\frac{h^2}{3} = 3.3 \times 10^{-5}$

The step-size floor. Shrinking h reduces truncation error but amplifies floating-point cancellation, so the total error bottoms out near $h \approx \sqrt{\epsilon_{\text{mach}}} \approx 10^{-8}$ for a central difference. Going smaller makes the answer *worse*.

The complex-step derivative dodges that entirely for real-analytic f :

$$f'(x) \approx \frac{\text{Im}[f(x + ih)]}{h}.$$

There is no subtraction of nearby values, so $h = 10^{-20}$ is fine and the result is accurate to machine precision.

Part 3 - Techniques for Evaluating Integrals

Integration is search, not mechanics. There is no algorithm that turns an arbitrary elementary integrand into an elementary antiderivative, because most integrands do not have one. So the working method is: **match the integrand against a short list of recognizable shapes**, and the list is what follows.

The routing order that solves most integrals. Try these in sequence, and stop at the first that fits.

- 1. Is it a table entry already, or a table entry after algebraic rewriting? (I1)
- 2. Is there an inner function whose derivative is also present, up to a constant? (I2)
- 3. Is it a product of two unlike types? (I3, I4)
- 4. Is it powers of trig functions? (I5)
- 5. Does it contain $\sqrt{a^2 \pm x^2}$ or $\sqrt{x^2 - a^2}$? (I6, after I7 if the quadratic is not centered)
- 6. Is it a rational function? (I9 if improper, then I8)
- 7. Is it rational in **sin** and **cos**? (I10)
- 8. Does it contain a root of a linear expression? (I11)
- 9. Is the answer non-elementary? (I18, I19, I20)

Always verify. Differentiating the answer is a complete, cheap, and independent check. It catches sign errors, missing chain-rule factors, and dropped constants, and it is the one habit that makes integration self-correcting.

I1. Reverse the derivative table

Every derivative rule read backwards is an integration rule.

Integral	Result
$\int x^n dx$	$\frac{x^{n+1}}{n+1} + C, n \neq -1$
$\int \frac{dx}{x}$	$\ln x + C$
$\int e^x dx$	$e^x + C$
$\int a^x dx$	$\frac{a^x}{\ln a} + C$
$\int \sin x dx$	$-\cos x + C$

Integral	Result
$\int \cos x \, dx$	$\sin x + C$
$\int \sec^2 x \, dx$	$\tan x + C$
$\int \sec x \tan x \, dx$	$\sec x + C$
$\int \tan x \, dx$	$\ln \sec x + C$
$\int \sec x \, dx$	$\ln \sec x + \tan x + C$
$\int \frac{dx}{\sqrt{a^2 - x^2}}$	$\arcsin \frac{x}{a} + C$
$\int \frac{dx}{a^2 + x^2}$	$\frac{1}{a} \arctan \frac{x}{a} + C$
$\int \frac{dx}{x\sqrt{x^2 - a^2}}$	$\frac{1}{a} \operatorname{arcsec} \frac{ x }{a} + C$

Example (algebra first).

Evaluate $\int \frac{x^3 + 2\sqrt{x}}{x} \, dx.$

Steps.

Calculation	Method
$\int \frac{x^3 + 2\sqrt{x}}{x} \, dx = \int (x^2 + 2x^{-1/2}) \, dx$	Split the fraction term by term (no quotient rule for integrals)
$= \frac{x^3}{3} + 2 \cdot \frac{x^{1/2}}{1/2}$	Power rule on each term
$= \frac{x^3}{3} + 4\sqrt{x} + C$	Simplify the coefficient

The $\frac{u'}{u}$ pattern. $\int \frac{f'(x)}{f(x)} \, dx = \ln|f(x)| + C$ is worth recognizing on sight; it is where $\int \tan x \, dx$ comes from.

MATH265 exam question (Q16.1a) — pull the constant out of the root, then it is pure power rule.

Evaluate $\int (x^2 - x)\sqrt{3x} \, dx$.

Calculation	Method
$\sqrt{3x} = \sqrt{3} x^{1/2}$	No substitution needed; rewrite the root algebraically
$(x^2 - x)\sqrt{3} x^{1/2} = \sqrt{3} (x^{5/2} - x^{3/2})$	Distribute
$\int x^{5/2} dx = \frac{2}{7} x^{7/2}, \int x^{3/2} dx = \frac{2}{5} x^{5/2}$	Power rule on each term
$\sqrt{3} \left(\frac{2}{7} x^{7/2} - \frac{2}{5} x^{5/2} \right) + C$	Combine into the final antiderivative

MATH265 exam question (Q16.1d) — expand before reaching for substitution.

Evaluate $\int (x^2 - 4)^2 dx$.

Calculation	Method
No inner-derivative factor is present, so expand instead of substituting $u = x^2 - 4$	Diagnosis: a substitution here would leave a stray x behind
$(x^2 - 4)^2 = x^4 - 8x^2 + 16$	Expand the square
$\frac{x^5}{5} - \frac{8x^3}{3} + 16x + C$	Integrate term by term (power rule)

The trap. Writing $\frac{(x^2 - 4)^3}{3}$ as if the chain rule ran backwards — that shortcut needs the inner derivative $2x$ to be present in the integrand, and it is not.

MATH265 exam question (Q16.2c) — split the fraction term by term before reaching for a technique.

Evaluate $\int \frac{x^3 + \sqrt{5x} - 4}{x^2} dx$, naming the technique.

Calculation	Method
Technique: algebraic rewriting, then the power rule	There is no quotient rule for integrals
$\frac{x^3}{x^2} = x; \frac{\sqrt{5x}}{x^2} = \frac{\sqrt{5} x^{1/2}}{x^2} = \sqrt{5} x^{-3/2}; \frac{-4}{x^2} = -4x^{-2}$	Split the fraction term by term

Calculation	Method
$\int x \, dx = \frac{x^2}{2}; \int \sqrt{5} x^{-3/2} \, dx = -2\sqrt{5} x^{-1/2};$ $\int -4x^{-2} \, dx = \frac{4}{x}$	Integrate each piece with the power rule
$\frac{x^2}{2} - \frac{2\sqrt{5}}{\sqrt{x}} + \frac{4}{x} + C$	Combine the three pieces

The trap. Trying to integrate the numerator and denominator separately — divide through first.

I2. u -substitution

Trigger. The integrand contains a composite function *and* (a constant multiple of) the derivative of its inner function.

Method. Set u = the inner function, compute $du = u'(x) \, dx$, and rewrite so that **no x remains**. If an x survives, either solve u back for x or the substitution was the wrong choice.

Example (indefinite).

Evaluate $\int 2x\sqrt{x^2 + 1} \, dx$.

Steps.

Calculation	Method
$u = x^2 + 1$	Identify the inner function
$du = 2x \, dx$	Differentiate; $2x \, dx$ is exactly what sits outside the root
$\int \sqrt{u} \, du = \int u^{1/2} \, du$	Rewrite in terms of u , no x remains
$\frac{2}{3} u^{3/2}$	Power rule
$\frac{2}{3} (x^2 + 1)^{3/2} + C$	Back-substitute $u = x^2 + 1$
$\frac{2}{3} \cdot \frac{3}{2} (x^2 + 1)^{1/2} \cdot 2x = 2x\sqrt{x^2 + 1}$	Verify by differentiating; matches the integrand

Example (definite: change the limits).

Evaluate $\int_0^{\pi/2} \sin^3 x \cos x \, dx$.

Calculation	Method
$u = \sin x, du = \cos x \, dx$	Identify the inner function and its differential
$x = 0 \Rightarrow u = 0$ and $x = \frac{\pi}{2} \Rightarrow u = 1$	Convert the limits rather than back-substituting
$\int_0^1 u^3 \, du$	Rewrite entirely in terms of u
$\left[\frac{u^4}{4}\right]_0^1 = \frac{1}{4}$	Evaluate with the power rule and FTC-2

Converting the limits in step 2 is safer than back-substituting, because it removes the chance of evaluating the old limits in the new variable.

Example (an x left over). $\int x\sqrt{x+1} \, dx$: take $u = x + 1$, so $x = u - 1$ and $dx = du$. Then $\int (u - 1)\sqrt{u} \, du = \int (u^{3/2} - u^{1/2}) \, du = \frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2} + C$.

MATH265 exam question (Q16.1e) — the same leftover- x pattern, as a definite integral.

Evaluate $\int_2^4 x\sqrt{x-1} \, dx$.

Calculation	Method
$u = x - 1$, so $x = u + 1$ and $dx = du$	Substitute the leftover- x pattern
$x = 2 \Rightarrow u = 1, x = 4 \Rightarrow u = 3$	Convert the limits rather than back-substituting
$\int_1^3 (u + 1)u^{1/2} \, du = \int_1^3 (u^{3/2} + u^{1/2}) \, du = \left[\frac{2}{5}u^{5/2} + \frac{2}{3}u^{3/2}\right]_1^3$	Expand and integrate with the power rule
Using $3^{5/2} = 9\sqrt{3}$ and $3^{3/2} = 3\sqrt{3}$: at $u = 3$ the bracket is $\frac{28\sqrt{3}}{5}$; at $u = 1$ it is $\frac{16}{15}$	Evaluate the bracket at each limit
$\frac{28\sqrt{3}}{5} - \frac{16}{15} = \frac{84\sqrt{3} - 16}{15} \approx 8.6328$	Subtract (FTC-2)

MATH265 exam question (Q16.2a) — solving for the exact piece that appears, rather than isolating dx alone.

Evaluate $\int \frac{\cos(\sqrt{2x})}{\sqrt{x}} dx$.

Calculation	Method
$u = \sqrt{2x} = \sqrt{2} x^{1/2}$	Let u be the inside of the cosine
$du = \frac{\sqrt{2}}{2\sqrt{x}} dx$	Differentiate u
$\frac{dx}{\sqrt{x}} = \sqrt{2} du$	Solve for exactly the combination that appears in the integrand, not dx alone
$\int \cos(u) \sqrt{2} du = \sqrt{2} \sin(u) + C = \sqrt{2} \sin(\sqrt{2x}) + C$	Rewrite and integrate, then back-substitute
$\sqrt{2} \cos(\sqrt{2x}) \cdot \frac{\sqrt{2}}{2\sqrt{x}} = \frac{\cos(\sqrt{2x})}{\sqrt{x}}$	Verify by differentiating; matches the original integrand

I3. Integration by parts

Trigger. A product of two functions of *different types* (a polynomial and an exponential, a log and a power, an inverse trig and a power).

Formula. From the product rule, $\int u dv = uv - \int v du$.

Choosing u : LIATE. Pick u as the first type that appears in this list, and dv as the rest.

Letter	Type
L	Logarithmic ($\ln x$)
I	Inverse trig ($\arctan x$)
A	Algebraic (x^n)
T	Trigonometric ($\sin x$)
E	Exponential (e^x)

The rationale: things early in the list get *simpler* when differentiated, and things late in the list stay manageable when integrated.

Example. Evaluate $\int x e^x dx$.

Steps.

Calculation	Method
$u = x, dv = e^x dx$	LIATE: x is Algebraic, e^x is Exponential
$du = dx, v = e^x$	Compute the remaining pieces
$xe^x - \int e^x dx$	Apply integration by parts: $\int u dv = uv - \int v du$
$xe^x - e^x + C = (x - 1)e^x + C$	Finish the remaining integral
$\frac{d}{dx}[(x - 1)e^x] = e^x + (x - 1)e^x = xe^x$	Verify by differentiating

Example (the " $dv = dx$ " trick).

Evaluate $\int \ln x dx$.

Calculation	Method
$u = \ln x, dv = dx$	Only one factor is present; use the " $dv = dx$ " trick
$du = \frac{dx}{x}, v = x$	Compute the remaining pieces
$x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int dx$	Apply integration by parts
$x \ln x - x + C$	Finish the remaining integral

The same trick gives $\int \arctan x dx = x \arctan x - \frac{1}{2} \ln(1 + x^2) + C$.

Example (cyclic, where the integral returns).

Evaluate $I = \int e^x \sin x dx$.

Calculation	Method
$I = e^x \sin x - \int e^x \cos x dx$	Parts once, with $u = \sin x, dv = e^x dx$
$\int e^x \cos x dx = e^x \cos x + \int e^x \sin x dx = e^x \cos x + I$	Parts again on the new integral, keeping the same type for u ($\cos x$)
$I = e^x \sin x - e^x \cos x - I$	Substitute the second result back into the first
$2I = e^x(\sin x - \cos x)$	Solve algebraically for I
$I = \frac{e^x(\sin x - \cos x)}{2} + C$	Divide by 2

The critical detail in step 2 is keeping the *same* choice of type for u both times. Switching (taking $u = e^x$ on the second pass) unwinds the first step and returns $I = I$.

I4. Tabular integration

Trigger. A polynomial times something that integrates repeatedly without getting worse (e^{ax} , $\sin ax$, $\cos ax$).

Method. Two columns: differentiate the polynomial down to 0, integrate the other factor the same number of times, then multiply along the diagonals with alternating signs $+, -, +, -$.

Example.

Evaluate $\int x^3 e^{2x} dx$.

Steps.

- Build the table.

Sign	Differentiate x^3	Integrate e^{2x}
+	x^3	$\frac{1}{2}e^{2x}$
−	$3x^2$	$\frac{1}{4}e^{2x}$
+	$6x$	$\frac{1}{8}e^{2x}$
−	6	$\frac{1}{16}e^{2x}$
	0	

- Multiply each row's derivative by the *next* row's integral, with the row's sign:

$$+ x^3 \cdot \frac{1}{2}e^{2x} - 3x^2 \cdot \frac{1}{4}e^{2x} + 6x \cdot \frac{1}{8}e^{2x} - 6 \cdot \frac{1}{16}e^{2x}.$$

- Collect: $e^{2x} \left(\frac{x^3}{2} - \frac{3x^2}{4} + \frac{3x}{4} - \frac{3}{8} \right) + C$.

- Verify by differentiating: the e^{2x} terms cancel in pairs and leave x^3e^{2x} .

This replaces four rounds of integration by parts with one table.

I5. Trigonometric integrals

Case A: $\int \sin^m x \cos^n x dx$ with an odd power. Peel one factor off the odd power, convert the rest with $\sin^2 + \cos^2 = 1$, and substitute.

Example.

Evaluate $\int \sin^3 x \cos^2 x \, dx$.

Calculation	Method
$\sin^2 x \cos^2 x \cdot \sin x$	The sine power is odd, so split off one $\sin x$
$\sin^2 x = 1 - \cos^2 x$	Convert the remaining even part with the Pythagorean identity
$-\int (1 - u^2)u^2 \, du$	Substitute $u = \cos x$, $du = -\sin x \, dx$
$-\int (u^2 - u^4) \, du = -\frac{u^3}{3} + \frac{u^5}{5}$	Expand and integrate
$-\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$	Back-substitute $u = \cos x$

Case B: both powers even. Use the half-angle identities $\sin^2 x = \frac{1 - \cos 2x}{2}$ and $\cos^2 x = \frac{1 + \cos 2x}{2}$ to lower the degree.

Example.

$$\int \sin^2 x \, dx = \int \frac{1 - \cos 2x}{2} \, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

Case C: $\int \sec^m x \tan^n x \, dx$. If the secant power is even, save $\sec^2 x$ and substitute $u = \tan x$. If the tangent power is odd, save $\sec x \tan x$ and substitute $u = \sec x$.

Example (the one to memorize).

$$\int \sec^3 x \, dx \text{ fits neither pattern, so use parts with } u = \sec x, \\ dv = \sec^2 x \, dx$$

. It is cyclic like the $e^x \sin x$ example, and solving for the integral gives

$$\int \sec^3 x \, dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C.$$

Case D: products of different frequencies. Use the product-to-sum identities.

$$\sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)],$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)],$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)].$$

Example.

$$\int \sin 3x \cos 5x \, dx = \frac{1}{2} \int [\sin(-2x) + \sin 8x] \, dx = \frac{\cos 2x}{4} - \frac{\cos 8x}{16} + C.$$

This is also the calculation behind the orthogonality relations in Fourier series.

MATH265 exam question (Q16.1b) — mismatched frequencies removed with a double-angle identity, then Case A.

Evaluate $\int \sin(2x) \cos x \, dx$.

Calculation	Method
$\sin(2x) = 2 \sin x \cos x \Rightarrow \int 2 \sin x \cos^2 x \, dx$	The mismatched arguments $2x$ and x block direct substitution, so apply the double-angle identity first
$-2 \int u^2 \, du$	Case A, odd power of sine: substitute $u = \cos x$, $du = -\sin x \, dx$
$-\frac{2}{3} \cos^3 x + C$	Integrate and back-substitute
$\frac{d}{dx} \left(-\frac{2}{3} \cos^3 x \right) = 2 \sin x \cos^2 x = \sin(2x) \cos x$	Verify by differentiating; confirmed

MATH265 exam question (Q16.2b) — Case C recognized directly, as a definite integral.

Evaluate $\int_0^{\pi/3} \tan x \sec^2 x \, dx$.

Calculation	Method
$u = \tan x, du = \sec^2 x \, dx$	Case C, even secant power: $\sec^2 x$ is exactly $\frac{d}{dx} \tan x$
$x = 0 \Rightarrow u = 0; x = \frac{\pi}{3} \Rightarrow u = \tan \frac{\pi}{3} = \sqrt{3}$	Convert the limits
$\int_0^{\sqrt{3}} u \, du = \left[\frac{u^2}{2} \right]_0^{\sqrt{3}} = \frac{3}{2}$	Rewrite in u and evaluate with FTC-2

MATH265 exam question (Q16.2d) — Case C with the hint already split out.

Evaluate $\int \sec^3 x \tan x \, dx$, using $\sec^3 x \tan x = \sec^2 x \cdot (\sec x \tan x)$.

Calculation	Method
$u = \sec x, du = \sec x \tan x \, dx$	The hint isolates the piece that becomes du
$\int \sec^2 x (\sec x \tan x \, dx) = \int u^2 \, du$	Rewrite entirely in u
$\frac{u^3}{3} + C = \frac{\sec^3 x}{3} + C$	Integrate and back-substitute
$\frac{d}{dx} \frac{\sec^3 x}{3} = \sec^2 x \cdot \sec x \tan x = \sec^3 x \tan x$	Verify by differentiating; confirmed

I6. Trigonometric substitution

Trigger. A quadratic under a root, in one of three shapes.

Integrand contains	Substitute	Identity used	dx
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 = \cos^2$	$a \cos \theta \, d\theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 = \sec^2$	$a \sec^2 \theta \, d\theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 - 1 = \tan^2$	$a \sec \theta \tan \theta \, d\theta$

Example.

Evaluate $\int \sqrt{9 - x^2} \, dx$.

Steps.

Calculation	Method
$x = 3 \sin \theta, dx = 3 \cos \theta \, d\theta$	Shape is $\sqrt{a^2 - x^2}$ with $a = 3$
$\sqrt{9 - 9 \sin^2 \theta} = 3\sqrt{\cos^2 \theta} = 3 \cos \theta$	Apply $1 - \sin^2 = \cos^2$ ($\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ so cosine is non-negative)
$\int 3 \cos \theta \cdot 3 \cos \theta \, d\theta = 9 \int \cos^2 \theta \, d\theta$	Substitute everything in
$9 \left(\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right)$	Half-angle identity (I5 Case B)

Calculation	Method
$\sin \theta = \frac{x}{3} \Rightarrow \theta = \arcsin \frac{x}{3};$ $\sin 2\theta = 2 \sin \theta \cos \theta = 2 \cdot \frac{x}{3} \cdot \frac{\sqrt{9-x^2}}{3}$	Convert back with a reference triangle
$\frac{9}{2} \arcsin \frac{x}{3} + \frac{x\sqrt{9-x^2}}{2} + C$	Combine into the final antiderivative

Example (a root in the denominator).

Evaluate $\int \frac{dx}{x^2 \sqrt{x^2 + 4}}.$

Calculation	Method
$x = 2 \tan \theta, dx = 2 \sec^2 \theta d\theta, \sqrt{x^2 + 4} = 2 \sec \theta$	Shape is $\sqrt{a^2 + x^2}$ with $a = 2$
$\int \frac{2 \sec^2 \theta d\theta}{4 \tan^2 \theta \cdot 2 \sec \theta} = \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} d\theta$	Substitute everything in
$\frac{\sec \theta}{\tan^2 \theta} = \frac{\cos \theta}{\sin^2 \theta}$	Convert to sines and cosines
$\frac{1}{4} \int \frac{dw}{w^2} = -\frac{1}{4 \sin \theta}$	Substitute $w = \sin \theta$ and integrate
$\tan \theta = \frac{x}{2} \Rightarrow \sin \theta = \frac{x}{\sqrt{x^2 + 4}}$	Read $\sin \theta$ off a reference triangle
$-\frac{\sqrt{x^2 + 4}}{4x} + C$	Back-substitute into the antiderivative

Step 5 is where marks are lost. Draw the triangle: opposite = x , adjacent = a (for the tangent case), hypotenuse = $\sqrt{x^2 + a^2}$, and read every trig function off it.

Hyperbolic alternative. $x = a \sinh t$ for $\sqrt{a^2 + x^2}$ and $x = a \cosh t$ for $\sqrt{x^2 - a^2}$ avoid the back-substitution triangle at the price of hyperbolic identities.

I7. Completing the square

Trigger. An irreducible quadratic that has a linear term, under a root or in a denominator.

Method. Rewrite $ax^2 + bx + c$ as $a(x + \frac{b}{2a})^2 + (c - \frac{b^2}{4a})$, then substitute $u = x + \frac{b}{2a}$ to reach a centered form the table (I1) or I6 can handle.

Example.

Evaluate $\int \frac{dx}{x^2 + 4x + 13}.$

Steps.

Calculation	Method
$x^2 + 4x + 13 = (x^2 + 4x + 4) + 9 = (x + 2)^2 + 9$	Complete the square
$u = x + 2, du = dx: \int \frac{du}{u^2 + 9}$	Substitute to center the quadratic
$\frac{1}{3} \arctan \frac{u}{3}$	Table entry (I1) with $a = 3$
$\frac{1}{3} \arctan \frac{x + 2}{3} + C$	Back-substitute $u = x + 2$

With a linear numerator. Split it into "a multiple of the derivative of the denominator" plus "a constant". For $\int \frac{2x + 7}{x^2 + 4x + 13} dx$: write $2x + 7 = (2x + 4) + 3$, so the first piece is a $\frac{u'}{u}$ log and the second is the arctangent above. The result is $\ln(x^2 + 4x + 13) + \arctan \frac{x + 2}{3} + C$.

18. Partial fractions

Trigger. A rational function $\frac{P(x)}{Q(x)}$ with $\deg P < \deg Q$ (otherwise do I9 first).

The four decomposition cases. Factor Q completely over the reals, then assign terms:

Factor of Q	Contributes
Distinct linear $(x - a)$	$\frac{A}{x - a}$
Repeated linear $(x - a)^k$	$\frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \cdots + \frac{A_k}{(x - a)^k}$
Distinct irreducible quadratic $(x^2 + bx + c)$	$\frac{Ax + B}{x^2 + bx + c}$
Repeated irreducible quadratic, power k	one such term for each power up to k

Example (distinct linear).

Evaluate $\int \frac{3x + 11}{x^2 - x - 6} dx.$

Steps.

Calculation	Method
$x^2 - x - 6 = (x - 3)(x + 2)$	Factor the denominator
$\frac{3x + 11}{(x - 3)(x + 2)} = \frac{A}{x - 3} + \frac{B}{x + 2}$	Set up the decomposition (distinct linear factors)
$3x + 11 = A(x + 2) + B(x - 3)$	Clear denominators
$x = 3: 20 = 5A \Rightarrow A = 4; \quad x = -2: 5 = -5B \Rightarrow B = -1$	Cover-up method: plug in strategic values
$4 \ln x - 3 - \ln x + 2 + C$	Integrate term by term

Example (irreducible quadratic).

Evaluate $\int \frac{2x + 3}{(x - 1)(x^2 + 1)} dx.$

Calculation	Method
$\frac{A}{x - 1} + \frac{Bx + C}{x^2 + 1}$	Set up with a linear numerator over the irreducible quadratic
$2x + 3 = A(x^2 + 1) + (Bx + C)(x - 1)$	Clear denominators
$x = 1: 5 = 2A \Rightarrow A = \frac{5}{2}$	Plug in the root of the linear factor
$0 = A + B \Rightarrow B = -\frac{5}{2}$	Match x^2 coefficients
$3 = A - C \Rightarrow C = \frac{5}{2} - 3 = -\frac{1}{2}$	Match constant terms
$\frac{5}{2} \ln x - 1 - \frac{5}{4} \ln(x^2 + 1) - \frac{1}{2} \arctan x + C$	Integrate the three pieces: log, $u = x^2 + 1$ log, and the arctangent entry

Splitting a $\frac{Bx+C}{x^2+1}$ term. Always break it into $\frac{B}{2} \cdot \frac{2x}{x^2+1}$ (a log) plus $\frac{C}{x^2+1}$ (an arctangent). Trying to integrate it whole is the usual dead end.

I9. Long division first

Trigger. A rational function whose numerator degree is greater than or equal to the denominator degree.

Example.

Evaluate $\int \frac{x^3}{x^2 + 1} dx.$

Steps.

Calculation	Method
Numerator degree 3 $\geq$ denominator degree 2	Diagnosis: partial fractions does not apply yet
$\frac{x^3}{x^2 + 1} = x - \frac{x}{x^2 + 1}$ (check: $x(x^2 + 1) - x = x^3$)	Long divide
$\frac{x^2}{2}$	Integrate the polynomial part
$-\frac{1}{2} \ln(x^2 + 1)$	Integrate the remainder with $u = x^2 + 1$
$\frac{x^2}{2} - \frac{1}{2} \ln(x^2 + 1) + C$	Combine both parts

I10. The Weierstrass substitution $t = \tan \frac{x}{2}$

Trigger. A rational function of **sin** x and **cos** x that none of the trig techniques reach.

The conversions.

$$\sin x = \frac{2t}{1 + t^2}, \quad \cos x = \frac{1 - t^2}{1 + t^2}, \quad dx = \frac{2 dt}{1 + t^2}.$$

This turns *any* such integrand into an ordinary rational function of t , which I8 then finishes. It is the universal fallback for this family, and it is usually messier than a clever alternative, so reach for it last.

Example.

Evaluate $\int \frac{dx}{1 + \sin x}.$

Steps.

Calculation	Method
$\int \frac{\frac{2 dt}{1+t^2}}{1 + \frac{2t}{1+t^2}}$	Substitute all three Weierstrass conversions
$\int \frac{2 dt}{1 + t^2 + 2t}$	Multiply numerator and denominator by $1 + t^2$

Calculation	Method
$1 + 2t + t^2 = (1 + t)^2$	Recognize the perfect square
$\int \frac{2 dt}{(1 + t)^2} = -\frac{2}{1 + t}$	Integrate with the power rule
$-\frac{2}{1 + \tan \frac{x}{2}} + C$	Back-substitute $t = \tan \frac{x}{2}$

Cross-check by another route. Multiplying the original integrand by $\frac{1-\sin x}{1-\sin x}$ gives $\frac{1-\sin x}{\cos^2 x} = \sec^2 x - \sec x \tan x$, whose integral is $\tan x - \sec x + C$. The two answers differ by a constant, which is exactly what antiderivatives are allowed to do.

I11. Rationalizing substitutions

Trigger. An n th root of a linear expression, or several roots of x with different indices.

Method. Set u equal to the root itself (or, for mixed indices, to $x^{1/L}$ where L is the least common multiple of the indices).

Example.

Evaluate $\int \frac{dx}{1 + \sqrt{x}}.$

Steps.

Calculation	Method
$u = \sqrt{x}$, so $x = u^2$ and $dx = 2u du$	Set u equal to the root itself
$\int \frac{2u}{1 + u} du$	Rewrite entirely in u
$\frac{2u}{1 + u} = 2 - \frac{2}{1 + u}$	Improper as a rational function, so divide (I9)
$2u - 2 \ln 1 + u $	Integrate
$2\sqrt{x} - 2 \ln(1 + \sqrt{x}) + C$	Back-substitute $u = \sqrt{x}$

Mixed indices. For $\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}}$ take $u = x^{1/6}$, since $\text{lcm}(2, 3) = 6$; then $\sqrt{x} = u^3$, $\sqrt[3]{x} = u^2$, and $dx = 6u^5 du$, leaving a rational function of u .

I12. Reduction formulas

Trigger. A power left symbolic, or a power high enough that repeated parts would be tedious.

Derivation. Apply integration by parts once and arrange the result to reference the same integral at a lower power.

Key formulas.

$$\begin{aligned}\int \sin^n x \, dx &= -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx, \\ \int x^n e^x \, dx &= x^n e^x - n \int x^{n-1} e^x \, dx, \\ \int \sec^n x \, dx &= \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx.\end{aligned}$$

Example.

Evaluate $\int \sin^4 x \, dx$.

Steps.

Calculation	Method
$-\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \int \sin^2 x \, dx$	Apply the reduction formula with $n = 4$
$\int \sin^2 x \, dx = \frac{x}{2} - \frac{\sin 2x}{4}$	Apply it again with $n = 2$ (or reuse the half-angle result)
$-\frac{\sin^3 x \cos x}{4} + \frac{3x}{8} - \frac{3 \sin 2x}{16} + C$	Combine the two pieces

Wallis's formula is the definite-integral version, and it collapses the recursion to a single product:

$$\int_0^{\pi/2} \sin^n x \, dx = \begin{cases} \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2} & n \text{ even} \\ \frac{(n-1)!!}{n!!} & n \text{ odd} \end{cases}$$

$$\text{So } \int_0^{\pi/2} \sin^4 x \, dx = \frac{3 \cdot 1}{4 \cdot 2} \cdot \frac{\pi}{2} = \frac{3\pi}{16}.$$

I13. Symmetry and periodicity

Trigger. A definite integral over an interval symmetric about 0 (or about the center of a period).

The rules.

$$\int_{-a}^a f(x) dx = 0 \text{ if } f \text{ is odd,} \quad \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f \text{ is even.}$$

Example.

Evaluate $\int_{-2}^2 (x^3 \cos x + x^2) dx.$

Steps.

Calculation	Method
$x^3 \cos x$ is odd (odd times even); x^2 is even	Split the integrand by parity
Odd part contributes 0	Symmetric-interval rule for an odd function; no antidifferentiation needed
$2 \int_0^2 x^2 dx = 2 \cdot \frac{8}{3}$	Even part doubles onto $[0, 2]$
$\frac{16}{3}$	Add the two parts

Periodicity. If f has period T , then $\int_a^{a+T} f = \int_0^T f$ for every a , so you may slide the window to wherever the algebra is easiest.

I14. The reflection (king) property

Trigger. A definite integral where substituting $x \mapsto a + b - x$ produces something that combines with the original.

Property. $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx.$

Example.

Evaluate $I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx.$

Steps.

Calculation	Method
Reflect with $x \mapsto \frac{\pi}{2} - x$	No standard technique applies cleanly, so try the king property

Calculation	Method
$I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx$	Under the reflection, $\sin x \rightarrow \cos x$ and $\cos x \rightarrow \sin x$
$2I = \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}$	Add the two expressions for I ; denominators match, so numerators add
$I = \frac{\pi}{4}$	Divide by 2

The antiderivative of the original integrand exists but is unpleasant; the reflection never computes it at all.

I15. The Fundamental Theorem of Calculus, part 2

Statement. If $F' = f$ on $[a, b]$ and f is continuous there, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Example.

Evaluate $\int_1^2 (3x^2 - 2x) dx$.

Steps.

Calculation	Method
$F(x) = x^3 - x^2$	Antidifferentiate
$F(2) = 8 - 4 = 4$	Evaluate at the top limit
$F(1) = 1 - 1 = 0$	Evaluate at the bottom limit
$4 - 0 = 4$	Subtract (FTC-2)

The trap: a discontinuity inside the interval. Writing $\int_{-1}^1 \frac{dx}{x^2} = \left[-\frac{1}{x}\right]_{-1}^1 = -2$ is wrong, and visibly so since the integrand is positive everywhere. The integrand blows up at 0 , which is inside the interval, so the theorem does not apply and the integral must be treated as improper (I17). It diverges.

Net change. $\int_a^b f'(x) dx = f(b) - f(a)$ is the same statement read as "the integral of a rate is the total change", which is where displacement, total growth, and accumulated cost all come from.

MATH265 exam question (Q16.1c) — symbolic limits are still just numbers once evaluated.

Evaluate $\int_a^b (x + \cos(2x))dx$.

Calculation	Method
$\int x dx = \frac{x^2}{2}; \int \cos(2x) dx = \frac{\sin(2x)}{2}$ (the $\frac{1}{2}$ from the inner derivative of $2x$)	Antidifferentiate term by term
$\left[\frac{x^2}{2} + \frac{\sin(2x)}{2} \right]_a^b$	Apply the limits
$\frac{b^2 - a^2}{2} + \frac{\sin(2b) - \sin(2a)}{2}$	Evaluate at b minus at a (FTC-2); still just a number once a, b are assigned

The trap. This is a **definite** integral, so no $+C$ — the letters a, b make it look symbolic, but it is a number as soon as they are assigned values.

MATH265 exam question (Q19.1) — a plain FTC-2 evaluation, dressed as an area-between-curves problem.

Find the area between $y = x$ **and** $y = 2 - x^2$.

Calculation	Method
$x = 2 - x^2 \implies x^2 + x - 2 = 0 = (x + 2)(x - 1) \implies x = -2, 1$	Intersect the two curves
At $x = 0$: parabola (2) sits above the line (0) on the whole interval	Test a point between the intersections; no split needed
$A = \int_{-2}^1 [(2 - x^2) - x] dx$	Set up top minus bottom
$\left[2x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-2}^1 = \frac{7}{6} - \left(-\frac{10}{3} \right) = \frac{27}{6} = \frac{9}{2}$	Antidifferentiate and evaluate (FTC-2)

The trap. Integrating line minus parabola returns $-\frac{9}{2}$; an area is positive, so a negative result means the top and bottom were swapped.

MATH265 exam question (Q19.3) — the "net change" reading of FTC-2, applied directly.

Water drains from a tank at $r(t) = 180 - 6t$ L/min for $0 \leq t \leq 50$. Find the amount that leaves during the first 15 minutes.

Calculation	Method
$\text{Amount} = \int_0^{15} (180 - 6t) dt$	A rate is given and a total is wanted; the net-change reading of FTC-2
$[180t - 3t^2]_0^{15} = 2700 - 675 = 2025$ litres	Antidifferentiate and evaluate
Rate falls linearly $180 \rightarrow 90$ L/min, averaging 135 L/min; $135 \times 15 = 2025$	Sanity check against the average rate

I16. Riemann sums from the definition

Trigger. The question says "using the definition of the definite integral".

Definition. With $\Delta x = \frac{b-a}{n}$ and $x_i = a + i\Delta x$,

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x.$$

Example.

Evaluate $\int_0^1 x^2 dx$ from the definition.

Steps.

Calculation	Method
$\Delta x = \frac{1}{n}, x_i = \frac{i}{n}$	Set up the Riemann sum from the definition
$\sum_{i=1}^n \left(\frac{i}{n}\right)^2 \frac{1}{n} = \frac{1}{n^3} \sum_{i=1}^n i^2$	Write out the sum $\sum f(x_i) \Delta x$
$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$	Use the closed form for the sum of squares
$\frac{n(n+1)(2n+1)}{6n^3} = \frac{2n^3 + 3n^2 + n}{6n^3}$	Simplify
$\frac{2}{6} = \frac{1}{3}$	Take the limit by degree comparison (L8)

The sums needed for this technique.

$$\sum_{i=1}^n 1 = n, \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2.$$

Read in the other direction, this technique is L19.

I17. Improper integrals and convergence tests

Trigger. An infinite limit of integration, or an integrand that blows up somewhere on the interval.

Method. Replace the offending endpoint with a variable and take a limit. If the blow-up is interior, split at it and require *both* pieces to converge independently.

Example (infinite interval).

Evaluate $\int_1^{\infty} \frac{dx}{x^2}.$

Steps.

Calculation	Method
$\lim_{b \rightarrow \infty} \int_1^b x^{-2} dx$	Replace the infinite limit with a variable and take a limit
$\left[-\frac{1}{x} \right]_1^b = -\frac{1}{b} + 1$	Antidifferentiate and apply the limits
$0 + 1 = 1$	Take the limit as $b \rightarrow \infty$
Converges to 1	The limit exists and is finite

Example (blow-up at an endpoint).

$\int_0^1 \frac{dx}{\sqrt{x}} = \lim_{a \rightarrow 0^+} [2\sqrt{x}]_a^1 = 2 - 0 = 2$, which converges even though the integrand is unbounded.

The p -tests, which settle most cases by inspection.

Integral	Converges when
$\int_1^{\infty} \frac{dx}{x^p}$	$p > 1$
$\int_0^1 \frac{dx}{x^p}$	$p < 1$

The two conditions are opposite, which is the point: at infinity you need fast decay, near zero you need mild blow-up.

Comparison test. If $0 \leq f \leq g$ and $\int g$ converges, so does $\int f$; if $\int f$ diverges, so does $\int g$.

Limit comparison test. If $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L$ with $0 < L < \infty$, then $\int f$ and $\int g$ do the same thing.

Example.

Does $\int_1^\infty \frac{dx}{\sqrt{x^3+1}}$ converge?

Calculation	Method
For large x , integrand $\sim x^{-3/2}$	Identify the dominant behavior
$\frac{x^{3/2}}{\sqrt{x^3+1}} \rightarrow 1$, finite and positive	Limit comparison test against $g = x^{-3/2}$
$\int_1^\infty x^{-3/2} dx$ converges since $p = \frac{3}{2} > 1$	Apply the p -test to the comparison integral
The given integral converges	Limit comparison test transfers convergence

I18. Numerical quadrature

Trigger. No elementary antiderivative exists (e^{-x^2} , $\frac{\sin x}{x}$, $\sqrt{1+x^4}$), or you only have data points.

The three rules, with $h = \frac{b-a}{n}$:

Rule	Formula	Error
Midpoint	$h \sum f(\bar{x}_i)$	$O(h^2)$
Trapezoid	$\frac{h}{2}[f_0 + 2f_1 + \cdots + 2f_{n-1} + f_n]$	$O(h^2)$
Simpson (n even)	$\frac{h}{3}[f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 4f_{n-1} + f_n]$	$O(h^4)$

Simpson's coefficient pattern is 1, 4, 2, 4, $\dots$, 4, 1: the endpoints get 1, odd indices get 4, even interior indices get 2.

Example.

Approximate $\int_0^1 e^{-x^2} dx$ with Simpson's rule and $n = 4$.

Steps.

Calculation	Method
$h = 0.25$; nodes 0, 0.25, 0.5, 0.75, 1	Set up the grid for $n = 4$
$f_0 = 1, f_1 = 0.9394131, f_2 = 0.7788008, f_3 = 0.5697828, f_4 = 0.3678794$	Evaluate $f(x) = e^{-x^2}$ at each node
$1 + 4(0.9394131) + 2(0.7788008) + 4(0.5697828) + 0.3678794 = 8.9622646$	Weighted sum, Simpson's 1, 4, 2, 4, 1 pattern
0.7468554	Multiply by $\frac{h}{3} = \frac{1}{12}$
True value 0.7468241; error $\approx 3.1 \times 10^{-5}$	Compare against the true value

Error bounds, which is what makes these usable rather than merely plausible:

$$|E_{\text{trap}}| \leq \frac{K(b-a)^3}{12n^2}, \quad |E_{\text{Simpson}}| \leq \frac{M(b-a)^5}{180n^4},$$

where $K \geq |f''|$ and $M \geq |f^{(4)}|$ on the interval.

Why Simpson wins. Doubling n cuts the trapezoid error by 4 and Simpson's by 16. Simpson is exact for cubics, despite being built from parabolas, which is the pleasant surprise in its error term.

I19. Term-by-term series integration

Trigger. The integrand has a known power series and no elementary antiderivative.

Method. Substitute the series, integrate term by term inside its radius of convergence, and truncate when the next term is below your tolerance.

Example.

Evaluate $\int_0^1 \frac{\sin x}{x} dx.$

Steps.

Calculation	Method
$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots$	Expand as a Maclaurin series
$\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \frac{x^6}{5040} + \dots$	Divide by x ; removes the apparent singularity at 0

Calculation	Method
$1 - \frac{1}{18} + \frac{1}{600} - \frac{1}{35280} + \dots$	Integrate term by term on $[0, 1]$
General term $\frac{(-1)^n}{(2n+1)(2n+1)!}$	Identify the pattern
$1 - 0.0555556 + 0.0016667 - 0.0000283 = 0.9460828$	Sum four terms
Truncation error $\approx 3 \times 10^{-7}$	Alternating series with decreasing terms bounds the error by the first omitted term

The true value is $\text{Si}(1) = 0.9460831$, inside that bound.

I20. Differentiation under the integral sign (Feynman's trick)

Trigger. A definite integral with a parameter in it, or one where you can *introduce* a parameter so that differentiating simplifies the integrand.

Method. Define $I(a)$, differentiate under the integral sign (Leibniz, D17), evaluate the easier integral, then integrate back in a and fix the constant using a value of a where I is known.

Example.

Evaluate $\int_0^1 \frac{x^a - 1}{\ln x} dx$ for $a > -1$.

Steps.

Calculation	Method
$I(0) = \int_0^1 0 dx = 0$	Name the integral $I(a)$ and note the easy anchor
$I'(a) = \int_0^1 \frac{x^a \ln x}{\ln x} dx = \int_0^1 x^a dx = \frac{1}{a+1}$	Differentiate under the integral sign w.r.t. a ; $\frac{\partial}{\partial a} x^a = x^a \ln x$ cancels the awkward $\ln x$
$I(a) = \ln(a+1) + C$	Integrate back in a
$0 = \ln 1 + C \Rightarrow C = 0$	Fix the constant with the anchor $I(0) = 0$
$I(a) = \ln(a+1)$	Final result

Check at $a = 1$: $\int_0^1 \frac{x-1}{\ln x} dx = \ln 2 \approx 0.693147$.

Second classic. $\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx = \ln \frac{b}{a}$, obtained the same way by differentiating with respect to a .

I21. Multiple integrals and change of variables

Iterated integrals (Fubini). A double integral over a region is computed as two single integrals, inner first. The inner limits may depend on the outer variable; the outer limits must be constants.

Example (swapping the order to make it doable).

Evaluate $\int_0^1 \int_x^1 e^{y^2} dy dx.$

Steps.

Calculation	Method
$\int e^{y^2} dy$ has no elementary antiderivative	Diagnosis: the given order of integration is a dead end
$0 \leq x \leq 1, x \leq y \leq 1$	Describe the region (the triangle below $y = 1$, above $y = x$)
$0 \leq y \leq 1, 0 \leq x \leq y$	Re-describe the same region with y outside
$\int_0^1 \int_0^y e^{y^2} dx dy = \int_0^1 y e^{y^2} dy$	Swap the order of integration
$\left[\frac{1}{2} e^{y^2} \right]_0^1$	Substitute $u = y^2$ and integrate
$\frac{e-1}{2}$	Evaluate at the bounds

Sketching the region in step 2 is not optional; the new limits cannot be read off the old ones by symbol shuffling.

Change of variables. With $x = g(u, v)$, $y = h(u, v)$,

$$\iint_R f(x, y) dA = \iint_S f(g, h) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv,$$

where the Jacobian is $\begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$.

The three standard coordinate systems and their factors.

System	Substitution	Area/volume element
Polar	$x = r \cos \theta, y = r \sin \theta$	$r dr d\theta$
Cylindrical	$x = r \cos \theta, y = r \sin \theta, z = z$	$r dr d\theta dz$
Spherical	$x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi$	$\rho^2 \sin \phi d\rho d\phi d\theta$

Forgetting the r or the $\rho^2 \sin \phi$ is the defining error of this technique.

Example (polar).

Evaluate $\iint_{x^2+y^2 \leq 1} \sqrt{x^2 + y^2} dA.$

Calculation	Method
Region is a disk; integrand depends only on distance	Diagnosis: polar coordinates are indicated
$\sqrt{x^2 + y^2} = r, dA = r dr d\theta$	Convert to polar
$0 \leq r \leq 1, 0 \leq \theta \leq 2\pi$	Set the limits for the unit disk
$\int_0^{2\pi} \int_0^1 r \cdot r dr d\theta = \int_0^{2\pi} \frac{1}{3} d\theta$	Integrate the inner variable r
$\frac{2\pi}{3}$	Integrate the outer variable θ

The Gaussian integral, which is a single integral solved by going up to two dimensions.

Calculation	Method
$I = \int_{-\infty}^{\infty} e^{-x^2} dx$; no elementary antiderivative exists	Name the integral (a single integral solved by going up to two dimensions)
$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA$	Square it and rename the second variable
$\int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta$	Convert to polar
$\int_0^{\infty} r e^{-r^2} dr = \frac{1}{2}$	The Jacobian's r makes the inner integral elementary ($u = r^2$)
$I^2 = 2\pi \cdot \frac{1}{2} = \pi \Rightarrow I = \sqrt{\pi}$	Integrate over θ and take the square root

I22. Vector-calculus shortcuts

Trigger. A line or surface integral where a theorem replaces the parametrization with an easier one.

Line integral, by parametrization. $\int_C f ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt.$

Fundamental theorem for line integrals. If $\mathbf{F} = \nabla f$ (conservative), then

$\int_C \mathbf{F} \cdot d\mathbf{r} = f(\text{end}) - f(\text{start})$, and the path is irrelevant. Test for conservativeness in the plane:
 $P_y = Q_x.$

Green's theorem converts a closed line integral into a double integral:

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Example.

Evaluate $\oint_C (y^2 \, dx + 3xy \, dy)$ around the unit circle, counterclockwise.

Calculation	Method
Direct parametrization gives a messy trig integral	Diagnosis: apply Green's theorem instead
$P = y^2, Q = 3xy$	Identify P and Q
$Q_x - P_y = 3y - 2y = y$	Compute the Green's-theorem integrand
$\iint_D y \, dA$ over the unit disk	Convert the closed line integral to a double integral
Value is 0	Disk is symmetric about $y = 0$ and y is odd in y (I13)

Stokes' theorem ($\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$) and the **divergence theorem** ($\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \nabla \cdot \mathbf{F} \, dV$) are the same trade in higher dimensions: swap a hard integral over a boundary for an easier one over the interior, or the reverse.

Example (divergence theorem). For $\mathbf{F} = \langle x, y, z \rangle$ over the unit sphere: $\nabla \cdot \mathbf{F} = 3$, so the flux is $3 \cdot \text{Vol} = 3 \cdot \frac{4\pi}{3} = 4\pi$, with no surface integral computed at all.

Part 4 - MATH265 Exam Questions: Precalculus, Sketching, Optimization, and Integration Applications

Parts 1-3 above weave individual **MATH265 exam question** examples into the technique that solves them. That covers every exam question whose subject is a limit, a derivative, or an integral technique proper. The questions below don't fit that technique-catalog shape - they are precalculus foundations, graph transformations, sketching-from-conditions, and the applied optimization/integration word problems - so they are carried over verbatim from the course question bank at [MATH265.md](#) (Part 5: every exam question, restated and solved), grouped the same way that file groups them.

[Q19.1](#) (area between a line and a parabola) and [Q19.3](#) (net change, water flowing from a tank) already appear above as worked examples under **I15. The Fundamental Theorem of Calculus, part 2**, so they are not repeated here.

1. Functions and precalculus foundations

Q1.1 - Exact value of a cosine

Question.

Give the exact value of $\cos\left(-\frac{\pi}{12}\right)$.

Calculation	Method
$\cos\left(-\frac{\pi}{12}\right) = \cos\left(\frac{\pi}{12}\right)$	Kill the minus sign: cosine is even, so $\cos(-x) = \cos x$
$\frac{\pi}{12} = 45^\circ - 30^\circ$	Recognise the angle: $\frac{\pi}{12} = 15^\circ$ is not one of the five memorized angles, but it's a difference of two that are
$\cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right) = \cos \frac{\pi}{4} \cos \frac{\pi}{6} + \sin \frac{\pi}{4} \sin \frac{\pi}{6}$	Apply the cosine difference formula $\cos(A - B) = \cos A \cos B + \sin A \sin B$ (the sign flips: a minus inside gives a plus outside)
$= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4}$	Substitute the five memorized trig values
$= \frac{\sqrt{6} + \sqrt{2}}{4}$	Combine over one common denominator

Answer.

$$\cos\left(-\frac{\pi}{12}\right) = \frac{\sqrt{6} + \sqrt{2}}{4} \approx 0.9659258263$$

Möbius: $(\text{sqrt}(6)+\text{sqrt}(2))/4$

Trap. Typing $\text{sqrt}(6)+\text{sqrt}(2)/4$ parses as $\sqrt{6} + \frac{\sqrt{2}}{4} \approx 2.80$, which is not even in range for a cosine. Bracket the whole numerator.

Q1.2 - Composite functions and their domains

Question.

Let $f(x) = 2x^2 - 5$ and $g(x) = \frac{x+5}{2x-9}$.

Find the composite functions and their domains.

- a. $f(g(x))$
- b. $g(f(x))$

Calculation	Method
$f(g(x)) = 2\left(\frac{x+5}{2x-9}\right)^2 - 5$	Part a, step 1: substitute g into f - everywhere f has an x , write the whole of $g(x)$
$= \frac{2(x+5)^2}{(2x-9)^2} - 5 = \frac{2(x+5)^2 - 5(2x-9)^2}{(2x-9)^2}$	Part a, step 2: expand into a single fraction
$2x-9 \neq 0 \implies x \neq \frac{9}{2}$	Part a, step 3: the domain - a composite keeps the restrictions of the inner function and of the result; here g needs $2x-9 \neq 0$, and f (a polynomial) restricts nothing

Answer a.

$$f(g(x)) = 2\left(\frac{x+5}{2x-9}\right)^2 - 5, \quad \text{domain } (-\infty, \frac{9}{2}) \cup (\frac{9}{2}, \infty)$$

Möbius: $2*((x+5)/(2*x-9))^2-5$, domain $x \neq 9/2$

Calculation	Method
$g(f(x)) = \frac{(2x^2-5)+5}{2(2x^2-5)-9}$	Part b, step 1: substitute f into g
$= \frac{2x^2}{4x^2-19}$	Part b, step 2: simplify - the numerator collapses because $-5+5=0$, and the denominator is $4x^2-10-9$

Calculation	Method
$4x^2 - 19 \neq 0 \implies x^2 \neq \frac{19}{4} \implies x \neq \pm \frac{\sqrt{19}}{2}$	Part b, step 3: the domain - f is a polynomial and restricts nothing, so only the result's non-zero-denominator requirement applies

Answer b.

$$g(f(x)) = \frac{2x^2}{4x^2 - 19}, \quad \text{domain } x \neq \pm \frac{\sqrt{19}}{2}$$

Möbius: $\left[\frac{2x^2}{(4x^2-19)} \right]$, domain $\left[x \neq \sqrt{19}/2 \text{ and } x \neq -\sqrt{19}/2 \right]$

Trap. The numerator simplifying to $2x^2$ tempts you to say the domain is all reals. It is not: the denominator still bans $\pm \frac{\sqrt{19}}{2}$.

Q1.3 - Which equations define a function of x

Question. Determine which of the following equations are well defined functions with independent variable x , and explain.

- a. $\frac{2y}{x^2 - 3|x|} = 1$
- b. $y^2 + x^2 = 10$

Calculation	Method
$2y = x^2 - 3 x \implies y = \frac{x^2 - 3 x }{2}$	Part a, step 1: solve for y by multiplying both sides by the denominator
Each x gives exactly one y (no $\pm$) $\implies$ this is a function	Part a, step 2: test the definition of a function
$x^2 - 3 x = 0 \implies x (x - 3) = 0 \implies x = 0, x = 3, x = -3$	Part a, step 3: state the domain - the original equation's denominator restriction survives

Answer a. Yes, it is a function of x :

$$y = \frac{x^2 - 3|x|}{2}, \quad \text{domain } x \neq 0, x \neq \pm 3$$

Möbius: $\left[(x^2 - 3 \cdot \text{abs}(x)) / 2 \right]$, domain $\left[x \neq 0, x \neq 3, x \neq -3 \right]$

Calculation	Method
$y^2 = 10 - x^2 \implies y = \pm \sqrt{10 - x^2}$	Part b, step 1: solve for y

Calculation	Method
At $x = 1$: $y = 3$ and $y = -3$ both satisfy the equation	Part b, step 2: test the definition - the $\pm$ is fatal, since one input now has two outputs
$x^2 + y^2 = 10$ is a circle of radius $\sqrt{10}$ centred at the origin	Part b, step 3: the graphical reason - a vertical line through $x = 1$ cuts the circle twice, failing the vertical line test

Answer b. No, it is **not** a function of x , because $y = \pm\sqrt{10 - x^2}$ assigns two values of y to each x in $(-\sqrt{10}, \sqrt{10})$.

Möbius: not a function; the relation is $x^2 + y^2 = 10$, a circle of radius $\sqrt{10}$

Q1.4 - Volume of a cone as a function of its radius

Question. A cone has height double its radius, so $h = 2r$. The volume of a cone is $V = \frac{\pi r^2 h}{3}$.

- a. Express V as a function of r .
- b. Find the domain.
- c. Compute V for $r = 3$.

Calculation	Method
$h = 2r$	Part a, step 1: write the constraint - "height is double the radius"
$V(r) = \frac{\pi r^2 (2r)}{3} = \frac{2\pi r^3}{3}$	Part a, step 2: eliminate h by substituting $h = 2r$ into the volume formula, leaving only r

Answer a.

$$V(r) = \frac{2\pi r^3}{3}$$

Möbius: $\frac{2\pi r^3}{3}$

Calculation	Method
$r > 0$, that is $(0, \infty)$	Part b: the domain - algebraically $\frac{2\pi r^3}{3}$ accepts every real number, but a radius is a physical length, so a cone requires strictly positive radius

Answer b.

$$r > 0, \quad \text{that is } (0, \infty)$$

Möbius: $r > 0$

Calculation	Method
$V(3) = \frac{2\pi(3)^3}{3} = \frac{2\pi \cdot 27}{3} = 18\pi$	Part c: evaluate the formula from part a at $r = 3$

Answer c.

$$V(3) = 18\pi \approx 56.55 \text{ cubic units}$$

Möbius: 18*pi

Trap. Leaving the answer as $\frac{54\pi}{3}$ is unsimplified, and answering **56.55** when the question did not ask for a decimal throws away the exact form.

2. Graph transformations

Q2.1 - Graph a shifted parabola by transformations

Question. Give a labeled graph of $g(x) = 3(x + 4)^2$ by starting from a basic function and applying transformations, explaining the procedure. No credit for any other method.

Calculation	Method
$y = x^2$	Step 1: name the parent function - the basic parabola, vertex at $(0, 0)$, opening upward, through $(\pm 1, 1)$ and $(\pm 2, 4)$
$g(x) = 3(x + 4)^2$ gives $a = 3, b = 1, h = -4, k = 0$	Step 2: match the general form $y = a f(b(x - h)) + k$
$y = x^2 \rightarrow y = (x + 4)^2$, vertex $(0, 0) \rightarrow (-4, 0)$	Step 3: horizontal shift, first - $(x + 4)$ means $h = -4$, so the graph moves left 4 (inside operations run backwards)
$y = (x + 4)^2 \rightarrow y = 3(x + 4)^2$	Step 4: vertical stretch, second - $a = 3$ is outside, multiplying every output by 3 without moving the vertex ($3 \times 0 = 0$)
No further change to the vertex or orientation	Step 5: no reflection and no vertical shift - $a = 3$ is positive (nothing flips) and $k = 0$ (nothing moves up or down)

Answer. Order of transformations: **shift left 4, then stretch vertically by 3.**

Key points to label on the sketch:

$$\text{vertex } (-4, 0), \quad (-3, 3), \quad (-5, 3), \quad (-2, 12), \quad (-6, 12)$$

The curve opens upward, is symmetric about the vertical line $x = -4$, and its axis of symmetry should be drawn and labeled.

Möbius: vertex $(-4, 0)$, axis of symmetry $x = -4$, function $3(x+4)^2$

Trap. Shifting right 4 because the sign says $+4$. Set the bracket to zero: $x + 4 = 0$ gives $x = -4$, which is where the vertex lands.

Q2.2 - Graph a reflected absolute value by transformations

Question. State the transformations, in order, applied to the basic graph of $|x|$ to obtain $f(x) = -3|x - 2|$, then sketch. No credit for any other method.

Calculation	Method
$y = x $	Step 1: name the parent function - a V shape, corner at $(0, 0)$, slope -1 on the left and $+1$ on the right
$f(x) = -3 x - 2 $ gives $a = -3, b = 1, h = 2, k = 0$	Step 2: match the general form
$y = x \rightarrow y = x - 2 $, corner $(0, 0) \rightarrow (2, 0)$	Step 3: horizontal shift, first $(x - 2)$ means $h = +2$, so the graph moves right 2
$y = x - 2 \rightarrow y = 3 x - 2 $	Step 4: vertical stretch by 3, second - the magnitude of a is 3, tripling outputs and narrowing the V
$y = 3 x - 2 \rightarrow y = -3 x - 2 $	Step 5: reflection in the x -axis, third - the sign of a is negative, so the V flips to open downward
No further vertical shift	Step 6: no vertical shift - $k = 0$, so the corner stays on the x -axis

Answer. Order: **shift right 2, stretch vertically by 3, reflect in the x -axis.**

Key points to label:

corner $(2, 0)$, $(1, -3)$, $(3, -3)$, $(0, -6)$, $(4, -6)$

The corner at $(2, 0)$ is the **maximum**, the arms have slopes $+3$ on the left and -3 on the right, and the function is continuous everywhere but not differentiable at $x = 2$.

Möbius: corner $(2, 0)$, function $-3\text{abs}(x-2)$

Trap. Reflecting before stretching gives the same picture here, but reflecting before *shifting* does not. Always do the inside horizontal move first.

5. Graphical interpretation and sketching from conditions

These three questions are marked on the *features* of your sketch, so the answer below lists exactly what a marker looks for.

Q5.1 - Sketch a function from four conditions

Question. Sketch the graph of a single function $f(x)$ satisfying all of:

- $\lim_{x \rightarrow 0} f(x)$ does not exist
- $f(0) = 0$
- $\lim_{x \rightarrow \infty} f(x) = -1$
- f is not differentiable at $x = -2$

Condition 1: the limit at 0 does not exist. Meaning: the left and right pieces approach different heights, so the graph **jumps** at $x = 0$. Draw a jump discontinuity: for instance the left piece rising to an open circle at $(0, 2)$ and the right piece starting at an open circle at $(0, -3)$.

Condition 2: $f(0) = 0$. Meaning: despite the jump, the function is defined at 0, and its value is exactly 0. Draw a **filled dot at the origin $(0, 0)$** , sitting on neither branch. This is the condition students drop; the two open circles need a third, solid point at $(0, 0)$.

Condition 3: $\lim_{x \rightarrow \infty} f(x) = -1$. Meaning: far to the right the graph flattens onto the horizontal line $y = -1$. Draw the dashed asymptote $y = -1$ and have the right branch approach it without crossing repeatedly.

Condition 4: not differentiable at $x = -2$. Meaning: the graph exists there but has no single tangent. The cheapest legal choice is a **sharp corner** at $x = -2$, like the vertex of a V. A cusp or a vertical tangent would also satisfy it; a hole or jump would too, but a corner is easiest to draw convincingly.

Answer: the checklist your sketch must show.

jump at $x = 0$; solid dot $(0, 0)$; dashed $y = -1$ as $x \rightarrow \infty$; corner at $x = -2$

Möbius: sketch question, no typed answer; the four features above are the marks.

Trap. Making $x = -2$ a hole *and* calling it non-differentiable is technically fine but weaker, since a marker wants to see you know a corner is continuous-but-not-differentiable.

Q5.2 - Interpret five conditions, then sketch

Question. Interpret each condition in terms of the graph of f , then sketch one function satisfying all of them.

- a. $\lim_{x \rightarrow \infty} f(x) = 3$

- b. $f(0) = 0$
- c. $\lim_{x \rightarrow -1} f(x) = 2$
- d. $\lim_{x \rightarrow 1} f(x) = \infty$
- e. f is continuous but not differentiable at $x = 0$

Interpretation a. As x grows without bound the outputs settle at height **3**. Graphically: a **horizontal asymptote** $y = 3$ on the right.

Interpretation b. The graph passes through the origin. Graphically: the point $(0, 0)$ is on the curve.

Interpretation c. Approaching $x = -1$ from either side the outputs approach **2**. Note the condition says nothing about $f(-1)$ itself, so the point may be filled at $(-1, 2)$, or a hole with the actual value elsewhere. Graphically: the curve funnels to height **2** at $x = -1$.

Interpretation d. As x approaches **1** the outputs increase without bound on both sides. Graphically: a **vertical asymptote** $x = 1$, with the curve going to $+\infty$ on *both* sides.

Interpretation e. Continuous at **0** means no break, so the limit exists and equals $f(0) = 0$. Not differentiable means no unique tangent line there. Graphically: a **corner** at the origin, like $|x|$.

Answer: the checklist your sketch must show.

dashed $y = 3$ on the right; through $(0, 0)$ with a corner; value **2** at $x = -1$; dashed $x = 1$ with $+\infty$ both sides

A workable shape: a V-corner at the origin, the left arm passing through height **2** at $x = -1$, the branch just left of $x = 1$ climbing to $+\infty$, and a separate branch for $x > 1$ dropping from $+\infty$ and flattening onto $y = 3$.

Möbius: sketch question; the five interpretations above are the marks.

Trap. Condition d says ∞ , not $-\infty$ and not "does not exist", so **both** sides must go up. Drawing a sign change at $x = 1$ contradicts the condition.

Q5.3 - Interpret six conditions, then sketch

Question. Give the graphical interpretation of each, then sketch $f(x)$.

- a. $f(0) = -3$
- b. $\lim_{x \rightarrow \infty} f(x) = -2$
- c. $\lim_{x \rightarrow 3} f(x) = \infty$
- d. $f'(x) < 0$ on $[3, \infty)$
- e. $f'(x) < 0$ on $(-\infty, -2)$
- f. $f''(x) > 0$ on $[4, \infty)$

Interpretation a. The curve passes through $(0, -3)$.

Interpretation b. Far to the right the curve levels off at height -2 : a horizontal asymptote $y = -2$.

Interpretation c. At $x = 3$ the outputs blow up to $+\infty$ from both sides: a vertical asymptote $x = 3$, upward on both sides.

Interpretation d. On $[3, \infty)$ the function is **decreasing**. Combined with c and b, the branch to the right of $x = 3$ starts at $+\infty$ and falls, levelling onto $y = -2$.

Interpretation e. On $(-\infty, -2)$ the function is also **decreasing**, so on the far left the curve falls as you read left to right.

Interpretation f. On $[4, \infty)$ the curve is **concave up**, holding water. This is consistent with d: a decreasing, concave-up branch flattens onto its asymptote from above, which is exactly how it must meet $y = -2$.

Step: check the conditions are consistent. Right branch: decreasing from $+\infty$ (d), concave up from $x = 4$ on (f), approaching $y = -2$ (b). These agree. Nothing is said about f' on $(-2, 3)$, so you are free there, which is where you route the curve through $(0, -3)$ (a) and up to the asymptote at $x = 3$ (c).

Answer: the checklist your sketch must show.

through $(0, -3)$; dashed $y = -2$; dashed $x = 3$ with $+\infty$ both sides

falling on $(-\infty, -2)$ and on $[3, \infty)$; concave up on $[4, \infty)$

A workable shape: on the far left the curve falls toward a local minimum near $x = -2$, rises through $(0, -3)$ and continues up to $+\infty$ as $x \rightarrow 3^-$; on the right of $x = 3$ it comes down from $+\infty$, decreasing throughout, and flattens onto $y = -2$ from above, curving upward from $x = 4$ on.

Möbius: sketch question; the six interpretations above are the marks.

Trap. Approaching $y = -2$ from *below* on the right. That branch is decreasing and concave up, so it must sit **above** -2 and settle down onto it.

6. Rates of change - interpretation

Q6.1 - Average and instantaneous rate of population change

Question. The population P of a city in year y is $P(y)$. Write a sentence in layman's terms explaining the meaning of:

- a. $\frac{P(15) - P(2)}{13} = 1431$
- b. $P'(10) = 14000$

Calculation	Method
$\frac{\Delta P}{\Delta y}$, with $13 = 15 - 2$	Part a, step 1: recognise the structure - a change in population divided by a change in years, i.e. a slope between two points on the population curve
Average rate of change = slope of the secant line	Part a, step 2: name it - a slope between two separated points
People per year	Part a, step 3: attach units - population divided by years

Answer a. Between year 2 and year 15, the city's population grew by an average of **1,431** people per year. That is the average over the whole 13-year stretch, so it says nothing about any single year; the population might have surged in some years and shrunk in others and still average this.

Möbius: average rate of change $\boxed{= 1431}$ people per year over $\boxed{[2, 15]}$

Calculation	Method
$P'(10)$ is a derivative evaluated at one instant	Part b, step 1: recognise the structure - not a difference over an interval
Instantaneous rate of change = slope of the tangent line at year 10	Part b, step 2: name it

Answer b. In year 10, the city's population was growing at a rate of **14,000** people per year at that instant. It is the growth rate at that moment, so if the city kept growing at exactly that pace for a full year it would add about **14,000** people, but the actual increase over that year may differ because the rate itself keeps changing.

Möbius: instantaneous rate of change $\boxed{= 14000}$ people per year at $\boxed{y=10}$

Trap. Saying "the population increased by 14,000 in year 10". A derivative is a rate, not a total. Part a is an actual average over 13 years; part b is a speed at a single instant.

9. Tangent lines

Q9.1 - Tangent line perpendicular to a given line

Question. Find the values of x where the tangent line to $f(x) = 3x^2 + 4x - 3$ is perpendicular to the line $y = 6x + 2$.

Calculation	Method
$m = 6$	Step 1: read the slope of the given line, in $y = mx + b$ form

Calculation	Method
$m_{\perp} = -\frac{1}{6}$	Step 2: convert perpendicular into a number - perpendicular slopes are negative reciprocals
$f'(x) = 6x + 4$	Step 3: the tangent slope is the derivative
$6x + 4 = -\frac{1}{6}$	Step 4: set the derivative equal to the required slope - the equation the question is really asking you to solve
$6x = -\frac{1}{6} - 4 = -\frac{1}{6} - \frac{24}{6} = -\frac{25}{6} \implies x = -\frac{25}{36}$	Step 5: solve for x

Answer.

$$x = -\frac{25}{36} \approx -0.6944$$

Möbius: -25/36

Check. $f'\left(-\frac{25}{36}\right) = 6\left(-\frac{25}{36}\right) + 4 = -\frac{25}{6} + \frac{24}{6} = -\frac{1}{6}$, and $6 \times \left(-\frac{1}{6}\right) = -1$, which confirms perpendicularity.

Trap. Setting $f'(x) = 6$, which finds where the tangent is **parallel** to the line, not perpendicular. The question asks for the negative reciprocal.

13. Newton's method

Q13.1 - Approximate a root with Newton's method

Question. Use Newton's method to approximate the solution of

$$x^4 + 2x - 5 = 0 \quad \text{in } [1, 2]$$

Calculation	Method
$f(1) = 1 + 2 - 5 = -2, \quad f(2) = 16 + 4 - 5 = 15$	Step 1: confirm a root exists in the interval - sign changes negative to positive, and f is continuous, so a root lies in $(1, 2)$ by the Intermediate Value Theorem
$f'(x) = 4x^3 + 2,$ $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^4 + 2x_n - 5}{4x_n^3 + 2}$	Step 2: write the derivative and the Newton iteration formula
$x_0 = 1$	Step 3: choose a starting value - an endpoint of the given interval

Calculation	Method
$x_1 = 1 - \frac{-2}{6} = 1 + \frac{1}{3} = 1.33333333$	Step 4: iterate, keeping all decimals
$x_2 = 1.33333333 - \frac{f(1.33333333)}{f'(1.33333333)} = 1.261290323$	Step 4 (continued): second iteration
$x_3 = 1.255964749$	Step 4 (continued): third iteration
$x_4 = 1.255937549$	Step 4 (continued): fourth iteration
$x_5 = 1.255937548$	Step 4 (continued): fifth iteration
x_4 and x_5 agree to eight decimal places	Step 5: stop when the digits stop moving - the method has converged

Answer.

$$x \approx 1.2559$$

To more places, $x \approx 1.255937548$.

Möbius: 1.2559

Check. $f(1.255937548) \approx 0$ to eleven decimal places, and **1.2559** lies inside $[1, 2]$ as required.

Trap. Rounding to four decimals *between* iterations. Newton's method converges by doubling correct digits each step, and rounding early throws that away. Keep the full display and round only the final answer.

14. Curve sketching with calculus

Q14.1 - Sketch two curves with a full calculus analysis

Question. Sketch each graph, stating all critical points, cusps, vertical asymptotes and points of inflection.

- a. $f(x) = \frac{x^2}{x^2 - 1}$
- b. $f(x) = x + \sin x$

Calculation	Method
$x^2 - 1 = 0 \implies x = \pm 1$; domain $x \neq \pm 1$	Part a, step 1: domain and vertical asymptotes - neither factor cancels, so both are genuine vertical asymptotes

Calculation	Method
$\lim_{x \rightarrow \pm\infty} \frac{x^2}{x^2 - 1} = 1 \implies y = 1$	Part a, step 2: horizontal asymptote - equal degrees, so take the ratio of leading coefficients
$f'(x) = \frac{2x(x^2 - 1) - x^2(2x)}{(x^2 - 1)^2} = \frac{2x^3 - 2x - 2x^3}{(x^2 - 1)^2} = \frac{-2x}{(x^2 - 1)^2}$	Part a, step 3: first derivative, by the quotient rule
$f(0) = \frac{0}{-1} = 0 \implies \text{critical point } (0, 0)$	Part a, step 4: critical points - $f'(x) = 0$ requires $-2x = 0$; f' is undefined at $x = \pm 1$, but so is f , so those are asymptotes, not critical points
$f' > 0$ for $x < 0$, $f' < 0$ for $x > 0$	Part a, step 5: classify it - the denominator is always positive, so the sign of f' is the sign of $-2x$; increasing then decreasing makes $(0, 0)$ a local maximum
$f''(x) = \frac{6x^2 + 2}{(x^2 - 1)^3}$	Part a, step 6: second derivative
$6x^2 + 2 = 0 \implies x^2 = -\frac{1}{3}$ (no real solution)	Part a, step 7: inflection points - an inflection needs $f'' = 0$ with a sign change, but the numerator never vanishes, so there are none; the sign change across $x = \pm 1$ doesn't count since f isn't defined there

Answer a.

Vertical asymptotes $x = -1$, $x = 1$; horizontal asymptote $y = 1$

Critical point $(0, 0)$, a local maximum; no cusps; no inflection points

Shape: three branches. The outer two sit above $y = 1$ and fall toward it as $|x|$ grows; the middle branch is a downward cap through $(0, 0)$, dropping to $-\infty$ at both $x = -1^+$ and $x = 1^-$.

Möbius: VA $\boxed{x=-1, x=1}$; HA $\boxed{y=1}$; critical point $\boxed{(0,0)}$ local max; inflection points: $\boxed{\text{none}}$

Calculation	Method
$x + \sin x$ is defined for all real x	Part b, step 1: domain - no asymptotes, no cusps
$f'(x) = 1 + \cos x$	Part b, step 2: first derivative

Calculation	Method
$1 + \cos x = 0 \implies \cos x = -1 \implies x = \pi + 2k\pi, \quad k \in \mathbb{Z}$	Part b, step 3: critical points
$f'(x) = 1 + \cos x \geq 0$ everywhere; $f(\pi) = \pi$, so (π, π) is a critical point	Part b, step 4: classify them - since $\cos x \geq -1$ always, f is increasing on all of $\mathbb{R}$ and never turns around; these are horizontal tangents, not maxima or minima
$f''(x) = -\sin x$; $-\sin x = 0 \implies x = k\pi$; $f(k\pi) = k\pi \implies$ inflection points $(k\pi, k\pi)$	Part b, step 5: second derivative and inflection points - the sign of $-\sin x$ genuinely alternates across each $k\pi$, so every $x = k\pi$ is an inflection point, lying on the line $y = x$
Concave up on $(\pi, 2\pi)$ and its 2π -translates; concave down on $(0, \pi)$ and its translates	Part b, step 6: concavity - $f'' = -\sin x > 0$ where $\sin x < 0$

Answer b.

No asymptotes, no cusps, no maxima or minima

Critical points at $x = \pi + 2k\pi$ (horizontal tangents on an increasing curve)

Inflection points at $(k\pi, k\pi)$ for every integer k

Shape: a staircase-like curve rising forever, weaving around the line $y = x$, touching it at every multiple of π , with a flat spot at $x = \pi, 3\pi, 5\pi, \dots$

Möbius: critical points $\boxed{x=\pi+2*k*\pi}$; inflection points $\boxed{(k*\pi, k*\pi)}$; no asymptotes; increasing everywhere

Trap. Calling $x = \pi$ a local maximum because $f'(\pi) = 0$. The first derivative does not change sign there, so it is neither a max nor a min.

15. Extreme values and optimization

Q15.1 - All maxima and minima on closed intervals

Question. Find all maxima and minima on the indicated intervals.

- a. $f(x) = 2x^{5/3} - 5x^{4/3}$ on $[-1, 20]$
- b. $f(x) = x + \cos x$ on $[-\pi, 2\pi]$

Calculation	Method
$f'(x) = \frac{10}{3}x^{2/3} - \frac{20}{3}x^{1/3}$	Part a, step 1: differentiate
$f'(x) = \frac{10}{3}x^{1/3}(x^{1/3} - 2)$	Part a, step 2: factor out the common $\frac{10}{3}x^{1/3}$
$x^{1/3} = 0 \implies x = 0; \quad x^{1/3} = 2 \implies x = 8$	Part a, step 3: find the critical numbers - both lie in $[-1, 20]$; f' is defined everywhere since $x^{1/3}$ and $x^{2/3}$ accept negatives
$f(-1) = 2(-1) - 5(1) = -7; \quad f(0) = 0;$ $f(8) = 2(32) - 5(16) = -16;$ $f(20) \approx 294.72 - 271.44 = 23.28$	Part a, step 4: evaluate at critical numbers and endpoints - $(-1)^{1/3} = -1$, so $(-1)^{5/3} = -1$ and $(-1)^{4/3} = 1$
$-7, 0, -16, 23.28$	Part a, step 5: compare the four values

Answer a.

Absolute maximum $f(20) = 2 \cdot 20^{5/3} - 5 \cdot 20^{4/3} \approx 23.281$ at $x = 20$

Absolute minimum $f(8) = -16$ at $x = 8$

Also present: a **local maximum** at $(0, 0)$, since f rises on $[-1, 0]$, falls on $[0, 8]$, then rises again on $[8, 20]$.

Möbius: max $2 \cdot 20^{5/3} - 5 \cdot 20^{4/3}$ at $x=20$; min -16 at $x=8$

Trap. Treating $(-1)^{4/3}$ as undefined or negative. Read it as $((-1)^{1/3})^4 = (-1)^4 = 1$; the even outer power makes it positive.

Calculation	Method
$f'(x) = 1 - \sin x$	Part b, step 1: differentiate
$1 - \sin x = 0 \implies \sin x = 1 \implies x = \frac{\pi}{2} + 2k\pi$	Part b, step 2: critical numbers - within $[-\pi, 2\pi]$ the only such value is $x = \frac{\pi}{2}$
$f'(x) = 1 - \sin x \geq 0$ everywhere	Part b, step 3: note the monotonicity - $\sin x \leq 1$ always, so f is increasing on the whole interval; the extremes must be the endpoints
$f(-\pi) = -\pi - 1 \approx -4.1416; \quad f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} \approx 1.5708;$ $f(2\pi) = 2\pi + 1 \approx 7.2832$	Part b, step 4: evaluate at the critical number and both endpoints

Answer b.

Absolute maximum $f(2\pi) = 2\pi + 1 \approx 7.283$ at $x = 2\pi$

Absolute minimum $f(-\pi) = -\pi - 1 \approx -4.142$ at $x = -\pi$

The critical number $x = \frac{\pi}{2}$ gives neither: it is a horizontal tangent on a curve that keeps rising.

Möbius: max $2\pi + 1$ at $x = 2\pi$; min $-\pi - 1$ at $x = -\pi$

Q15.2 - Rectangle of least perimeter

Question. Find the dimensions of the rectangle of area 220 cm^2 with the smallest perimeter, and give that perimeter.

Calculation	Method
Constraint: $xy = 220$; Minimize: $P = 2x + 2y$	Step 1: name the variables and write both equations
$y = \frac{220}{x} \implies P(x) = 2x + \frac{440}{x}, \quad x > 0$	Step 2: use the constraint to eliminate one variable
$P'(x) = 2 - \frac{440}{x^2}$	Step 3: differentiate
$2 = \frac{440}{x^2} \implies x^2 = 220 \implies x = \sqrt{220} = 2\sqrt{55}$	Step 4: set to zero and solve - only the positive root is physical
$P''(x) = \frac{880}{x^3} > 0$ for $x > 0$	Step 5: confirm it is a minimum - concave up everywhere on the domain
$y = \frac{220}{\sqrt{220}} = \sqrt{220} = 2\sqrt{55}$	Step 6: find the other dimension - the rectangle is a square, the general result for fixed area and least perimeter
$P = 4\sqrt{220} = 8\sqrt{55}$	Step 7: compute the perimeter

Answer.

Dimensions $2\sqrt{55} \times 2\sqrt{55} \approx 14.83 \text{ cm} \times 14.83 \text{ cm}$

Minimum perimeter $8\sqrt{55} \approx 59.33 \text{ cm}$

Möbius: side $2\sqrt{55}$, perimeter $8\sqrt{55}$

Trap. Answering with the side length when the question also explicitly asks "What is the perimeter?". Both are marked.

Q15.3 - Absolute extremes by the Extreme Value Theorem

Question. Use the Extreme Value Theorem to find the absolute extreme values of

$$f(x) = \frac{1}{x^2 + 1} \quad \text{on } [-1, 1]$$

Calculation	Method
$x^2 + 1 \geq 1 > 0$, so f is continuous on all of $\mathbb{R}$, in particular on $[-1, 1]$	Step 1: check the theorem applies - the Extreme Value Theorem needs a continuous function on a closed, bounded interval, so f attains both an absolute max and min there
$f'(x) = -(x^2 + 1)^{-2}(2x) = \frac{-2x}{(x^2 + 1)^2}$	Step 2: differentiate - write $f(x) = (x^2 + 1)^{-1}$ and use the chain rule
$-2x = 0 \implies x = 0$	Step 3: find the critical numbers - the denominator is never zero, so f' is defined everywhere
$f(-1) = \frac{1}{2}, \quad f(0) = \frac{1}{1} = 1, \quad f(1) = \frac{1}{2}$	Step 4: evaluate at the critical number and both endpoints
$\frac{1}{2}, 1, \frac{1}{2}$	Step 5: compare the three values

Answer.

Absolute maximum $f(0) = 1$ at $x = 0$

Absolute minimum $f(-1) = f(1) = \frac{1}{2}$ at both $x = -1$ and $x = 1$

Möbius: max 1 at x=0; min 1/2 at x=-1 and x=1

Note. The minimum is attained at two points. Reporting only one endpoint is an incomplete answer.

Q15.4 - Cheapest rectangular box

Question. A rectangular box has base length twice its width and volume 120 cm^3 . Material costs \$1.20 per cm^2 ; the lid costs \$1.50 per cm^2 . Find the dimensions of the cheapest box and the minimum cost.

Calculation	Method
width = w , length = $2w$, height = h	Step 1: name the variables using the constraint immediately
$V = (2w)(w)(h) = 2w^2h = 120 \implies h = \frac{60}{w^2}$	Step 2: write the volume constraint

Calculation	Method
base: $2w^2 \text{ cm}^2$ @ \$1.20; four sides: $6wh \text{ cm}^2$ @ \$1.20; lid: $2w^2 \text{ cm}^2$ @ \$1.50	Step 3: itemise the surfaces, keeping the lid separate - this separation is the entire difficulty of the problem
$C = 1.20(2w^2 + 6wh) + 1.50(2w^2) = 5.4w^2 + 7.2wh$	Step 4: write the cost function
$C(w) = 5.4w^2 + 7.2w \cdot \frac{60}{w^2} = 5.4w^2 + \frac{432}{w}, \quad w > 0$	Step 5: eliminate h
$C'(w) = 10.8w - \frac{432}{w^2} = 0 \implies w^3 = 40 \implies w = \sqrt[3]{40} = 2\sqrt[3]{5} \approx 3.4200$	Step 6: differentiate and solve
$C''(w) = 10.8 + \frac{864}{w^3} > 0$ for $w > 0$	Step 7: confirm it is a minimum - concave up on the whole domain; numerically $C''(\sqrt[3]{40}) = 32.4 > 0$
length = $2w = 2\sqrt[3]{40} \approx 6.8399$; height = $\frac{60}{w^2} = 3\sqrt[3]{5} \approx 5.1299$	Step 8: recover the other dimensions
$C = 5.4(40^{2/3}) + \frac{432}{40^{1/3}} = \frac{324\sqrt[3]{25}}{5} \approx 189.476$	Step 9: compute the minimum cost

Answer.

$$\text{width } \sqrt[3]{40} \approx 3.42 \text{ cm, length } 2\sqrt[3]{40} \approx 6.84 \text{ cm, height } 3\sqrt[3]{5} \approx 5.13 \text{ cm}$$

Minimum cost: approximately \$189.48

Möbius: width $\boxed{40^{1/3}}$, length $\boxed{2 \cdot 40^{1/3}}$, height $\boxed{3 \cdot 5^{1/3}}$, cost $\boxed{324 \cdot 5^{2/3}/5}$ or $\boxed{189.48}$

Trap. Charging the lid at \$1.20 as well, or forgetting the base entirely. The box has a base at \$1.20, four sides at \$1.20, and a lid at \$1.50: six faces, two prices.

Q15.5 - Absolute extremes of a function with a cusp

Question. Find the absolute extreme values of

$$f(x) = (x^2 + 2x)^{2/3} \quad \text{on } [-2, 3]$$

Calculation	Method
$f'(x) = \frac{2}{3}(x^2 + 2x)^{-1/3}(2x + 2) = \frac{2(2x + 2)}{3\sqrt[3]{x^2 + 2x}}$	Step 1: differentiate with the chain rule

Calculation	Method
$2x + 2 = 0 \implies x = -1$	Step 2: critical numbers where $f' = 0$
$x^2 + 2x = 0 \implies x(x + 2) = 0 \implies x = 0, x = -2$	Step 3: critical numbers where f' is undefined - the denominator vanishes when the inside is zero; f is still defined there (equals 0), so these are cusps and count as critical numbers
$f(-2) = 0; \quad f(-1) = (-1)^{2/3} = 1; \quad f(0) = 0; \quad f(3) = 15^{2/3} \approx 6.0822$	Step 4: evaluate at every critical number and both endpoints - the two-thirds power is a cube root squared, so the result is never negative
0, 1, 0, 6.0822	Step 5: compare the four values

Answer.

Absolute maximum $f(3) = 15^{2/3} \approx 6.082$ at $x = 3$

Absolute minimum 0, attained at both $x = -2$ and $x = 0$

There is also a **local maximum** of 1 at $x = -1$, sitting between the two cusps.

Möbius: max $15^{(2/3)}$ at $x=3$; min 0 at $x=-2$ and $x=0$

Trap. Missing $x = 0$ and $x = -2$ because f' is undefined there rather than zero. A critical number is any point where f' is zero **or** undefined while f is defined, and here those points supply the minimum.

17. Properties of the definite integral

All three of these use the same bounding property, so learn it once:

$$\text{if } m \leq f(x) \leq M \text{ on } [a, b], \text{ then } m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$$

Q17.1 - Interval containing a secant integral

Question. Use the properties of the integral to find an interval containing

$$\int_0^{\pi/3} \sec x dx$$

Calculation	Method
$[a, b] = \left[0, \frac{\pi}{3}\right], \quad b - a = \frac{\pi}{3}$	Step 1: identify the interval and its width

Calculation	Method
$\sec 0 = 1$ (minimum), $\sec \frac{\pi}{3} = 2$ (maximum)	Step 2: find the extreme values of $\sec x$ on that interval - cosine decreases from 1 to $\frac{1}{2}$, so its reciprocal $\sec x$ increases from 1 to 2 (the interval stops short of $\pi/2$, so $\sec x$ stays finite)
$1 \cdot \frac{\pi}{3} \leq \int_0^{\pi/3} \sec x \, dx \leq 2 \cdot \frac{\pi}{3}$	Step 3: apply the bounding property $m(b-a) \leq \int_a^b f \, dx \leq M(b-a)$

Answer.

$$\frac{\pi}{3} \leq \int_0^{\pi/3} \sec x \, dx \leq \frac{2\pi}{3}$$

Numerically $1.047 \leq I \leq 2.094$.

Möbius: interval $[\pi/3, 2\pi/3]$

Verification. The exact value is $\ln(2 + \sqrt{3}) \approx 1.3170$, which does sit inside $[1.047, 2.094]$.

Q17.2 - Estimate an integral from its properties

Question. Use the properties of the definite integral to estimate

$$\int_{-1}^1 \frac{1}{x^2 + 1} \, dx$$

Calculation	Method
$[a, b] = [-1, 1], \quad b - a = 2$	Step 1: identify the interval and its width - the width is 2, not 1; this is the most common slip on this question
maximum $f(0) = 1$, minimum $f(\pm 1) = \frac{1}{2}$	Step 2: find the extreme values of the integrand (same function as Q15.3) - $f(x) = \frac{1}{x^2 + 1}$ is largest where x^2 is smallest
$\frac{1}{2}(2) \leq \int_{-1}^1 \frac{dx}{x^2 + 1} \leq 1(2)$	Step 3: apply the bounding property

Answer.

$$1 \leq \int_{-1}^1 \frac{1}{x^2 + 1} \, dx \leq 2$$

Möbius: interval $[1, 2]$

Verification. The exact value is $\arctan(1) - \arctan(-1) = \frac{\pi}{2} \approx 1.5708$, comfortably inside $[1, 2]$.

Q17.3 - Bound an integral from its properties

Question. Use the properties of definite integrals to bound

$$\int_{-2}^3 (x^2 + 2x)^{2/3} dx$$

Calculation	Method
$[a, b] = [-2, 3], \quad b - a = 5$	Step 1: identify the interval and its width
minimum 0 at $x = -2$ and $x = 0$, maximum $15^{2/3}$ at $x = 3$	Step 2: find the extreme values of the integrand - exactly the function from Q15.5 on exactly that interval, so the work is already done
$0 \cdot 5 \leq \int_{-2}^3 (x^2 + 2x)^{2/3} dx \leq 15^{2/3} \cdot 5$	Step 3: apply the bounding property

Answer.

$$0 \leq \int_{-2}^3 (x^2 + 2x)^{2/3} dx \leq 5 \cdot 15^{2/3} \approx 30.41$$

Möbius: interval $[0, 5 \cdot 15^{2/3}]$

Verification. The true value is about **10.67**, which lies inside $[0, 30.41]$. The bound is loose because the integrand only reaches its maximum at the single right endpoint, which is normal for this method: it gives a guaranteed range, not a sharp estimate.

Trap. Concluding the lower bound is negative. The exponent $\frac{2}{3}$ is an even power of a cube root, so the integrand is never negative, and **0** is the correct floor.

Q19.2 - Area between a cosine and a horizontal line

Question. Find the area between $y = \cos x$ and the horizontal line $y = \frac{1}{\sqrt{2}}$ on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Calculation	Method
$\cos x = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \implies x = \pm \frac{\pi}{4}$	Step 1: find where the curves cross - both crossings lie strictly inside the given interval, so the integral must be split
$x = 0 : \cos 0 = 1 > \frac{1}{\sqrt{2}} \implies \text{cosine on top on } \left(-\frac{\pi}{4}, \frac{\pi}{4}\right);$ $x = \frac{\pi}{2} : \cos \frac{\pi}{2} = 0 < \frac{1}{\sqrt{2}} \implies \text{line on top on } \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$	Step 2: decide which is on top in each piece by testing one point in each region (by symmetry the same holds on $\left(-\frac{\pi}{2}, -\frac{\pi}{4}\right)$)
$A = \int_{-\pi/2}^{-\pi/4} \left(\frac{1}{\sqrt{2}} - \cos x\right) dx + \int_{-\pi/4}^{\pi/4} \left(\cos x - \frac{1}{\sqrt{2}}\right) dx + \int_{\pi/4}^{\pi/2} \left(\frac{1}{\sqrt{2}} - \cos x\right) dx$	Step 3: write the total area as three integrals
$\left[\sin x - \frac{x}{\sqrt{2}}\right]_{-\pi/4}^{\pi/4} = \sqrt{2} - \frac{\pi}{2\sqrt{2}} \approx 0.30349$	Step 4: evaluate the middle piece using antiderivative $\sin x - \frac{x}{\sqrt{2}}$
$\left[\frac{x}{\sqrt{2}} - \sin x\right]_{\pi/4}^{\pi/2} = \frac{\pi}{4\sqrt{2}} - 1 + \frac{\sqrt{2}}{2} \approx 0.26247$	Step 5: evaluate the right piece using antiderivative $\frac{x}{\sqrt{2}} - \sin x$
left piece = 0.26247	Step 6: the left piece equals the right by symmetry - both $\cos x$ and the constant line are even, so the region is symmetric about the y -axis

Calculation	Method
$A = \left(\sqrt{2} - \frac{\pi}{2\sqrt{2}} \right) + 2 \left(\frac{\pi}{4\sqrt{2}} - 1 + \frac{\sqrt{2}}{2} \right) = \sqrt{2} + \sqrt{2} - 2$	Step 7: add all three pieces - the π terms cancel exactly between the middle and outer pieces

Answer.

$$A = 2\sqrt{2} - 2 \approx 0.8284 \text{ square units}$$

Möbius: $2*\text{sqrt}(2)-2$

Note on reading the question. Because an explicit interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is given, the answer is the **total** area over that whole interval, which is $2\sqrt{2} - 2$. If a version of this question instead asks only for the region **enclosed** by the two curves, that is the middle piece alone:

$$\sqrt{2} - \frac{\pi}{2\sqrt{2}} = \sqrt{2} - \frac{\pi\sqrt{2}}{4} \approx 0.3035$$

Read which one is wanted before you set up.

Trap. Integrating $\cos x - \frac{1}{\sqrt{2}}$ straight across $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ without splitting. That returns $2 - \frac{\pi}{\sqrt{2}} \approx -0.221$, a signed value in which the outer negative regions cancel part of the middle positive region.

Q19.4 - Work, raising one end of a chain

Question. A chain lying on the ground is 10 m long with mass 80 kg. Find the work required to raise one end to a height of 6 m.

Calculation	Method
$\rho = \frac{80 \text{ kg}}{10 \text{ m}} = 8 \text{ kg/m}$	Step 1: find the linear density
Only the top 6 m of chain is lifted; the remaining 4 m stays on the ground	Step 2: picture the final configuration - this is why the integral runs to 6, not to 10
$dm = \rho dy = 8 dy;$ $dW = (dm)gy = 8(9.8)y dy = 78.4 y dy$	Step 3: slice by final height y , $0 \leq y \leq 6$ - the piece ending at height y started on the ground, so it's lifted a distance y
$W = \int_0^6 78.4 y dy = 78.4 \left[\frac{y^2}{2} \right]_0^6 = 78.4 \cdot 18$	Step 4: integrate

Answer.

$$W = 1411.2 \text{ J}$$

Möbius: 1411.2 (using $g = 9.8$)

Note. With $g = 9.81$ the answer is **1412.64 J**. State which g you used.

Trap. Integrating from **0** to **10**, which treats the whole chain as lifted and gives **3920 J**. Only 6 m of chain leaves the ground.

Q19.5 - Work, stretching a spring

Question. 20 ft-lb of work stretches a spring 1 ft beyond its natural length.

- a. Find the spring constant.
- b. Find the work needed to stretch it 2 ft beyond natural length.

Calculation	Method
$F(x) = kx, \quad W = \int_0^d kx \, dx = \frac{1}{2}kd^2$	Part a, step 1: write Hooke's Law and the work integral
$20 = \frac{1}{2}k(1)^2 = \frac{k}{2}$	Part a, step 2: substitute the given data, $W = 20$ and $d = 1$
$k = 40$	Part a, step 3: solve

Answer a.

$$k = 40 \text{ lb/ft}$$

Möbius: 40

Calculation	Method
$W = \int_0^2 40x \, dx = [20x^2]_0^2 = 20(4)$	Part b, step 1: use the same formula with $d = 2$

Answer b.

$$W = 80 \text{ ft-lb}$$

Möbius: 80

Note. Doubling the stretch **quadruples** the work, because $W \propto d^2$. Answering 40 ft-lb by assuming proportionality is the trap this question is built to catch.

Q19.6 - Average temperature of a metal rod

Question. A rod 6 m long has temperature $3x$ degrees centigrade at distance x metres from one end. Find the average temperature.

Calculation	Method
$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$	Step 1: write the average value formula
$f(x) = 3x, \quad [a, b] = [0, 6], \quad b - a = 6$	Step 2: identify the pieces
$\int_0^6 3x dx = \left[\frac{3x^2}{2} \right]_0^6 = \frac{3(36)}{2} = 54$	Step 3: integrate
$f_{\text{avg}} = \frac{54}{6} = 9$	Step 4: divide by the width

Answer.

$$9^\circ \text{C}$$

Möbius: 9

Sanity check. The temperature runs linearly from **0** at one end to **18** at the other, and the average of a linear function is its midpoint value: $\frac{0+18}{2} = 9$. Confirmed.

Q19.7 - Find k from an average value

Question. Find a positive number k such that the average value of $f(x) = \frac{5}{x^2}$ over the interval between 1 and k is **32**.

Calculation	Method
$\frac{1}{k-1} \int_1^k \frac{5}{x^2} dx = 32$	Step 1: write the average value equation
$\int_1^k 5x^{-2} dx = \left[-\frac{5}{x} \right]_1^k = -\frac{5}{k} + 5 = 5 \left(1 - \frac{1}{k} \right)$	Step 2: integrate
$5 \left(\frac{k-1}{k} \right)$	Step 3: simplify the numerator into a single fraction - the step that makes the whole problem collapse
$\frac{1}{k-1} \cdot \frac{5(k-1)}{k} = \frac{5}{k}$	Step 4: divide by the width and watch $(k-1)$ cancel
$\frac{5}{k} = 32 \implies k = \frac{5}{32}$	Step 5: solve

Answer.

$$k = \frac{5}{32} = 0.15625$$

Möbius: 5/32

Note. k is positive, as required, but it is **less than 1**, so the interval is really $\left[\frac{5}{32}, 1\right]$. The phrase "between 1 and k " allows this, and the algebra is identical either way: reversing the limits flips the sign of both the integral and the width, and the two flips cancel.

Verification. On $\left[\frac{5}{32}, 1\right]$, the average of $\frac{5}{x^2}$ is exactly **32**, confirmed by direct computation.

Trap. Rejecting $k = \frac{5}{32}$ because it is smaller than 1 and hunting for a larger root. There is no other solution: the average simplifies to exactly $\frac{5}{k}$, which is one-to-one.

Q19.8 - Displacement and distance travelled

Question. A particle moves with velocity $v(t) = t^2 - 3t + 2$ m/s.

- a. Find the displacement on $[0, 3]$.
- b. Find the distance travelled on $[0, 3]$.

Calculation	Method
$s = \int_0^3 (t^2 - 3t + 2) dt$	Part a, step 1: displacement is the plain integral of velocity - signs are kept, so backward motion subtracts
$= \left[\frac{t^3}{3} - \frac{3t^2}{2} + 2t \right]_0^3 = 9 - \frac{27}{2} + 6 = \frac{3}{2}$	Part a, step 2: antidifferentiate and evaluate

Answer a.

$$\text{Displacement} = \frac{3}{2} = 1.5 \text{ m}$$

Möbius: 3/2

Calculation	Method
$d = \int_0^3 v(t) dt$	Part b, step 1: distance needs the absolute value
$t^2 - 3t + 2 = (t - 1)(t - 2) = 0 \implies t = 1, t = 2$	Part b, step 2: find where the velocity changes sign - both lie inside $[0, 3]$, so split there
$t = 0.5 : v > 0 \quad t = 1.5 : v < 0 \quad t = 2.5 : v > 0$	Part b, step 3: determine the sign on each piece - the particle moves forward, reverses, then moves forward again
$\int_0^1 v dt = \frac{5}{6}; \quad \int_1^2 v dt = -\frac{1}{6}; \quad \int_2^3 v dt = \frac{5}{6}$	Part b, step 4: integrate each piece

Calculation	Method
$d = \frac{5}{6} + \left -\frac{1}{6} \right + \frac{5}{6} = \frac{11}{6}$	Part b, step 5: add the absolute values

Answer b.

$$\text{Distance} = \frac{11}{6} \approx 1.833 \text{ m}$$

Möbius: 11/6

Check. The three signed pieces sum to $\frac{5}{6} - \frac{1}{6} + \frac{5}{6} = \frac{9}{6} = \frac{3}{2}$, matching part a. Distance always exceeds displacement when the velocity changes sign, and $\frac{11}{6} > \frac{3}{2}$ as expected.

Q19.9 - Sprinter with piecewise acceleration

Question. A sprinter in a 100 m race accelerates at 4 m/s^2 for 2 seconds, then accelerates at zero for the rest of the race. Find her time for the race.

Calculation	Method
$v(t) = \int 4 dt = 4t \implies v(2) = 8 \text{ m/s}$	Step 1: phase 1 velocity - integrate the acceleration, with $v(0) = 0$ from a standing start
$d_1 = \int_0^2 4t dt = [2t^2]_0^2 = 8 \text{ m}$	Step 2: phase 1 distance - integrate the velocity over the first 2 seconds
$v = 8 \text{ m/s}$ for the rest of the race	Step 3: phase 2 setup - zero acceleration means constant velocity, held at the value reached at the end of phase 1
$100 - 8 = 92 \text{ m}$	Step 4: remaining distance
$t_2 = \frac{92}{8} = 11.5 \text{ s}$	Step 5: time for phase 2
$T = 2 + 11.5 = 13.5 \text{ s}$	Step 6: total time

Answer.

$$T = 13.5 \text{ seconds}$$

Möbius: 13.5

Trap. Dividing the full 100 m by 8 m/s to get 12.5 s. That ignores the 8 m already covered while accelerating, and it ignores that she was slower than 8 m/s for those first 2 seconds.

Closing: the two habits that matter most

1. Verify every antiderivative by differentiating it.
2. Substitute before you choose a technique.